

Astr 101 Astronomy 101

Numbered Transcript

ASTR 101 • Dr. Brian Keating

Contents

Lecture 1. Looking Upward

Lecture 2. The Astronomer's Toolkit

Lecture 3. Measurements and Mysteries

Lecture 4. The Life of Stars

Lecture 5. Galaxies and Gravity

Lecture 6. Distance and Dark Matter

Lecture 7. Beginning to End

Lecture 8. Beyond Earth

Lecture 1: Looking Upward

At a glance — Keating opens the tour of the universe — looking upward and why astronomy matters.

- 1.1 Welcome, everybody, to what I hope and promise will be one of the most interesting and out-of-this-world courses you may have ever taken. This is astronomy. I've taught here before. This is my second course. There's planned to be a third course, but this course in particular is really the bedrock foundation of everything I cover, not only in my teaching in the university system, but also in my life and my whole worldview. Astronomy has been a part of my life since I was 12 years old.
- 1.2 And I'm going to take you on a journey through the universe throughout this course, and hopefully whet your appetite for what is really humanity's oldest science. And it's a science we can all do, and we probably all have done it in one way or another, just with the simple telescopes that we're born with in our eyes. And today, I'm going to take you on a vast tour of astronomy's history and also take you on a little bit of a glimpse of what we're going to cover in the next seven lectures. And I hope you're going to enjoy this ride through the cosmos.
- 1.3 So, I'm Brian Keating. I'm a cosmologist, which doesn't mean I do hair and makeup, as many people think. But there is a similarity between the term cosmology and cosmetology. As you may know, it means basically the beautiful world, the beautiful universe that we're surrounded by. And you may have also heard when somebody tells you that they're studying astronomy, that there's a likelihood that they might get confused. And in fact, on the way out here, I was asked what I was doing on the plane out to come to teach at Peterson Academy. And I said, I'm going to teach about astronomy.
- 1.4 And they said, "That's great. I'm a Gemini. What can you tell me about my horoscope?" And I used to say, "Well, you know," when people asked me such things, "I'm not an astrologer, I'm an astronomer." But now I really lean into it. "Oh, you're a Gemini? Oh, I have some bad news for you. You'd better stay home all next week. It's not going to look good for you." So astronomy and astrology are actually intimately connected. And we'll see how the early astrologers paved the way for people like me and my colleagues to do the astronomical work that we do to this day.
- 1.5 I study the universe using the tools of modern astronomy, which have to do with telescopes located on some of the tallest mountains in the world, sometimes shooting rockets into space with telescopes on them. But most of the time, I'm spending my life thinking about ways to improve how we measure the universe. The word astronomy comes from the Greek words astron, nomos, which basically means star-naming or star culture. So I like to think of us as being part of a cult. And I want to welcome you to this cult of people that study the universe.

- 1.6 And it's a really, really wonderful group of people, and there's going to be so many characters that I introduce to you along the way. Astronomy is a story of people. It's a story of places and times and the revolution that was unlocked by first contemplating where we exist in the universe and how what we know about the universe can actually inform us about who we are as a people. And so astronomy, as I said, is the oldest science. For me, it's the oldest science I've been involved with.
- 1.7 One night when I was 12 years old, I was living in New York state, town called Dobbs Ferry, and I couldn't sleep. It was three in the morning, and I woke up and I looked outside, and the Moon was glowing so bright, so close to the horizon. I couldn't contemplate how it could be so bright, and it seemed like I could just touch it. And I looked next to the Moon; there was another bright light, almost as if a piece of the Moon had broken off, but it wasn't.
- 1.8 And it wasn't Southwest Flight 1086 that I came in on yesterday. It was actually the planet Jupiter. But I didn't know that, for this was 1984. Google had not been invented yet, and it wouldn't be for 15 more years. So I had to wait until the Sunday New York Times came out. And in the Sunday New York Times, there was printed a column that showed the pattern of what the sky would look like. They don't even print that anymore. In fact, most newspapers give much more coverage to astrology than astronomy.
- 1.9 You almost never hear about astronomical things, even though on occasions like tonight, as we are recording this, there's going to be a partial eclipse of the Moon. And I would bet most people in the world don't know about it. But after this course, you'll know not just about eclipses, you'll know about the wonderful world of the heavens, starting with our solar system, extending out to the realm of the comets, past the realm of the planets, to other solar systems, and even the beginning of time itself, for astronomy encompasses all these things.
- 1.10 And when I found out that you could see a planet with your naked eye, I was transfixed and I begged and borrowed. I didn't steal. But I did start working my first job at age 12 in a delicatessen called the Venice Deli outside of Dobbs Ferry, New York. And after about six weeks, I could afford to buy one of these beauties, a telescope just like this one, a small refracting telescope. And we'll spend a lot of time talking about what technology was needed to invent a way that we could describe the universe.
- 1.11 Shown here is a picture of Galileo's sketch of the Moon. What's remarkable about astronomy is that it is the only science where you can not only see the things that great scientists of the past saw, but you can feel what they felt when they made the discoveries that they made, the visceral sense of looking at the Moon through a tiny telescope from anywhere on Earth. Anywhere on Earth, you can see the same exact craters that Galileo saw.

- 1.12** You can see the moons of the planet Jupiter, which was the planet that I saw, and know that it, too, is another world—and how, we will see. That discovery of moons around Jupiter transformed humanity. Not just astronomy, not just some nerds working away and looking up at the sky, or mathematicians, but human civilization was transformed forever. Effectively, the telescope is like a lever that can move the universe if you have a sensitive mind enough to perceive it.
- 1.13** So this is a sketch of what Galileo saw. Now, what's interesting, I'll show you later, the most prominent crater on this diagram, it doesn't exist. There's no crater that's that big on the Moon's surface. And if you've ever glimpsed it through a small telescope, you'll know that. We'll see soon that the crater is much smaller. So why did Galileo, who was the first scientist to use the scientific method, why did he depict it such? He was a great artist. He did it because he wanted to convey what he felt when he saw it.
- 1.14** And that's what I mean. You, unless you have \$12 billion, cannot feel what it was like to feel the discovery of the Higgs boson. You can't do it. You can't look through the James Webb telescope even if you wanted to. But you can feel the discovery of the first time that the Moon was not a perfect sphere, unblemished and made of crystalline ethereal material. But, no, it was pockmarked. It had mountains. It had valleys. It had lava flows. And Galileo knew all of this and more.
- 1.15** The journey from amateur to professional is really blurred. You don't have to be a professional to actually contribute to modern astronomy. And I'll describe ways that average everyday citizens, laypeople without even degrees yet, are making contributions to our understanding of our cosmos. And it's fascinating because again, you can't do that if you want to discover a new neutrino or you want to discover some fact about DNA or RNA catalysis or something. You can't do it. It's not impossible that you can become professional in those fields. But to make contributions, you can do that tonight, effectively.
- 1.16** I also have a co-author, a pretty famous co-author. His name is Galileo. So with my colleagues that I met through the course of my career, we recorded the first-ever audiobook. And this is not a plug for my books; thank goodness, they've been well received. But to think that a kid from Dobbs Ferry, New York, who really wasn't slated to be some scientist, could eventually make the first-ever audiobook, 22 hours long, you know. Suffering from insomnia, it'll help you fall asleep.
- 1.17** With the famous Carlo Rovelli and Nobel Prize-winning physicist Frank Wilczek and many other people, that we could record this book that literally changed the course of human history. For this was the book that first proposed, with data and evidence and rational philosophical argument, that the Earth went around the Sun and not the other way around. This was only existing in written form for four centuries until my colleagues and I made it into an audiobook that you can listen to. So this is just kind of tickles me that my journey to astronomy was launched by Galileo, my great hero.

- 1.18** And we'll talk a lot about him because he's one of the most fascinating characters in literature, in history, a wonderful writer, an incredible human being, and we're going to encounter him and many, many other characters. So I said a little bit about astrology. Once I went out when I was dating my wife, we went to an astrologer. OK, I'm going to admit it, I'm among friends now. She took me to an astrologer. I wanted to, you know, continue the courtship. So I didn't say, "I think it's nonsense. I'm an astronomer. I can't be seen here." I said, "OK, we'll go and get my horoscope read."
- 1.19** So we went there. Lady reads my horoscope. "This is going to happen. You're going to get this thing, promotion, and, you know, you're going to get married, whatever." And then, and I said to her, "Well, you asked me when we came in, what's your sign? And I said, well, I'm a Gemini." And she said, "Oh, yeah, so this is what's gonna happen to you." And I said at the end of it, I was like, "I just wanna make sure— I'm born in September. That's Gemini, right?" "Oh, no, no, no, that's Virgo. But don't worry, the same things are gonna happen anyway." So it didn't make a difference.
- 1.20** And that's the hallmark of something that's not scientific. Scientific hypotheses have to be testable; they have to not be provable. They have to be able to be proven wrong, or as we say, falsifiable. That's a hallmark of science. If you say anything can happen, that's not scientific at all. We won't cover cosmetology at all, as I say. Although the prefix cosmos, as I said, refers to beauty. And certainly our universe, if nothing else, is beautiful.
- 1.21** One thing I love about astronomy is that it's a safe space. It's free of politics, it's enduring, it's everlasting, it's eternal in a way that our worldly events are not. And so it gives us a safe space to think about things, the biggest picture things, the things that we normally don't get time to think about. And I'm not an exception. I'm a professional astronomer, I'm a cosmologist. I don't get to spend as much time thinking about the ultimate issues, contemplating where do we come from, where is the universe heading, and what is it all about?
- 1.22** The meaning questions. We kind of lose track of that in our day-to-day lives. Taking the kids to school, going to work, we don't get a chance to think about it. But here is a safe space to think about these questions and more. And in the chat, in the discussion, in the social network of Peterson, I interact with you all in it to basically answer questions and continue the dialogue because it's so special and important to me.
- 1.23** To reiterate what I said, the astronomer Galileo was the first person to really conduct science using what we call the scientific method, which is a process, a systematic process, to ensure that what you're doing is correct and to suspect at every moment that you could be wrong. That's the difference between scientists and people that practice, and people that practice, say, astrology, or charlatans that attempt to sell you some sort of snake oil or some false promise. They can't be proven wrong; they can only be proven right.

- 1.24 And when you encounter such individuals, you should always ask, how would you know that you're wrong? It could be some yogi or some real cult leader. How would you know if you're wrong? If you can't, if you can say, "well, I'm not wrong," if you hear them say that, or "I can't be proven wrong," then you know you're not dealing with something with science.
- 1.25 Now, there are many things that can't be proven. I can't prove to you that there's not a purple unicorn on the moon of Saturn called Enceladus. I can't prove to you right now. But technology could be created. In fact, soon we will go to Enceladus and we can look at the North Pole and see if there's a purple unicorn there. So what is not falsifiable today may someday become falsifiable, and therefore it can be part of science, even though that's kind of a frivolous example.
- 1.26 But other examples that have been around for thousands of years, in some cases longer, they have to be subject to claims that could be used to refute them, not just confirm them. I like to use this sort of logo or emblem of what we're trying to do in this course. We're trying to comprehend how our universe came into existence, which will be the subject of Lecture 7. Cosmology is a thriving part of astronomy, but I have a whole other course on that, which you're welcome to take at Peterson Academy.
- 1.27 There are objects in the universe made of matter, energy. Those forms of matter and energy eventually formed life. How that process took place, we'll discuss potential ways that it could take place on other planets called exoplanets in Lecture 8. We'll talk about what it means to be alive. And other courses in Peterson talk about consciousness, the brain and the mind, and how we contemplate these things. But it's really astounding that a science can do this now.
- 1.28 Literally, 100 years ago, none of this was known. None of what we'll talk about is known. It's a cutting-edge science, and yet it's the oldest science that exists, as I say. So it's a fascinating blend of new and old, and it challenges the human mind to think in ways that we're not used to thinking. And that's what makes it so much fun and such a thrill to get to do what I get to do every day of my life.
- 1.29 Despite the stereotype, scientists are people. Scientists are normal, everyday people. Often it's said that, how do you know if a scientist is outgoing? The answer, he looks at your shoes when he talks to you. But in reality, astronomers and other forms of scientists, my collaborators, are ordinary human beings. They're not only brilliant, they're not only courageous, some of these people are the most interesting characters that I've ever encountered, whether in real life or in literature, or in my scientific studies and scientific history.
- 1.30 And we'll encounter each of these individuals as we go through the course. They revolutionized our scientific understanding, but more than that, they changed what it meant to be human, to be part of the only civilization, the only form of animal, the only species that can contemplate these questions; and not only do that, but do that with

such character and flair, I find it irresistible. So I'll be talking about their personalities, and I hope you'll come to enjoy them and get to know them as friends, as I have.

- 1.31 So the very first person to sort of really think quantitatively about astronomical things, as well as many other things, creating the laws of logic, discovering that dolphins and whales are mammals, thousands of years ago, 2,000 years ago, was Aristotle. And one of the great things I love to point out are that the great men and women that we're going to meet not only had brilliant ideas, they were deeply flawed and had huge blunders and foibles and incredible oopsie moments that we're going to talk about.
- 1.32 For example, Aristotle—he proposed that the Earth is a sphere. That's pretty cool. A lot of people think, oh, Columbus had proven—no, no, no. Aristotle knew that, literally 2,000 years before. He had reasons for that that we'll encounter in just a moment. But as brilliant as he was, he also failed to do very simple forms of what we call due diligence, or evidence-collecting. He thought that women had fewer teeth than men. I mean, he could have asked Mrs. Aristotle, right? He could have said, "Could you open your mouth?" Maybe he had a daughter or what. He never did that.
- 1.33 He also claimed that heavier objects fall faster than lighter ones. Totally untrue. That took 1,600 more years to be disproven. He could have just gone out and done a measurement or done a thought experiment, as Galileo did, as we'll talk about. Cecilia Payne-Gaposchkin, she was the first person under extreme duress to really claim the process correctly of how stars burned. This wasn't known until less than 100 years ago. She was one of the first people to get a PhD from Harvard. And her work changed how we look at stars, not just as these pinpricks of light, but what they're actually doing is undergoing nuclear fusion, which was a process not even known when she conjectured it. Just a brilliant, brilliant scientist.
- 1.34 Encounter people that were so incredibly brilliant that their intellects got in the way of their very lives. Giordano Bruno famously claimed that every star was associated with another planet, and on those planets there were other worlds and people even. And that got him into a little bit of trouble with the Catholic Church. They told him, "Let's not talk about that so much." He kept talking about it, and they kept worrying about things that are natural to worry about.
- 1.35 On those planets, could, for example, the Catholic Church was concerned, could Jesus Christ have been to every one of those planets? "Stop talking about that, Bruno." They wouldn't let him talk about it. Eventually, they burned him at the stake in the year 1600. And he was right in such a way, but he was also wrong in a different type of way, in a more important way. And it cost him his life. Galileo, too, as we'll talk about, spent the last nine years of his life in prison for the heretical suggestion that the Earth goes around the Sun.
- 1.36 These are incredible individuals. These are people with courage. They knew that these things would happen. You don't see this in other forms of science. You don't see this in

zoology, perhaps, people willing to die for their claims. So it shows you the nexus between what we're doing and its importance in philosophy, in theology, and in our daily lives. And no other science can match what astronomy has to offer.

- 1.37** Caroline Herschel, she was the first female scientist in history who was a professional scientist, in that she got paid to do scientific research. Incredible. She helped to discover numerous comets, nebulae, things like asteroids and other objects. And for her work, she was awarded the first female to be awarded the Royal Astronomical Society's gold medal. At a time in the 17 and 1800s when women had no education, certainly could not contribute to most intellectual discourse. But astronomy was for everybody.
- 1.38** So the origins of astronomy is what we turn to next. We all can do it. We've all seen the Moon travel alongside of us, haven't we, in a car? It seems like it's not even moving. It's centered on us. We've seen these pinpricks of light. Some places that are dark, you can see the Milky Way itself. But to ancient people, not astronomers yet, thousands, tens of thousands, perhaps even as long as 100,000 years ago, people were grappling with, "What is this set of objects that seem to be centered on the Earth, these luminaries that illuminate the night sky?", in the years and millennia before Netflix and TikTok.
- 1.39** So the prehistoric origins of astronomy have been noted to go back, as I say, tens of thousands of years. The primitive people were tracking the objects in the night sky; the constellations seemed to change from season to season. The position and phase of the Moon would change. The Sun's appearance would not change. Year after year, decade, century, millennium after millennium. And later civilizations would associate these astronomical objects with events here on Earth. We joke, oh, well, "How could you believe the Sun was a god?" Well, the Sun's pretty important, I have to tell you. And it really is dominating our existence. Without the Sun, we wouldn't exist.
- 1.40** So they're right in a certain sense, that the Sun has this incredible importance. It certainly looks like it's going around the Earth. And I challenge almost any layperson, and even some of my students, "Prove to me," sometimes I'll say it. "Prove to me that the Earth goes around the Sun." They can't do it. Most even professional astronomers are ignorant of the history of astronomy because they're so anxious to get on to do the new cool discoveries and make news and headlines. Not all, but some.
- 1.41** So astronomy is the oldest science. Why? Because we're born with the only instruments we need, the eye. And the eye is the only organ in the human body that has a piece of the brain inside of it. And it's a piece of the brain called the retina. And the retina is the only part of the brain that's outside the cranial skull, the skull itself. And that is important because you have this magical connection between the optics, the lenses, the iris, that we'll talk about the structure of the eye. That connects to your brain, which then allows you to make thoughts, and those thoughts then allow you to transport, teleport yourself, to make experiments in time and in space throughout the entire universe.

- 1.42 No other science can do that. Do you realize how cool that is? And it's been done for literally tens, some say 100,000 years. This is Stonehenge with the Milky Way galaxy in the background. They used the positions of stars not just for calendars, but very important. When do I plant a crop? I mean, this is in the days before you could go to Whole Foods and pick up a \$20 loaf of bread. You had to know exactly when to plant. When is the flood season going to come? When is the most optimal time to plant these things? Because your very life depended upon it.
- 1.43 So being an astronomer carried great favor back then. There's a position in every kingdom that we know about for astronomers from the ancient Egyptians, 5,000 years ago, up until right now. I have a friend. His name is Lord Martin Rees. He is the royal astronomer of the United Kingdom. So he's the royal astronomer. I once asked him, "What do you do, Martin?" And he said, "I read the Queen her horoscope." No, he didn't. He was joking about that. But he was responsible for what takes place right outside of London, in Greenwich, England.
- 1.44 The meridian is a fictitious circle that goes around the circumference of the Earth that demarcates where time begins for all humans on Earth to reckon their clocks to. Nowadays, friends like Bill Phillips of mine at the National Institute of Standards and Technology can measure time not using the Earth's rotation, but by using the properties of atoms cooled to within a few millionths of a degree of absolute zero. Clocks that lose one second every trillion years. So far much more accurate than any ancient royal astronomer could ever contemplate.
- 1.45 And yet, and yet they did incredible work with these primitive observatories. We're not even sure exactly what, say, Stonehenge was for. There are certain properties of the pyramids we're just learning about—how they were oriented, what their positions were modeled after. And they all had to do with astronomical objects—constellations, stars, solstices, equinoxes. And it was very powerful in ancient Judea and Israel. It used to be important, for it used to be mandatory for two people to come to the court to declare that they saw the new Moon.
- 1.46 Why? Because Judaism and Islam are based on lunar calendars, Chinese calendar as well. But in Judaism, the signs and the seasons and the holidays are based upon when the new Moon starts, in April typically, for the holiday of Passover that commemorates the exodus from Egypt, which also occurred during, some say, an eclipse, an astronomical object, an astronomical event. So these things go back thousands of years, and the foundation of whole religions observed by literally more than a billion people on Earth. To this very day, I have students come to me and say, "Oh, Professor Keating, I'm going to have to—I might have to miss the class on Monday because it might be the beginning of Ramadan."
- 1.47 Why is that? Because it's a lunar calendar and it changes throughout the year. So we have to make accommodations to this very day for astronomical objects that impact the lives of, as I say, billions of people on Earth. So what were these primitive

calendars? Well, we're not going to study so much of the history of astronomy because there's way too much to get to, although it would be a fascinating course to teach here. But I'll say the following, that biblical astronomy was concerned with what was observable.

- 1.48** Again, the telescope wasn't invented until about the year 1600, when Bruno, as I mentioned, was burned at the stake. The telescope was the first technology used for astronomy other than the human eye and some very simple pieces of technology that I'll be talking about next lecture. So the Bible contains within it a cosmological statement as the very first statement of the foundations of Judaism, Christianity, of these two great world religions. The beginning statement is a statement about cosmology, the Big Bang, if you will, according to some people.
- 1.49** "In the beginning." The beginning of what? The beginning of the whole universe. "God created the heavens and the Earth and said, let there be light." We'll be talking a lot about light. Astronomy is, I hope to convince you in the course of this course, the most fascinating subject that you can study, but it's also, in some sense, the most difficult to study because we cannot really do experiments. As much as I make fun of my zoology or biology friends, they can take a fruit fly and do something to it, you know, expose it to some citric acid. I don't know. I have no idea what biologists do. I should go over there and see what they do.
- 1.50** But they can take some fruit fly and expose it to citric acid or salt or something like that, and see by comparison with something that they don't call the control subject, a fly that they don't expose to it, they can see what happens to the two of them. That's a proper experiment. They can do a population study of different species. We cannot do that. We cannot go to the surface of the Sun, change the temperature, and say, "What does that do to the sunspots on the surface of the Sun?" And it would have an effect, but we can't do an experiment.
- 1.51** We have to wait for information to come to us in basically three different forms. Electromagnetic radiation, which is a fancy way of saying light, but light of all wavelengths. I primarily study what's called radio wavelengths or microwave wavelengths in my research. Number two, we have things that crash into us on Earth every now and then, or we go to other objects, like the Moon, and collect physical samples. I'll be giving each and every one of you a meteorite later on in the course, and I'll talk about the importance of meteorites, not only for the enjoyment of space and everything around us, but for the possibility that they may have brought life to Earth. That's our final lecture.
- 1.52** And for those of you watching at home, you can also get your own meteorite at [briankeating.com /PA](http://briankeating.com/PA) for Peterson Academy. So you guys locally will get one in the next few lectures. That's number two. We have those two things, electromagnetic radiation and meteors or fragments of the Moon that were brought back by astronauts or rovers that go to comets and asteroids. Number three, things like neutrinos, gravitational

waves, long-wave, long-range propagation phenomena that can go through the entire cosmos. That's it. We don't get a chance to change the temperature of an asteroid and see what that would do to potential amino acids that are inside of it.

- 1.53** So it's very challenging. As humanity progressed from the ancient biblical conception that the Earth was certainly the center of the universe, we came to the Greeks who also believed the Earth was the center of the universe. And they started to give reasons why—physical reasons why—scientific hypotheses that could be tested and, if they were tested then, would have been completely falsified. But that's a good thing, because that is science. Remember, science is that which can be proven wrong, not which can only be proven right. Those are charlatans.
- 1.54** Here's an ancient Greek astronomical instrument so complicated there's no way I could figure out how to use it. It's some form of ancient slide rule for those of you that remember those things, probably not many of you. It can predict the positions of where the Sun, Moon, and some planets were. Now, the word planet is a Greek word, from the Greek etymology of something that travels or moves. Where do we see that? A plane. Plane. Same route. Wanderer. Moving object.
- 1.55** Planets are the only objects besides the Sun and the Moon that seem to move against the background of fixed stars. They thought those stars were somehow glued to the firmament, which was fixed and unchanging. This object was very important for one of the oldest and most important things in humanity's technological advance was timekeeping. Because if you could keep track of time, you could accurately navigate across the globe. If you could have accurate navigation, you could have commerce.
- 1.56** And it's still said that a tremendous amount of the world's commerce still goes through London, right near Greenwich, England, where the prime meridian is this fictitious circle that demarcates where time's origin is for every single human being on Earth. Before that, people had their own time zones, own clocks, own watches. You'd say you're going to meet me somewhere. I'm going to provide you 7,000 bushels of grain on this day, at this time, and you might not show up. So it made commerce possible. What made commerce possible? Timekeeping. What made timekeeping possible? Astronomy.
- 1.57** It impacts your life in a way that's hardly perceptible to the average person because it's so foundational to what we do and what our technology and what our technological society has become, thanks to astronomers. Fast forward a few hundred thousand years almost, in Persia, the Babylonians, they had their own approach to science that was very different from the Greeks. It really was isolated in many ways, and so they developed in parallel. And many of the contributions of these Persian and early medieval Islamic astronomers was to formalize things in terms of mathematics and in terms of nomenclature.

- 1.58** What do we call constellations? What do we call the different stars? How do we track them? They were also quite big on things like the zodiac and constellations. Now, what is the zodiac? Why do people care about the zodiac? Well, the zodiac was important only in the sense that, say, the Sun is important, right? The Sun dominates our life. It gives a possibility for life on Earth, as far as we know. And the Sun's position amidst the constellations it travels through—there's 12 constellations that are recognized as zodiac constellations.
- 1.59** Your zodiac sign—I'm a Virgo—that means the Sun was in the constellation Virgo on the day in September when I was born. That's all it means. Now, how could you see that? Think about it. How did they know the Sun was in the constellation Virgo if the Sun is there? Have you ever been able to see a constellation when the Sun is out? Of course not. So it required exquisite keeping of time and of noticing these patterns that repeat every season, every year, to know that, six months earlier, the constellation Virgo is visible in March.
- 1.60** And so they could keep track of that time, and they could say with great accuracy when the Sun would be there again. So astrology did have these unexpected benefits, much to my wife's delight. And the astrological encounters that these ancient people had gave rise to a rich, valuable data set for astronomers to later use. It's much the same way that alchemists provided knowledge and evidence for chemical reactions. And we'll actually be doing some chemical reactions in the next lecture, talking about how observations of those very few things that we have access to, electromagnetic radiation being the most dominant one, most important one, we can learn not only what these things look like in deep space, but what they're made of.
- 1.61** And this has ramifications throughout all of science. So astrology benefited astronomers, and alchemists benefited chemists, physicists, and even have influence to this very day, from things like the names of objects and the stars or chemicals, to the properties of their reactions, to the positions and data that could then be used by proper astronomers. And the lines got blurred, going from Greek astronomy, where there was a lot concern with the work of, as I said, Aristotle, but then dominated for 2,000 years nearly.
- 1.62** They were so persuasive, their minds were so astute, that their wrong arguments lasted for many generations. And it's not clear if it had to occur that way. Maybe in some parallel universe, what I talk about in my cosmology lectures, that such a thing would not have had to be possible. But at least in our universe, as it occurred, we needed to kind of go through this long period of time to discover the actual laws and to overthrow, as perhaps the most important lesson, that these brilliant scientists were not only stunningly right, but they were sometimes brilliantly wrong.
- 1.63** And even as we'll see in the case of Einstein, even some of his blunders that he said were fatal, they came back to be true. So that's what makes it so delightful, so interesting, that great men and women can make great discoveries and they can also

be wrong in the most stupendous ways. The Roman gods had a great influence on astronomy. Again, not scientific, astrological, but it actually plays into everything in our lives today. The names of these planets, Mercury, Mars, Venus, they're named after Roman gods.

- 1.64** Mars, we've been to; the Moon, we've been to; Venus, we've even been to. Not we, me personally, but spacecraft from Earth made by human beings have been to other worlds. Absolutely mind-blowing. Don't forget, the first human beings ever to fly, for 197 feet, were Orville and Wilbur Wright in 1903. 66 years later, two men walked on the Moon. 67 years later, this image was taken from Mars' surface. Where will we be in the next 60 years of astronomy?
- 1.65** This is the largest volcano in the known solar system. It's called Olympus Mons. It's about 50,000 feet tall. It's almost twice as tall as Mount Everest. It's almost half the span of the United States in diameter. And the way and the reasons for those unusual features will be discussed when we talk about the effects of gravity in astronomy. Here's a surface of Venus. Venus, we know, is a hellish world. It's almost the same size as the Earth. It has an atmosphere and has a rocky surface. But that's kind of where the similarities end.
- 1.66** Its atmosphere is almost entirely carbon dioxide, leading to a runaway greenhouse effect, making the surface so hot that the Soviet spacecraft that took this image melted 15 minutes after this picture was taken, about 600 degrees Celsius. So it melted and all it was able to do is transmit a couple of images. There have been other satellites that have orbited around—spacecraft, I should say—that have orbited around Venus, taken pictures.
- 1.67** But again, we want to make note of the fact that, even though they were wrong, even though astrology is complete bunk in terms of scientific accuracy, still influences us to this very day. We're here on a Tuesday. And in French, which is a Romance or Roman-based language, Latin-based language, the word for Tuesday is mardi, like Mardi Gras, Fat Tuesday. Mardi is named after Mars. Here are the other days of the week.
- 1.68** The only one that we use in English that's not named after a Roman god is Thursday, which is named after the Viking god Thor, Thor's day. Obviously, Sunday, Sun; Monday, Moon day; and everything else is named after something from a Roman god or, in the case of Thursday, a Viking god. Now and back to the Greeks, this cosmology went through several iterations to get to a more modern conception. Aristotle was the first, as I said, not only that women have fewer teeth than men, which is, of course, wrong and kind of, could have been confirmed by asking his wife to open her mouth.
- 1.69** But he was accurate in the sense that he claimed the Earth was spherical. Now the reason that he was right was wrong. It's not round for the reason that he said it was, which is basically that the most perfect shape is a sphere and things tend to fall towards the middle. And because of that, the most compact perfect shape is a sphere.

That was his reasoning. That's not right. It forms due to the gravitational effects of, in Newtonian physics that we'll discuss, how objects can form in a minimal energy, most compact configuration.

- 1.70** But he gave reasons for it at least, and then he gave evidence for it. He gave evidence for the spherical Earth. And here are just a couple of pieces of evidence. So, as I say, you see different stars at different seasons in the universe. If the Earth were flat, for example, you would see the same stars from anywhere on Earth. Also, when there was an eclipse of the Moon, which is occurring tonight at about 7:30 tonight—not for those of you watching at home, but I do mention when these events occur on my website, briankeating.com/PA for Peterson Academy, so I'll keep you abreast and informed of all the kinds of cool sightings that you can see in astronomy.
- 1.71** So during an eclipse you'll see a little bit of the shadow of the Earth with the notion that the Earth is casting a shadow on the Moon. If the lights coming from that direction are the Sun and the Earth is my hand, it's casting a shadow on my face. And we'll demonstrate that using these tools called orreries. It's hard to spell but good for Scrabble. There's a lot of good words in Scrabble like syzygy and many other things that I hope that you'll at least take away in your next game of Scrabble. And you'll credit me with those discoveries.
- 1.72** So the question of, once the Earth was known to be spherical or thought to be spherical, could you prove it? Could you measure the only important number that characterizes a sphere, which is its radius? Yes. That was done a few hundred years later by a Greek scientist named Eratosthenes. And here's how he did it. He went, or had helpers, scientific, we call them graduate student teaching assistants now. But he had colleagues that would go to a different place on Earth. He was in Greece. He sent somebody to Alexandria on the solstice, the summer solstice.
- 1.73** Solstice literally means Sun stationary, solstice stationary, and that's the longest day of the year. And in the northern hemisphere, it occurs in June. And he made a measurement of a shadow cast by two identical sticks, one in Alexandria, in Egypt, and one in Greece. And the one in Alexandria, the shadow was almost nonexistent, suggesting that the Sun was exactly overhead. Exactly, precisely casting almost no shadow. And in Greece, the shadow had a slight angle to it.
- 1.74** Doing some simple geometry, he was able to calculate the circumference of the Earth in units of the distance between Alexandria and Greece, which somebody could just walk off. So they got it in terms of feet. And by the way, a lot of the things that we connote and still use to this day, like feet and the second, those are all based on these ancient astronomical or physical measurements. The foot, and really the cubit, was literally the length of the arm or the foot of the Pharaoh 3,000 years ago. That's how we get the word foot, which is obviously changed from the hieroglyphics or Greek or Egyptian alphabet to what we use today.

- 1.75** So they measured it in terms of what they called stadiums or stadia. And they got a value that corresponds to about 12% larger than we actually know the Earth's circumference to be today. This highlights an important distinction about science. Science is never done. You must agree with me that measuring this fact about the Earth to an accuracy of about 10 or 12% is astounding. It is infinitely better than saying the Earth is flat. Do we agree with that? That the Earth is flat is wrong.
- 1.76** But if I tell you the Earth is a perfect sphere, that's also wrong. The Earth is not a perfect sphere, not just because it has mountains on it that deviate from perfect sphericity, although that would be enough to falsify the claim that the Earth is a perfect sphere. But it also has distortions due to the fact that it formed from this molten state and is rotating, so it bulges out at the equator and gets a little bit squished at the poles. So it has a shape called an ellipsoidal shape. It's very small, but it's actually more technically true to say that.
- 1.77** Now even that is not perfectly right, because to get the actual shape of the Earth, you need to measure the height at every point on the Earth's surface compared to the ray. OK, the point is that, if you say the Earth is flat, you're wrong. If you say it's a perfect sphere, you're wrong, but you're much, much, much, much less wrong, and so on. So, science is not about getting the answer perfectly right. It's about saying the answer and then saying, "By how much could I be wrong?" That's how you know you're talking to a scientist.
- 1.78** They tell you some number, they will say to you, "You weigh less than 1,000 pounds." Thanks a lot, that's great. But if they say you weigh exactly 152 pounds and 36 grams or something like that, I'm mixing up units, 3 ounces, then they're much more accurate and they're much more precise. OK, so when we get long past the classical Greek astronomy, we came into the Babylonian and Persian era. Not only did they bring us the constellations and their names and, and mathematics that we know and use to this very day, like algebra—that is an Arabic name— they brought us the first sort of simple tools to use to actually do data collection in astronomy.
- 1.79** Positions and crude measurements of time. They did that with tools that we call the astrolabe and the sextant and other objects. We're not going to spend too much time on it. They also brought us the names of the constellations, how they use them for the purposes of calendrical activities. When did a certain holiday start? When would a king come into office? For example, back in the classical and ancient world, if you had a contract, you're going to deliver, you know, such and such grain, or I'm going to give you so many head of oxen. You would say, "In the year of such-and-such's rule."
- 1.80** King Cyrus' rule. Alexander the Great's rule. You would say, "I would give you this on this third month," which they would name after one of the zodiac constellations. So back then, your birthday would be basically a month. You know, they wouldn't get it down to the day, typically, but the contracts would be in the month of a certain rule of the king, when a given constellation was where the Sun was. So astronomers are very

important for commerce. Kings, when they would be inaugurated, like election season, it wouldn't be on January 20th, it would be in the month of Nisan, you know, or the beginning of their calendar.

- 1.81** And they would use that based on the astronomical sign that was where the Sun was at that time. I say that they're not important, but they're useful because they delineate certain boundaries on the globe, right? If an alien is looking at the Earth, as we'll talk about in Lecture 8, they won't know that we're in Florida, or, you know, they'll just say you're on the land part of the surface of a world that's mostly 70% water and has certain mountains. You know, they won't know about our arbitrary distinctions. But once they come to visit, hopefully they'll be coming in peace.
- 1.82** We'll talk about that at the very end of the course—I say that's a cliffhanger—that they will perhaps make use of those in some other way, but you know, to take us to their leader or take them to our leader. So, the constellations change throughout the seasons. Some of them are visible all the time. Like, the Big Dipper in the northern hemisphere is almost always visible from everywhere in the northern hemisphere. The Southern Cross is visible in the southern hemisphere. And these have different names.
- 1.83** Again, we astronomers use them because if I want to say, "Oh, we're going to talk about exoplanets in Lectures 7 and 8, we're going to talk about how life could spring up on another planet," we have to say, well, where is that planet? Oh, it's located around a star, and it's the 51st star ever named, or 51st brightest star in the constellation Pegasus. And that planet that's been discovered that we'll talk about was the first planet discovered. So it is given the letter B. The star has the letter A, usually. And the planet, first one to be discovered, this one happens to be, as we'll talk about, one of the most Earth-like planets in the known, in our local galaxy at least.
- 1.84** Moving forward to more modern era, the Renaissance, we encounter some very interesting characters, including Tycho Brahe, Johannes Kepler, and still they were the last of the naked-eye astronomers only. They had these tools like sextants and astrolabes. But really, the first people to conjecture scientifically testable hypotheses that are falsifiable—that's the hallmark of good science—was Nicolaus Copernicus. So he had suggested, made a version of the solar system which didn't feature the Earth at the center.
- 1.85** And it featured the Sun, as you see there. Sol is the Sun, and it was a basic structure where the planets moved out and we were the third planet from the Sun, as we are, as we know now. But back then, this was a wild conjecture and there was no real evidence for it. So he had made these kind of conjectures based upon a principle known as Occam's Razor. Simplicity is truth, that the simplest explanation for something is usually the correct answer.
- 1.86** So, instead of the Sun being a god that can actually change the fate of life on Earth and its position on the day you were born, even though we have twins at home, and they

couldn't be more different, and the fact that they're 15 minutes separated in time should suggest that they have the exact same horoscopes, but they're radically different. So my kids can falsify astrology usually. But this basic structure could not really be proven or motivated for another three and a half centuries.

- 1.87** So the Ptolemaic model, which is really what Aristotle also believed, which is what the Church believed and why they were so vehemently opposed to people like Bruno and Galileo, this was that the Sun was, was really just orbiting around the Earth, which, by the way, is what it looks like. And it's very hard for an average person, and even, as I said, some of my students who are brilliant, to prove that that's true or to even recount the historical fact pattern that led us to believe that to be so.
- 1.88** So one of the most important successes of the early incarnation of the heliocentric model was that it explained what's called retrograde motion, which is that, for certain planets like Mars, Mars will orbit and make its way across the solar system, the constellations in the zodiac, and then at different points, it'll stop and go the opposite direction. That's called retrograde. This happens for the inner planets, primarily. And it's a phenomenon caused by the fact that the Earth and Mars are in orbit around the Sun. And as shown here, because Earth is closer to the Sun, it orbits faster.
- 1.89** So sometimes you'll see the Sun, it'll be close to Mars, what's called opposition, and for the months preceding it, it'll look like Mars is going in one direction. And as the Earth passes by Mars, because it's faster, it will look like Mars is moving backwards amidst the fixed constellations of the zodiac behind it. That can only be explained simply in the heliocentric Sun-centered universe. It could be explained in the Ptolemaic or Earth-centered cosmos only if you added extra orbital patterns called epicycles to the other planets. And that became very cumbersome and very difficult to maintain.
- 1.90** So this is what it looked like, say, in 2018. It goes from April, then it goes May 28 and June 28. So it looks like Mars is making this, like, hula hoop-type behavior, but nothing's happening to Mars. It's just going around, and we're going around it, and because of our relative motion differential, we catch up to it. So it looks like it's going in front of us, and then we catch up and then it goes behind. That can only be explained simply in the heliocentric model of Copernicus.
- 1.91** So we went from the Superdome model, and the biblical creation story talks about the Big Bang, if you will, or the origin of the universe, origin of light, origin of the Sun and Moon on the fourth day for some reason. And eventually you come to this notion that the heavens are like this extra layer of waters. They're called the upper waters, and the lower waters. And the lower waters are like our atmosphere. And the upper waters are supposedly the realm of the heaven, of the fixed stars. And then there are holes in the firmament and that lets rain in, and all sorts of very contrived things that we know now not to be true.

- 1.92 But it was the best that they could do with the tools that they had, which was really their mind and certain aspects of philosophy and so forth. So the planets, meaning those that move in Greek, wanderers, those are actually opaque, just like the Earth. They can actually make transits in front of other objects like the Sun. Tonight, the Moon—an amazing day that we happen to be convening, a very fortunate—astrologers would be very happy that we started this class today. Today is not only an eclipse of the Moon, but also the planet Saturn is being eclipsed and what's called occulted.
- 1.93 When a planet goes behind the Moon, the Moon is actually blocking the planet Saturn for a little bit. And you can actually learn a lot about it because it's like a giant filter in the sky. The Moon provides this great filter. And so astronomers learn a lot about these objects during transits, eclipses, and occultations as we'll discuss. So stars, as Cecilia Payne-Gaposchkin said many centuries later, are balls of hydrogen and helium fusing together via the process of nuclear fusion that releases tremendous, almost unimaginable energy that we'll discuss soon.
- 1.94 And the Sun is a star just like that. And all the other stars are exactly the same, but they're so far away, we don't resolve them as disks. We can't see them in two-dimensional fashion. They just look like these zero-dimensional points of light because they're so far away. So, by the, after the age of Galileo, we now began to realize that the Sun is the center of our solar system. What's ironic is that, no matter how many times brilliant men and women make a mistake, they'll still make the exact same mistake a few decades or centuries later.
- 1.95 And that gives me humility because it may mean that the concepts of cosmology that I talk about in my Peterson Academy cosmology class, that those advanced-frontier ideas are going to be laughed at just the same way that we laugh at, say, the Earth-centered universe cosmology. But that's OK. You can't do astronomy if you're afraid to be embarrassed by later discoveries. The goal should be that you're eventually proven wrong.
- 1.96 They thought the Earth was flat. Then they thought the Earth was the center of the universe. Then they thought the Sun was the center of the universe. Soon we'll see the Milky Way galaxy was the center of the universe. Then we'll see maybe, just maybe, we're the center of what's called the multiverse. That's an advertisement, more for my Peterson cosmology class.
- 1.97 So, the last of the naked-eye astronomers did amazing work. People like Tycho Brahe, Johannes Kepler. But they still made blunders. Kepler was the first to prove that these planets seem to go in elliptical orbits, not circular orbits, as we'll discuss. The reason why was given by Isaac Newton a century later. But he conjectured it and gave laws that describe elliptical orbits of planets, which is 100% correct. But then he should have quit while he was ahead because then he said, well, there's a reason for that.

- 1.98** And the reason is that the planets—there's only five known planets at that time, including the Earth—the planets are in an arrangement corresponding exactly to the structure of what's called the Platonic solids. And those were the reasons that they were in these, and it has nothing to do with that. There's no relationship whatsoever between these ancient Greek solids and the orbits of the planets. But Kepler, for all of his brilliance, made another blunder, and this happens all the time.
- 1.99** Tycho Brahe was the first to do something really astounding which we take for granted to this day. We'll encounter meteorites in the last lecture, and meteors. Meteor is the object as it comes through the atmosphere of the Earth. We know that it gets so hot, heated up by the reentry, or the entry force of impacting at 20,000 miles per hour, the fluid that we call the Earth's atmosphere, it causes it to ignite, burn up and dissociate. Well, 400 years ago, it was thought perhaps comets are the exact same thing. In other words, perhaps comets are inside the atmosphere of the Earth, just like meteors.
- 1.100** This was thought for thousands of years until Tycho, old Tycho, who had a fake nose made of metal, another character—I don't remember if he lost it in a duel or how exactly he lost it, maybe some disease—but he kept on doing his work. He made measurements of the position of the comet from different places on Earth and showed that the distance had to be many, many times the distance to the Moon, which had been estimated very roughly in terms of, what? The distance around the Earth.
- 1.101** So first, remember, Eratosthenes measured the distance, 46,000 kilometers. It's closer to 40,000 kilometers. Then they could make some estimates, given the curvature of the Earth's shadow on the Moon, of how far away the Moon was, and then they could make an estimate of how much farther away a comet appeared to be against the fixed stars, using what's called parallax that we'll discuss in the next lecture.
- 1.102** Up through Renaissance astronomy, we encounter my hero, Galileo. A hero again because he was brilliant, but also made incredible blunders. That he had such incredible writing abilities, persuasive abilities, he was a salesman, he was a magician in many ways. He did incredible work, but he also made incredible blunders. Eventually, even though he was right about the fact that the Earth goes around the Sun and not the other way around, his reasons for why that was true are completely wrong.
- 1.103** He claimed the fact that the Earth has tides, high tide and low tide four times a day, that was a proof. Just as if you take a glass of water and spin it around your head and rotate it, they knew the Earth was rotating by that point; but as it rotated and revolved around the Sun, that was causing the sloshing of the tides. It has nothing to do with that. Correct for the wrong reason. Right, but for a blundered conjecture.
- 1.104** There's me at Galileo's prison, as I took a year ago. This is outside of Florence, Italy. You can see the Duomo in the background, Renaissance Italy on display. And he was doing observations up until his dying day in 1642, having been in prison, not for studying. The Church—it's a popular misconception that the Church put him in jail because he studied

the Copernican system of heliocentrism. Actually, he was allowed to study it all he wanted to, but he was forbidden to teach it.

- 1.105** So he wrote his final two books in Italian, which was the common parlance. His first books and all the other books that he wrote about astronomy and physics were written in Latin, which was not considered teaching. The Church was totally fine. Even the Pope Barberini and the Pope, Pope Urban VIII, they didn't mind him learning and studying. They thought that was for the greater glory of God. And Galileo believed in God. He was a very devout Catholic, so it's often shown as this war against religion and science. That's total nonsense.
- 1.106** There's me in his office with the sketch of the Moon that I showed you at the very beginning. The Moon connected my first love of astronomy and connects me to Galileo to this very day through his books and writings. Again, that crater doesn't exist because he wasn't trying to depict it perfectly accurately. He wanted to show you what it felt like, because anybody else had a telescope like his.
- 1.107** Kepler, as I said, he predicted the laws of planetary motion that we're going to talk about. And we'll show that it can be derived from Newton's laws. And Newton's laws describe the exact mathematical justification for these elliptical orbits that we know the planets move on. And to this very day, just to conclude lecture number one, there's a beautiful connection between modern astronomy and the search for life in the cosmos far beyond the Earth's surface. And it's from a satellite that bears Kepler's name. Four hundred years later, he could scarcely imagine such a thing.
- 1.108** That's the Kepler satellite. And the Kepler space telescope took images of over 2,600 planets. Many of them, if not most of them, could potentially support life. And we'll have to discuss in the last lecture, as an ultimate cliffhanger, what it would mean both for life to exist somewhere else, for the study of astronomy, but also for the study of what makes us human beings. And that is a teaser for upcoming lectures.
- 1.109** So this course will take us from the beginnings of the Earth's history out to the beginning of the universe's history. And along the way, we're going to encounter these incredible, incredible characters, discoveries, conjectures, blunders, missteps, and brilliancies. In the next lecture, we'll talk about what do astronomers like me, like myself and my colleagues, what do we actually do and how do we actually do it? Thank you very much.

Lecture 2: The Astronomer's Toolkit

At a glance — The astronomer's toolkit — how modern astronomy is actually done.

- 2.1 So, now we turn to how modern astronomy is done. And the distinction between the ancient astronomers and modern astronomers really comes down to the types and size of their tools, starting with the oldest tool of all, the human eye, connected to the human brain. And we'll explore that in great detail. And in fact, there are ways that you can actually improve and enhance the God-given, or nature-given, if you will, telescopes that we're all born with.
- 2.2 The astronomer's craft is actually the most ancient of all scientific professions. Here's a painting from Vermeer depicting what he called "The Astronomer" from back in 1668, just a mere 50 or 60 years after the invention of the very first telescope. But not only is the astronomer responsible for using telescopes like the modern telescope that we have here, the type of telescope that launched my obsession and later career with astronomy, but also other tools, including ideas, brains, and later computers that play a tremendous role in our understanding of the universe around us.
- 2.3 This painting, I love it so much because it celebrates how astronomy is actually done. And it's typically done as much during the daytime, studying, thinking, hypothesizing, and sometimes correcting your hypotheses during the day away from the telescope, as it is done in front of a telescope. So astronomy is challenging because we only can learn about things by their propagation of information from them to us. The Earth is about 8 light minutes away from the Sun. Light travels faster than anything that we know about, and yet it's still so slow compared to the vastness of the cosmos that it takes a mere, that it takes 8 minutes, 8 long minutes, for the Sun to transmit the information and the heat and the light to us here on Earth.
- 2.4 So that means that if the Sun were to disappear right now, we wouldn't know about it for 8 minutes, because nothing, not even gravity or information, can travel faster than the speed of light. So some objects emit light, like stars, and then clusters of stars we'll call galaxies, we'll talk about those. And then others reflect light, planets and our Moon. We'll discuss those in turn as well. So light is an electromagnetic phenomenon. There are four forces of nature, four fundamental forces of physics. There's the nuclear forces, two of those. There's gravity, and then there's electromagnetism.
- 2.5 And electromagnetism is what is responsible for light, or what we call electromagnetic radiation. All forms of electromagnetic radiation, including visible light, travel at this speed, which is the fastest speed anything can travel, roughly 300,000 kilometers per second, or 186,000 miles per second, if you prefer US-specific units. Light we know to have these dualistic, this duality of properties. It's both a particle and a wave. In a particle form, we call it photons. And in a wave form, we'll talk about its characteristics in terms of its period, frequency, wavelength, et cetera.

- 2.6** So, waves propagate. Typically we think of them propagating in mediums or in media such as air waves. Sound waves that I'm speaking now are pressure and density fluctuations in a surrounding atmosphere. Water waves in the ocean are propagation of disturbances in height and energy that travel with the speed along the ocean surface and sometimes even down below. But electromagnetic waves are unique. They don't require a medium. For a long time, this stymied astronomers and physicists. How could it be possible for light, which is a wave, to travel through the universe if it's not supported by some sort of medium?
- 2.7** And in fact, they had a name for that medium. It was called the ether or the luminiferous ether, in that it supported lumen, or light. That was ruled out in experiments in the late 1800s at my alma mater, Case Western University, by two scientists, Michelson and Morley. And it was shown that there is no evidence for the ether. And that was what's called a null result. Again, science is concerned with what can be disproven. They disproved, Michelson and Morley disproved the fact that light, which is a wave, needed a medium. It doesn't need a medium. It is sort of self-propagating.
- 2.8** So you describe a wave in terms of its properties—its frequency, which is how many cycles per second; its period, how long it is between successive wave crests or wave troughs; and its speed, as well as its amplitude. How much energy is there in a wave? The speed of the wave is simply the product of its wavelength times its frequency. It's denoted by the symbol C . And you're no doubt familiar with the C in the same C in Einstein and perhaps the most famous equation in all of science, $E=MC^2$. That C stands for celeritas, which is Latin for speed.
- 2.9** And it's appropriate because the speed of light is the fastest speed that anything can propagate at. So here are some typical wavelengths, frequencies of different disturbances, waves, ocean waves, sound waves, and light waves. Visible light represents a very tiny fraction of an infinite electromagnetic spectrum, from zero frequency, if you will, or infinite wavelength, which is in the radio regime of the electromagnetic spectrum, all the way up to gamma rays, which are incredibly energetic, powerful, and in fact can be dangerous.
- 2.10** Light waves are described by three typical properties—the frequency or the spectrum of the light, the amplitude or the intensity of the light. And there's a third property which we won't get into in this class, but is really the way the bread is buttered around the Keating household. It's called polarization. I study the polarization of the most ancient light in the universe, called the cosmic microwave background radiation, and discussed extensively in my Peterson Academy cosmology class. We won't discuss it too much. The property known as polarization, it's the electromagnetic field's orientation in space.
- 2.11** The energy of a particle, in this case a photon, is related to its frequency via a very famous relationship from quantum mechanics. This is Planck's law. Planck's law, he

didn't call it Planck's law, that would be kind of egotistical. But it was later called Planck's law. And it features a constant of nature that's true on Venus and it's true in the Andromeda galaxy as we'll encounter it—all throughout the entire universe. It's a constant of, it's a constant of nature, like the effects of gravity, the gravitational force, the electric charge.

- 2.12** The Planck's constant lowercase h represents a tiny number that converts frequency, which is how many oscillations per second a wave of light has, to an energy. It's an interconversion between those two quantities. That light can be both a particle and a wave is unusual, surprising, perhaps uncomfortable to our normal everyday existence. We almost never see the effects of either the particle nature of light or the wave-like nature of light. But it is in fact possible for both manifestations, particle and wave, to be manifest in the human eyeball. And we'll talk about that in just a moment.
- 2.13** So light is a fundamentally quantum mechanical object that can be described as either a particle or a wave, depending on the circumstance, typically depending on the size scale of the interaction you're considering. We're considering light going through telescopes. We'll really focus it, no pun intended, on its wave-like characteristics. If we were talking about quantum mechanics, when we do discuss the little bit of quantum mechanics that we'll use in this course, we'll talk about some of its particle-like properties.
- 2.14** So our very first telescope, in fact, the very first scientific instrument, is the human eye. And the human eye is a telescope. And it's a magnificently constructed, some say designed, instrument for unraveling and discovering the cosmos. And we're going to take us through it and we're actually going to take apart a human eyeball, thanks to one of you donors. No, we didn't need a human eyeball donor. We can get this at many scientific supply stores. But I want to highlight the similarities and some of the differences between the eye, the human eye, and a telescope.
- 2.15** And the most simple telescope, the telescope that everybody has, because we all have cell phones, even if you don't have a telescope, although I encourage you to get a telescope, if not for you, for any child in your life. For \$50, you may inspire your child or your grandchild to become an astronomer like me or like my colleagues. And it's really the cheapest form of literal time. It's the cheapest form of teleportation known to exist. And we'll talk about telescopes in just a bit. But a camera, like in your cell phone camera, is exactly a form of a telescope.
- 2.16** Telescope means distance viewer. And we don't normally think of our eye as a telescope, but that's exactly what it is. All telescopes have the same sort of functionality. They're designed to collect light, focus it to make an image, and then later on connect it to some sort of instrumentation detector system, eventually to our brains, a human brain, in order to do things with it; or in some cases, animal brains actually have primitive—some very primitive objects go back literally hundreds of millions of years that connect the—have similar properties of an octopus from 100

million years ago, are ancient relatives of octopi, or octopuses, I think, is the correct term, that connect their eye and their ocular structure to even the human eye.

- 2.17** It's pretty amazing that these things are all connected, even though we couldn't be more different. So the eye has these functions. The functions are carried out by structures. The structures are instantiated, not in silicon or in glass, but by biological tissue. And that's what makes it so astounding. And that we can connect the biological tissue and the functional parts of this instrument to our brain, which is also a computer, is really quite astounding. Nothing else that we know about does that besides biological eyes connected to biological brains.
- 2.18** So, shown are the major features of the human eyeball. We have an entry into the eye which is called the pupil. It's protected by this outer layer here called the cornea. And the pupil is the aperture that allows light into and defines effectively the field of view of the eyeball, in this case a human eyeball. Now, this structure, we can take this apart so you can see the inside of it depicted here with all sorts of arteries and veins. And there's a tremendous amount of, if you like, it's impossible not to think about how incredible the human eyeball is.
- 2.19** The light is brought in through that pupil, the protective cornea. We can take this off. And I should note that all telescopes, even this telescope, even your cell phone camera, it could have 48 megapixels or whatever. They all have tremendous limitations, too. Either biological— as we age, we get cataracts, we get deterioration of the eye structure— or there are limitations in every camera. They can't see perfect color dispersion; it can't see polarization properties, et cetera. We'll talk about the limitations.
- 2.20** Again, so we can take this out. This is called the iris. So this is this beautifully colored object, but it also is unique to each person has a different iris, a different pattern of color and morphology of the eye. Then we get to the first real, I mean, these do have ocular—the cornea does have an optical focusing component to it, which is why, when you get LASIK surgery, you're really shaving away with high-powered lasers invented by one of the subjects of my second and third books, Donna Strickland and her collaborators, that use very high-intensity, very high-intensity short bursts of lasers to reshape the cornea.
- 2.21** Not the lens to make it focus slightly differently, to correct for visual defects like nearsightedness or astigmatism. But then you have the lens itself, which in the human biological structures is actually flexible. Unlike if you try to squish and squash the telescope lens, you'll shatter it. So this is kind of miraculous, if you will, that this object is biological and we can stretch and squish it by these muscles that surround the eye. And so the pupil can be constricted for purposes of limiting the amount of light and intensity that come into the eye.

- 2.22** During a dark night, you want as much light to be coming in as possible. In a bright room like this, you want as little light or you can constrict it. So again, that's done by muscles, by flesh and blood connected to the brain, which is somehow able to do this thousands of times per second. And you're getting many, many gigabytes of information equivalent into the human eye. The resolution of the human eye in terms of megapixels is off the charts. And no modern cell phone camera can match it.
- 2.23** This is not a real, this is not a solid structure. This represents the fluid inside the eye that keeps the pressure regulated inside the eye to keep the lens in shape. So this is just representing fluid. It's not actually solid like this. There can be objects and deterioration of it. You have so-called floaters where you see weird after-images if you look at something bright. Sometimes that's degeneration of what's called the vitreous humor inside the eye. And those are the major structures until you get to the retina.
- 2.24** And of course, there's also something called the optic nerve. The optic nerve is responsible for what's called the blind spot. So everybody has a blind spot. It's not just the name of a movie or some political bias that we have. We actually have a region where the retina has no cells. Because you actually have to get the data out through a literal type of fiber optic that goes directly from the retina, which is the detector material in the back of the eye where the light is focused, going right to the brain, the optic nerve. So that's another limitation that cameras don't have. They have other limitations that we don't have.
- 2.25** But this detector system is actually part of the human brain. And it's outside of the skull. It's in the back of your squishy eyeball there. And there's all sorts of techniques that we use as astronomers. We can enhance it. We can enhance our night vision by waiting to so-called dark adapt our eyes. We can avoid bright white light and only have red light where our eye is less sensitive to. And that will improve our ability to see faint objects through telescopes or through cameras or binoculars.
- 2.26** And the human eye can also be trained to do what's called averted vision or peripheral vision, in that you spend so much time looking straight ahead, what's called your fovea, that you actually get less sensitive directly head-on than you are at the periphery. You can actually detect more subtle movements from the sides of your eyes. And you can detect lower light levels from your averted vision, the side of your retina, if you like, that doesn't get as much exposure to bright intensity, accidentally looking at the Sun, you know, or a laser or something like that.
- 2.27** So we can actually model the human eye and understand there's many different types of defects and many different types of physicians and other technicians that study the eye and make glasses. And it's one of the oldest professions in the world. In fact, the profession of astronomy and the use of telescopes was enabled by the fact that we have eye defects. And it was actually the first lens makers in Holland that designed the very first lenses to use for eyeglasses to correct vision problems, as I'll demonstrate.

- 2.28** And those corrections then led a very bright individual named Hans Lippershey to finally take two lenses and put them together, rather than just use one lens for each eye, as we normally do with eyeglasses, and he put two of them together and then looked at things and he saw that he could make distant objects look bigger. And that was called a telescope, tele-distance scope vision. Galileo then did something which old Hans didn't do. He went like this and looked up at the night sky for the first, took six years for somebody to do that. It's kind of incredible.
- 2.29** It's like, you get a cell phone. Everyone uses a cell phone to take pictures of their kids and no one ever takes a picture of a sunset or something. I can't, I don't understand why that never happened. It took six years to do that. Finally, Galileo did it and he made these radical discoveries that changed modern science, philosophy, and theology forever. So, as amazing as the human eye is, it's not perfect, as no structure can possibly be. In particular, we aren't able to see some of the aspects of the electromagnetic spectrum that other animals can see even, and certainly that cameras can see.
- 2.30** So, some of the properties we can't see—for example, we can't see infrared light or ultraviolet light. There are lots of creatures that can. We can't see polarization very well, like most objects can, like many biological species can, but it serves us quite well. And we can actually see this incredible world around us in stunning detail. Now that leads again, we talked about Vermeer's Astronomer and of course the most famous picture in all of popular imagination and popular culture happens to do, besides the Mona Lisa, I suppose, is Van Gogh's Starry Night.
- 2.31** Now this depicts this beautiful, transcendent, heavenly experience that he's experiencing. But a lot of art historians even suggest that he might have had macular degeneration or had some eye defects himself that caused him—actually, he was trying to represent the objects not as whimsical creations, but actually as he saw them. So it's a byproduct, this beautiful painting might be a byproduct of some of his optical deficiencies. And by the way, how do you know the difference—what is the difference between an ophthalmologist and an optometrist? About \$100,000 a year. That's a joke for my optometrist and ophthalmologist friends out there.
- 2.32** So, telescopes, far-lookers, they really revolutionized science as we know it because it took science from being this qualitative thing to a very, very quantitative subject. There are two different types of telescopes we'll describe and I'll build primitive versions of them on the table here. But again, go to—my dream is to get Peterson Academy-branded telescopes. I don't have those yet, but you can go on any store and get one, \$50. You almost can't go wrong. They're almost all as good as or better than Galileo or Isaac Newton could have ever dreamed of.
- 2.33** So a refracting telescope is the simplest one. It's very similar to the human eye. In fact, the human eye only has one lens. And most telescopes will have at least one lens, probably two lenses. The very simplest lens is called a convex lens. And these types of

lenses focus light to a single point. Now, this is not ideal. This only works for one specific wavelength, and it only works perfectly in a very limited situation, in a very limited physical situation. In fact, it was very difficult for lens makers to make lenses in this particular configuration for many hundreds of years or dozens of years.

2.34 And in fact, the very first use of eyeglasses was precipitated by the printing press, by Gutenberg in the 15th century, was able to print for the first time typesetting that was standard throughout the whole world. So, if you had a copy of the Gutenberg Bible, you could take it from Italy, you could take it to Spain, or take it anywhere you liked and ask, "Could you read this font at a distance of 20 feet?" And if you could, you had perfect vision, and that's how they measured it. And then if you didn't have perfect vision, well, they'd get you a set of lenses, and they'd correct your nearsightedness, your farsightedness, with those lenses.

2.35 And it's ironic that the Bible was used as a type of eye test or eye chart, like we go to when we used to go to the DMV. You know, there's eye charts. Instead of an eye chart, they would now—now it costs \$100 million for a single page from the Gutenberg Bible. But back then, they had a couple of them around, and you'd use it. And then if you didn't have it, you'd get an supplemental eyeglass made for yourself. And then, ironically, the Church was upset when Galileo used the telescope to provide evidence that the Earth was not the center of the solar system, which was the whole universe.

2.36 This has two lenses instead of the one lens of our eye or the one lens that I showed you before. And it has a variety of features. This is called a Keplerian telescope. Kepler was a friend of Galileo's, but they were also rivals. They wanted to both get credit for different things. Kepler would be more famous in certain circumstances than Galileo, and vice versa. So Galileo actually didn't share his monopoly that he had on building telescopes. He wasn't the first to build it, but he improved the quality of the telescopes made by the Dutch and others by a factor of 10. He made them bigger with better glass material and fewer defects. And so they were priceless, in some sense.

2.37 And then he did what many people do, especially professors. He tried to make a lot of money with them, but he didn't want to give away the secrets on how to make the telescope. So what he would do is, he would offer to build the telescopes himself and then he sold them to powerful politicians. Now, why was this useful? Well, transport yourself back to 1610 and 1611 in Venice, which was always at war with different states, and Rome and throughout Europe. They would be often besieged by ships coming through the Venetian lagoons. If you've ever been there, it's quite lovely.

2.38 If you had a telescope, you could see a boat coming to attack you three days before somebody on land without a telescope could see it. So it's basically like, if there's a stealth bomber coming towards you, you could turn off the stealth of it whenever you wanted to. It was a very powerful military instrument. And for this discovery, he sold it to the Venetian Senate, and they gave him what every professor wants: lifetime tenure. And that was great.

- 2.39 So, how did the telescope work? Well, he was able to make these two lenses work in a specific way to make an image. And you see what's happening here, the actual image, the objects that you're looking at, is this thing that's labeled number four. It's this black arrow. OK, it doesn't look that great, but remember, you're looking from the side of it. So imagine there's an actual bow and arrow. There's an arrow sticking there, and it's physically as big as that object shown in black in number four.
- 2.40 With these three lenses, two from the telescope and one from your eye, you will see an image that appears to be as big as the pink number six image, but it'll be upside down. If you ever look through a telescope, it inverts up and down. And that's done because of the tracing of these rays through the telescope. Now, we can dim the lights now and we can build a tiny telescope over here that will demonstrate these properties.
- 2.41 So there's many ways to build a so-called Keplerian telescope. But as you see here, we have these rays shooting out of this telescope, we have these lasers shooting out of this apparatus over here. And we're going to put lenses in front of it, and then the lenses, depending on their orientation and their distance from one another, will make a focus. First they'll make a focus, but actually you either want it to make a focus and then diverge it afterwards, so that the rays will come into your eye parallel as shown in number two above. So you'll put, you can put these lenses in to correct, depending on which distance you are from the lens.
- 2.42 So it's a little bit faint. But if you were to put your eyeball, which is a lens like this here, depending on how you arrange these guys here, you could actually get it to focus in your eye, represented here. So the light would then appear to be coming from a much larger object and it would be upside down. You see, this ray enters at the top, or where my finger is here, but it emerges from the bottom. So it's inverting the image so you'll see it as it's upside down. If you look at a spotting scope with a telescope at something on the ground, you'll see it reversed. It'll be demonstrated upside down.
- 2.43 And it's actually, if you think about it, quite amazing, because this happens too without these lenses, right? If I remove these lenses here, ignore this part past there, but say your retina is here, say this one, this one is not a lens, it's just, oops, this, I'm going to use this as a stand-in for the retina, just a flat surface or curved surface. So if you think about it, we are always seeing the world upside down. Our lens is actually taking the light vertically and then it's going inverted into our eye. So, physically, your images are upside down in my retina.
- 2.44 So our brain, which is connected and has a part of it outside of the cranial vault, the skull, is smart enough to correct it and turn things right side up with respect to gravity. Now that's partially due to evolution and how we've evolved to, our brains have evolved to process things, but physically we're seeing the world completely upside down. This is a refracting telescope. Refraction is a property of bending light through transmission.

- 2.45** There's another type of telescope which was created by Isaac Newton called the reflecting telescope. He did an incredible series of experiments over several days during the bubonic plague period and later in the 1700s in his laboratory outside of Cambridge, England. And he would do things with prisms. So Isaac Newton realized that the refracting telescope has certain other limitations. The most dominant one is that it focuses different colors at different places on the retina or on a detector system. That's called chromatic aberration.
- 2.46** Remember, science is finding out where you're wrong or what is a deficiency of what you would otherwise think is perfect. He realized that it's possible to have a telescope that focuses light, that magnifies the image, but keeps every color in perfect focus. Unlike pieces of glass, it's impossible to have a piece of glass that focuses every wavelength exactly at the same point. And that was the impetus for him to design the so-called reflecting telescope.
- 2.47** So if we look here, we can see how a reflecting telescope works. You see that light beams come in; they bounce off this parabolic or spherical surface, concave surface; they bounce off this concave surface, and then they come to a focus as well. And if you make the ray tracing and make the image trace the image path through, you see that this is actually upright. This top image comes and focuses at the top here, and then it goes through a focus and you can put another lens in at that point.
- 2.48** Now, a convex lens, sorry, a convex mirror, will do the opposite. It'll disperse it back outside like this. And this will keep—the ray on top will also go to the top, and the ray on bottom will also go to the ray on the bottom. And you all have these in your cars. Your rearview mirrors always come with this thing that says "objects in the mirror are closer than they appear." And that's because it's a convex mirror that's taking light rays. You wouldn't want to have an upside down image of a car, ambulance about to pass you, right? So it's therefore used in a convex fashion. So it's the opposite of the way things are for lenses.
- 2.49** So a convex lens focuses images upside down, and a convex mirror keeps them right side up, but it minifies them and makes them look smaller, so they're actually closer than they appear in the mirror. So there's two different major types of telescopes. And nowadays we primarily use only reflecting telescopes, although I do use refracting telescopes for my cosmology research. But every single reflecting telescope, from the biggest ones to the smallest ones, eventually goes through some sort of camera, which means that every reflective telescope goes through a lens at some point.
- 2.50** It could be the human eyeball that's going through, or it could be a CCD camera, which is coupled to a small refractor system like this. To make a reflecting telescope, this shape here is a parabola. And you make these ray-tracing diagrams shown here. The pink line is coming from above, from some great distance. It takes all images and focuses them to a single point, regardless of their color, so it doesn't suffer from this chromatic aberration.

- 2.51** It has other deficiencies, but otherwise it is far superior in that you can actually make a telescope much, much larger, which is very good because the amount of resolution that a telescope has and the sensitivity of a telescope is dependent on how much light gets let in. Our eyeballs have a very small pupil that lets light in. And to get more magnification and get more sensitivity, you want more photons coming in. If you think of photons as these little drops of rain, which are particles of light, particles of energy of a specific wavelength and energy, coming into a telescope, the more you can collect, the more sensitive and the better resolved the image will be.
- 2.52** And it's physically difficult to make a giant lens. We have a telescope in our observatory that's six meters across. It's a mirror. And to make, it's impossible to make a lens that big. It would be astronomically heavy and cumbersome and you'd have to support it. And eventually the glass over large distances will start to sag. That will distort the curvature, that will make imperfections and so-called aberrations to the image. So you don't want that.
- 2.53** So, a lens of about about 1 meter in diameter is about the biggest lens ever made for astronomical purposes. And we have telescopes in space that are six times bigger than that. Here's an image made by my hero. This is Galileo, and he's doing something incredible here. It's the first example of what's called data science, that the presentation of data was used in a way to make a scientific argument. This is actually an image of what's called the Seven Sisters, or if you drive one of these, it's called Subaru. It's not a constellation. It's called an asterism.
- 2.54** An asterism is a smaller collection of stars. The Big Dipper is not a constellation. It's a collection of seven stars that make the shape of a Big Dipper, part of a bigger constellation called the Great Bear, Ursa Major. Same with the Little Dipper, Ursa Minor. But this is called the Seven Sisters, because in antiquity they thought if you could see the seven stars here, really it's six stars, that you had perfect vision. And they were also related to the Pleiades, which were the Greek goddesses that were, I believe, the nurses of Dionysus, who was, of course, the God of the grapes and of wine. So he was very important to the Greeks and to many of my fraternity brothers.
- 2.55** The stars that are shown here are in two different, are shown in two different types of depictions. One is sort of open stars, six-pointed sheriff's badges or something like that. And then there are these closed dark stars. What do you think he was trying to illustrate here? Well, I'll tell you. The stars that are shown open are those that are visible to the naked eye, that you could count. There's not seven of them, there's six of them. But for some reason they've been called—I think it's to fit the narrative of the seven sisters of Dionysus' maids or nurses.
- 2.56** Every other star is black. You could only see with the aid of a telescope. And Galileo went further, and he said all this stuff here, even that blue glow in the background, which we call the nebula now, he said that was also individual, those were also individual stars. But his telescope, he realized, wasn't big enough to resolve them into

individual points of light. So just as he resolved the six stars into hundreds, he thought also that blue glow could be resolved. And he called that "the unraveling of the nature of our galaxy," which he called the Milky Way, which we still call the Milky Way.

- 2.57** But like many things, he was right and wrong at the same time. It is true that 96 of the hundred stars can only be seen, or 94 of the hundred stars can only be seen with the aid of a telescope. But no matter how much magnification you have, you'll never be able to resolve the glowing blue and orange into stars, because it's not made of stars. It's glowing gas and dust, reflections of the stars that are newborn, infant stars in this cluster called the Pleiades Cluster, and they reflect the light.
- 2.58** And how we know that they're reflecting the light rather than emitting the light will become clear, no pun intended, when we discuss spectroscopy in just a bit. So a telescope that's larger can resolve finer and finer details. If you've ever seen at a great distance on a desert highway, you've seen a light coming towards you on a highway. At first, you can't tell if it's a motorcycle with a single light or if it's a car really far away until it gets closer. Remember, a telescope makes things look closer no matter what.
- 2.59** So a bigger telescope will be able to resolve two things as being distinct objects from a greater distance or, equivalently, if they're stars, you can separate into finer and finer detail, ranging from radio telescopes that have a equivalent resolution capacity, as if the telescope is 100 km across, ranging all the way down to the human pupil, which can go from a maximum of about a centimeter, if you're really, really in a dark place, down to 2 millimeters in a bright environment like this. So that's the aperture of different types of optical systems.
- 2.60** The largest ones on Earth are reflecting telescopes. And even in space there are reflecting telescopes, like the James Webb space telescope. The largest refractor is about one meter, which is very low, a 36-inch telescope. And that doesn't even rank in the top 1,000 telescopes on Earth. But for a time, it was one of the biggest telescopes on Earth. Now, we haven't really explained how we use telescopes. And one of the first applications of the telescope was to prove that indeed the Earth goes around the Sun and not the other way around. Or if you like, to falsify the idea that the Earth is stationary and the Sun goes around the Earth.
- 2.61** And that was done by looking at a tiny deflection due to the motion of the Earth and the trajectory of light coming through a telescope. That's only a tiny fraction of a few, about a hundredth of a degree or even less, that occurs in a periodic elliptical motion as the Earth orbits around the Sun. That's only possible if the Earth is orbiting around the Sun and not if the Sun is orbiting around the Earth. This effect would not happen.
- 2.62** And so it took from the conjectures of Copernicus in the mid-1500s to 200 years almost later. So it was proven that the Earth goes around the Sun and not the other way around. And that's called an effect called aberration, stellar aberration, an effect due to

the fact that the Earth is physically in motion. Wasn't proven by Galileo. He had great ideas. Copernicus, certainly, it wasn't proven until the 1700s.

- 2.63** Nowadays the telescopes keep getting bigger. So when I bought my first telescope about this size when I was 12 years old, the first thing I wanted to do is get a bigger telescope. And nowadays these things are amazing because, unlike in 1984, you can put a cell phone, attach a cell phone, take images of it, put it on Instagram, TikTok, become famous, watch eclipses and occultations and do all sorts of cool stuff that I couldn't even dream of.
- 2.64** But the very first thing you do when you get a telescope is start thinking about your next telescope. And you want it to be bigger and better because you get what's called aperture fever. The bigger the aperture, the finer detail you can see and the fainter that you can see. It turns out that in an extremely dark environment, scientists published in the journal *Nature*, that the human eye can see a single photon. It's incredible. In other words, a single quantum of energy. The human eye is magnificent.
- 2.65** But in most circumstances, it's not the greatest because it doesn't preserve information. It can only see a limited range of color. It can't see polarization, it has different optical artifacts, but still it's amazing. And I was treated in 2019 to visit one of the largest telescopes on Earth, six and a half meters across. So this is 24.5, 25 feet across. Imagine a mirror as wide as this room. Now it's made up of sub-panels, mirrors, but they were—it's the largest telescope on Earth. It's bigger by a large margin, about twice as big as the Hubble space telescope.
- 2.66** And they actually attach an eyepiece to it. Most telescopes, professional telescopes, have no way of the human eye ever looking through it. But this one they built and they had the capability that you could look through it and you could see this incredible image called the Carina Nebula. And for a kid, it was, you know, for a kid who dreamed of becoming an astronomer, like kids dream about going into the NBA, this was probably one of the highlights of my life in many ways, certainly my professional life, seeing this nebula.
- 2.67** Because, in almost all cases, you almost can never see color even with a moderately sized telescope, because the human eye has these two different components of the retina, rods and cones. Cones, letter C. Mnemonic. Mnemonic, remember, is what's responsible for color vision. But color vision comes at the cost of sensitivity. You're actually more sensitive to black and white for the rods. But the rods are a more primitive system shared by more and more species going back, as I said, perhaps millions of years. And so rods are much more sensitive.
- 2.68** And so you tend to see things through a low-light environment. You tend to only see their intensity in black and white. But with a huge six and a half, 25-foot diameter telescope, now you can actually see the color of this object, and it's red and orange and green. It was such a beautiful sight. It really looked like when you looked at this object

through the Hubble space telescope. Now these folks down in Chile, this is on a mountain in Chile, about 9,000-feet elevation in what's called the Atacama Desert of northern Chile in the Andes mountain range, they are building not one, but two telescopes that are going to be four times the diameter of this enormous other telescope.

- 2.69** So almost 100-foot diameter telescope called the Giant Magellan telescope. It's one of the most ambitious, expensive, and promising scientific instruments ever designed. Hopefully it'll be online in the next five or so years. One of the major innovations in astronomical history was the realization that the properties of the atom could be revealed through a telescope. In other words, the chemical composition of an object produces something called a spectrum, a unique chemical fingerprint that is specific to the element or molecule comprising the object.
- 2.70** So too if you look out at a distant star or a nebula, as we just discussed, you can examine the composition of that object without going there. Spectroscopy is like a poor scientist time machine teleportation device, allowing you to go thousands or millions of light years away in an instant and examine not only the beautiful intensity of the image, but actually know what it's physically made of, as if you went there, took a sample of it, and came back with it.
- 2.71** And the first to do this, in the 1800s, is shown here. And this is from a great German scientist named Joseph von Fraunhofer. That's a sketch that he made of the first spectrum that he took of the Sun. He looked with a telescope, took light, and dispersed it through a series of prisms. And the prisms disperse the different colors into different angles. And then the angles are projected through a telescope onto a piece of paper, and then he colored in the piece of paper.
- 2.72** Now, it's not a perfect rainbow. And the fact that it's not a perfect rainbow is indicated by these dark bands on it. Those dark bands are characteristics of the chemical composition of the surface of the Sun that he was looking at. So this allowed, for the very first time, the discovery of a new element. Not on Earth, not in a chemistry lab somewhere or an alchemist's lair, but on the surface of the Sun. In other words, we discovered the element helium on the Sun.
- 2.73** I always joke with my students. They ask, "How did he do that?" I said, "Well, they went at night, they went to the Sun." Now, it can be done without time travel, without space travel. Looking at the Sun through a telescope, going through what's called a diffraction grating, or in this case, a prism. So I've given you all diffraction gratings. You can put them on now, please. And we'll look around at different objects in the room and see what different colors they have.
- 2.74** Everyone got their prescription glasses on? Cost a lot for me to get them at Lenscrafters. What these are, are called diffraction gratings. They're not prisms. They're tiny little rules and scrapes in a piece of plastic that then take light of different

wavelengths and focus them at different distances or direct them to different distances, depending on their color, through interference patterns that we call diffraction.

- 2.75** And so what you'll see—if you have these, you can actually get them on the Internet for a few cents each— and you can then judge what types of spectra of different objects are being produced and therefore where and what the chemical composition of the object you're looking at is. So we have a couple of different things up here on stage. The first one is this Bunsen or this propane torch, which is really glowing in the blue. But if you look very carefully, I'm not going to get too close to you with this, but you can see that it makes copies of a rainbow with tiny little black lines at different places in it.
- 2.76** That's absorption spectra. There's quantum mechanical transitions of energy from different energy levels to different energy levels that is causing absorption and emission at different wavelengths. So that's what this burner does. And we'll play around with this later to show you, when you ignite different chemicals, they also glow with a different, with a different spectral characteristic.
- 2.77** Here is a lighter, just a normal BIC lighter. And if you see this, it makes a very, very clear image, almost perfect rainbow with almost no dark or light bands in it. And it's very yellow, as you see. And it's very closely matched to the human eye. The human eye evolved to be very sensitive to the light that is most predominant, which is yellow light, because we live near a yellow-type star. And so this sensitivity is actually showing you that there's actually information or chemical transitions that are coming at all different wavelengths, because this is what's called a blackbody.
- 2.78** A blackbody is an object that emits all different wavelengths as a function of its temperature. This is about 2 or 3 or 4,000 degrees above absolute zero. And it's emitting characteristic yellow spectra, very close to what the Sun is like, although the Sun also has these gaps or missing pieces of it as well. Lastly, we see a laser like this here. I'll put this one on the screen here. It's a different type of red laser. This will never make a rainbow pattern whatsoever because it's only producing a single frequency. So it goes through the diffraction grating at the exact same angle. No matter what way I turn it, rotate it.
- 2.79** It's kind of fun, a little bit trippy. Won't spend too much time on this, or you guys will get all hypnotized by it. But you only see the red light here, no matter what diffraction grating it goes through. That's because it's a single or monochromatic single-wavelength light source. All right, that's it for the demos.
- 2.80** So this was an innovation that took place in the 18th century and even into the 19th century, to realize that light is comprised of a spectrum, a rainbow of different colors, including colors that you can't see. One of the greatest scientists of all time, William Herschel, whose sister, again, Caroline Herschel, was the very first female to ever get

paid to be a professional scientist—they did an interesting experiment where they took light from the Sun and they put it through a prism. The prism also diffracts light of different colors and makes a rainbow pattern. We'll talk about that later on.

2.81 But what they then did is, they took this light source and they did something quite astounding. So they took light from the Sun. They were inside of a room. The room had a tiny little slit or aperture. You can see, I'm just like van Gogh and Vermeer in terms of my prowess. So light would come in, then it would go through a prism, and the light would come in. So Caroline and William would do experiments with these different types of prisms and so forth. And then they had a brilliant idea, which is to take a thermometer. And they took the thermometer and they lowered the thermometer down into a region.

2.82 So here's the spectrum. They had a piece of, of cardboard, black cardboard that they could see, or white cardboard, rather, on the other side of the room. So light would come in and it would disperse into different colors, the red at the top, blue at the bottom. And then they lowered a thermometer down, just a little bit, where there was no red light visible to the human eye. And they saw the thermometer had its mercury, the thermometer had mercury, and when they got it close to where there was just about to be red, the mercury started to rise, which meant that they were detecting infrared, longer than red visible wavelength light.

2.83 That is what we call heat. So infrared light can be felt even if you can't see it. And you can prove that to yourself by going outside on a sunny day, closing your eyes. You can basically find the Sun with your hand, roughly. It's not a great telescope, but you're feeling, you're obviously not seeing light from the Sun. But some animals can basically, effectively, see infrared light using very similar techniques. And that's because the wavelength of light in the red, which is slightly longer than 700 nanometers, that's what the longest part of red light is, say, 1000 nanometers or 1 micron, 1 millionth of a meter in length, is about the size of most chemical molecules, like our skin molecules and atoms that make up our skin.

2.84 And so we can actually sense that because it's agitating, causing those molecules to warm up. And that's how we detect infrared light, even though we don't have infrared sensitivity in our eyes. And the production of this infrared light is done by atoms. In quantum mechanics, the transition of an atom involves the motion and physical transition of an electron in an atom from a higher energy state to a lower energy state will cause it to emit light, and the converse process will cause it to absorb light.

2.85 And so, depending on the composition of the material, whether it's hydrogen, calcium, oxygen, whatever, we can measure based on the transition and knowing the laws of quantum mechanics very precisely, in particular the Schrodinger equation, which we will cover in another course for Peterson Academy. The transitions between these different atomic energy levels are responsible for specific absorption and emission features that give rise to what's called the chemical fingerprint of each individual atom.

- 2.86** So here are some transitions in the Sun, the hydrogen from the Sun, and eventually the Carina Nebula. The object that I saw through a 6-meter telescope, diameter telescope, looks just like this. It's bright red, there's some blue, you can see even some yellow. And so you could imagine that this is coming from all different types of composition in this stellar nursery, where stars are going to be born and are going to be reborn, if you will, out of the dead embers and remnants of a previous star that exploded, called a supernova. We'll talk about that later.
- 2.87** So light can be absorbed or emitted in materials, and you can actually measure stuff in the foreground and in the background. And that's how they measured the, I told you that the light that surrounds it, that's glowing, that Galileo thought could be resolved into individual stars in the Pleiades, was shown not to be from individual stars when it had the exact same spectrum as the stars that were next to it. In other words, the gas and dust in the nebula, which is Latin for cloud, the nebula acted like little mirrors that were reflecting the light.
- 2.88** And you could tell it was the same exact composition because it had the exact same spectrum. OK, so every element has a unique spectrum. Shown here, hydrogen, helium, oxygen, neon, iron. We'll go through some chemical reactions. We'll show how they glow at different colors when we do our chemistry experiments in a little bit. Now, the other thing that spectroscopy is used for is to not only measure the position and the color and the composition of static objects. But it could be used to measure the velocity or the dynamics of objects in motion.
- 2.89** This was first observed by Christian Doppler, who shared his hometown of Salzburg, Austria, with a very famous manipulator of sound by the name of Wolfgang Amadeus Mozart. They both lived in Salzburg, Austria. And what Doppler showed was that a moving object causes the wavelengths of light or sound to stack up in the direction it's moving and to become more and more dispersed the farther away you are from it, its direction of motion. So an object coming towards you will have a blueshift or higher pitch, higher frequency light or sound, or moving away from you, lower.
- 2.90** We've all seen that with ambulances. You've heard them coming towards you and away from you. And we will demonstrate later on how this effect, the Doppler shift, is used to not only measure the position and motion of planets, the position and motion of nebulae and stars, but actually that galaxies, with a handful of exceptions only, that hundreds of billions of galaxies are all moving away from the Milky Way galaxy, from us. There's either something very wrong with us—we need some better cosmic deodorant or something like that—or there's another conclusion much, much more dramatic, that we live in an expanding universe. And that's a cliffhanger for lecture number seven.
- 2.91** So you'll see, in the spectrum of light, the entire absorption features in this case will be shifted to the left or to the right, to higher frequency or lower frequency, depending on if they're moving towards you or away from you. And that size of the shift in microns or

nanometers is directly related to the object's velocity relative to the fastest velocity that we spoke about earlier, the speed of light.

- 2.92** The next major invention, after Doppler and after spectroscopy by Fraunhofer, was the advent of photography. And photography in its modern incarnation, we all have very high-resolution cameras and telescopes in our pockets with very high, very sensitive detectors attached to them. But the very first images ever made in astronomy were pretty crude. They were of the brightest objects you could see outside of the Earth and the Sun, and that was the Moon. So the first astronomical image was only taken about 190 years ago, and that was of the Moon.
- 2.93** And just a few years later, they were able to capture images of not only the Moon, which is very bright and relatively easy to take pictures of, but of the stars, et cetera. So it's kind of cute. Remember, the first sketches of an astronomical object were the Moon and the stars, Pleiades and the Moon, and they did the exact same thing. They just keep doing these things over and over again. We'll see it again and the advent of CCD astronomy in a second.
- 2.94** What was great about photography is that it has memory; it stores things, unlike the human eye, which gets refreshed every time you look away and look at something differently. And, yes, you could do sketches, but that was all affected by the bias of the sketch, the abilities of the artist, as shown here, right. Not so great. But later on, the combination of photography and spectroscopy led to this unbelievable conclusion that the universe is larger than the Milky Way galaxy and that that universe is made of galaxies and they're all expanding away from us, which led to the innovation that we now call the Big Bang Theory, and we'll spend part of Lecture 7 discussing.
- 2.95** In 1976, the very first digital image was taken. Again, Galileo sketched the Moon. First photographs ever in the 1830s were of the Moon, and 100 years later, the first digital photos with just a—this would be like a 0.01 megapixel camera. This wasn't very impressive, but nowadays we have telescopes and cameras that I'll talk about in later lectures that have gigapixels, billions of pixels. The resolution is unparalleled. They can also detect single pixels.
- 2.96** They can also detect single photons, and they can do so across an incredibly wide wavelength range and even in different polarization states as well, something the human eye is just incapable of doing. In our next lecture, lecture three, we're going to start to encounter these astronomical objects that are right in our cosmic backyard.

Lecture 3: Measurements and Mysteries

At a glance — Measurements and mysteries across the cosmos.

- 3.1 Welcome back to our tour of the universe. And now we're going to meet our cosmic neighbors just around the corner from us. We're going to start with the, perhaps the fundamental, most challenging problem in astronomy, in addition to the fact that we can't actually do experiments. As a science, as a scientist, we'd love to be able to do them. We really can't do them.
- 3.2 But the next challenge that we face is that we have no ability to really go out from the Earth beyond the solar system's confines, as we've gone so far, and measure how far away something is. We can actually go to the—we can actually go to the nearest celestial objects. We've been to the Moon, we've been to Mars and even some of the outer planets, we've actually explored there, landed spacecraft on it, as we'll discuss. But beyond that, it's almost impossible to venture out much further.
- 3.3 In fact, the farthest something has ever gone is a spacecraft called Voyager 2, launched before almost all of us were born in the early 1970s. And it's one light day away from Earth. That's it. And the nearest star, as we'll discuss, is over four light years away from the Earth, so 1200 more times farther away. It's just incredible how vast the cosmos is. But to get to the stars, we have to first start with our own solar system.
- 3.4 Solving this problem has vexed astronomers since antiquity. When we look up at the beautiful night sky, it's a two-dimensional representation of what we know is actually three-dimensional, so we have no sense of depth perception. Right here you see a picture of a comet, which we've already discussed and we'll continue to discuss. Were once thought to be atmospheric phenomena like meteors and clouds, but then were immediately shown by Tycho Brahe and others to exist far beyond the orbit of the Moon.
- 3.5 And yet, for a long time, we couldn't tell if an object like a comet, which is relatively close by to the Earth, just a few times the distance to the Moon in some cases, versus a galaxy. And there's actually shown in the lower left corner of this slide, it's a little bit cut off, is the tip of the Andromeda Galaxy, which is located not in our solar system and it's not in our galaxy. It's really the first-ever galaxy outside of the Milky Way to ever be identified. And the light that we see now from that galaxy left its environs 3 million years ago. It's been traveling at the speed of light for 3 million years to get to our eyes, cameras, and telescopes.
- 3.6 And yet they kind of appear the same, a white, fuzzy smudge. So how can you tell how far away something is? Classic problem in astronomy. Everything else in the universe that we see is in our own galaxy, in some cases, in our own solar system. It's actually impossible to tell, just based on the size or the brightness of something, how far away it

is without understanding many more properties of the phenomenon that you're looking at.

- 3.7** For example, the Andromeda Galaxy is about 12 times the angular size of the Moon. So you might think, "Oh, it's much closer than the Moon," but in fact, we know it's millions of light years away, whereas the Moon is only 1 light second away, about 1.5 light seconds away from the Earth. And we've been there, haven't been to the Andromeda Galaxy. It'll be a long time before we get there. So it's always been a vexing problem. Are the stars uniformly distributed throughout space, the objects that we see, or are they clumped together? It wasn't known.
- 3.8** And in fact, we're still learning about the properties of the most distant objects in the universe that we can see out to billions of light years away. They're still teaching us valuable lessons about the structure, evolution, future, and past of our cosmos. The nearest cosmic neighbor is our Moon, and we learned about the Moon in early times. We learned to measure how far away it was in terms of its distance, in terms of the distance of things we could measure on Earth.
- 3.9** And we found over time that it was possible to measure the distance of the Moon to a precision of just a few hundred or thousand miles, which was quite astounding, given that the Moon is about 250,000 miles away from the Earth. Once we determine that distance, it then served as a tool, a ruler, that you could measure other objects' distances away from the Earth in terms of the Earth-Moon distance. And of course, the Earth-Moon distance was based on the Earth's own properties, the circumference radius, however we want to describe it, which was first measured by Eratosthenes in the year 200 BCE.
- 3.10** So we've been able to make measurements, to do these distance measurements, but they've improved over time. We can actually understand how far away the Moon is as a way of explaining why the Moon appears to move with us when we're driving on Earth. That's only possible because the Moon is farther away from Earth than a typical distance that we can traverse. And even the circumference of the Earth, it's about 10 times the circumference of the Earth away from the Earth, in terms of its distance from us.
- 3.11** So we can actually use the Earth-Moon distance to then measure things in terms of the Earth and the Sun, for example, and the other planets in our solar system. You see what we're doing; we're building a ladder that takes us from local measurements on Earth that you could physically walk from Alexandria to Greece. And they did that and rode horses and whatnot. And they could measure distances in terms of the human body, how many feet in terms of, say, the ruler, the king, the pharaoh. That's where the word foot comes from. It's the length of pharaoh's foot from 1300 BCE.
- 3.12** So those things are historical. And we can measure those in terms of features like that. Those are human-centric. They're not universal. If there are other species of intelligent

life in the universe, they're not going to know what a foot is. So we'd have to define things in terms of universal fundamental constants. And later on we would do that in terms of things like the speed of light. That's why I'll talk about things in terms of the light travel time needed to traverse some distance. The speed of light is a universal constant here, the same here on Earth as it is way out in the rest of the universe.

- 3.13** We use different units for different purposes in astronomy. The foot, the meter, the mile. Those aren't very convenient when you're talking about things that are millions of light years away. So we'll use different units. The first such unit is called the astronomical unit, and it's the distance from the Earth to the Sun. That distance, you may know, in miles, is 93 million miles. And it's also known in light travel time as about 8.3 light minutes. So we can measure things in those terms, terms that are consistent throughout the whole universe, not specific to some pharaoh who died 1400 years ago or more.
- 3.14** I mentioned in Lecture 1 that this interesting character, Tycho Brahe, one of the last of the naked-eye astronomers, could determine that comets were not in our atmosphere by the fact that they were much farther away as seen from Earth than were, then was the Moon as seen from Earth in two different locations. And this was done using a type of measurement called parallax, which we'll discuss in just a second. And we'll actually do a demonstration of how parallax can be used to make measurements of distance.
- 3.15** It's using trigonometry or geometry to measure angles, and then angles can combine with lengths to measure distances. And that was almost as good as it could be for many hundreds of years in astronomy. Once you know the position of the Sun, you can start to examine the properties of other astronomical phenomena that take place in the universe. So we know here, we have this handy model of our solar system here. Not to scale. If it was to scale, the Sun would be at the center and the closest planet would be a tiny little grain of sand and it would be way off in the parking lot here.
- 3.16** But we can turn this on and kind of, this is an, this is a model of the solar system, typically known as an orrery, and it used to be just made of metal. Now it has motors and gears and there's even light in here that projects out different colors if I, if I roll this little wheel here. And obviously this is not accurate, but it gives some properties that are somewhat accurate. The positions of the planets, the order of the planets, starting from Mercury, named after the Roman god, the fleet-footed Roman god, is the planet that orbits the fastest because it's the closest.
- 3.17** We'll explain why that's true, both in terms of Kepler's laws and then later prove it via Newton's laws, later on in the class. There was just a collision of Mercury with Venus here. So it's not exactly to scale. So Venus comes next, the Earth, and then Mars, Jupiter, and then Saturn, and then you go even farther out. This is the planet Uranus, which looks kind of strange compared to Saturn. And I'll explain why later on. And then lastly, the planet Neptune.

- 3.18** Now, when I was a kid, there was another planet that would have been added out here, Pluto, that's no longer considered a planet by the International Astronomical Union. I'll explain why those union members made that decision to demote Pluto from being a planet. And I'll talk to you about the astronomer friend of mine who is actually responsible for demoting killing. He has a book called, Michael Brown, eminent astronomer at Caltech, has a book—How I Killed Pluto and Why It Had It Coming.
- 3.19** So we measure things in terms of their distances, not necessarily knowing the absolute distance, but measuring it relative to things that we can measure on Earth. So we make this ladder. The first rung of the ladder is literally our feet, our bodies, and measurements on Earth. We extend that out to the heavens, and we'll see how far out we can go with that technique. But you see here, there's a possibility for certain planets. I'll put all the outer planets on one side just for clarity.
- 3.20** And I'll put the planet—here's Venus. And I'll show you the planet— here's Earth, looking at Venus. And Venus is transiting from the perspective of the Earth, is transiting across the Sun. Now, obviously, the scale model is way off, but the diagram over here shows the actual circular spot is the planet Venus going in front of the Sun. So this is a form of eclipse, but it's a very, very partial eclipse, because Venus is so small, even seen from the closest approach to the Earth. We can use this to measure the distance of the Earth to Venus, and then Venus to the Sun. And then from that, we can combine those two and measure the distance to the Sun itself.
- 3.21** And that's how astronomers did it for hundreds of years before the advent of more precise tools that were then used, that we'll explain in just a moment. So this type of phenomena happens very rarely. It happens twice in a decade, say, and then it doesn't happen for another 90 years or something like that. So these alignments, which is another one—one benefit of taking this class is you'll win at Scrabble because this type of event is called a syzygy, which has a lot of Ys and Zs and so forth.
- 3.22** And a syzygy just means an alignment of two or more astronomical objects as perceived from the Earth. So another type of syzygy would be a solar eclipse, a lunar eclipse. These are transits, and only the inner planets can transit. Only the inner planets, as we see them, can transit. We can't see Mars transiting from Earth. So this was another sort of advent of the telescope because you can't see this transit without filters and so forth, and tools that Galileo and others had developed. You couldn't see it. So it wasn't really used to make a proof of it.
- 3.23** The other thing you can see from this model here is that when Venus does make a, not a perfect transit, but even when it's close by, it will be observed as a crescent, which can also only happen for an object that's closer to the Sun than the Earth is. So the Venus can appear as a crescent, and in fact, it often does and is quite bright. It looks like a mini Moon. And it's the brightest object other than the Sun in the sky at that time or the Moon at that time. So the Moon can make a crescent because it can sometimes

be closer to the Sun than the Earth is, but the outer planets can't. So we only see them illuminated more or less fully.

- 3.24** OK, so that was some of the partial justification for why the Sun was at the center of the solar system, which for many thousands of years was also considered the center of the universe. So you can see these things. You can start to make plots of the planetary orbits. A plot is just another name for a model. This is a model in three dimensions, which you can make two-dimensional models as well. And you can measure it down to fractions of tens of meters.
- 3.25** So we know the distance of the Sun in terms, with a precision of about the diameter of this room, and we can actually do even better than that. And at a certain level, you can start to notice departures from perfect circularity or perfect ellipses. And that will start to tell you not about the planets themselves, but about the force of gravity itself. We'll talk about that very briefly when we discuss the large-scale structure of our universe.
- 3.26** So nowadays you don't need to wait for a transit of the planet Venus and then use Kepler's laws, et cetera. You can actually instantaneously almost measure the distance within a couple of minutes, at least. You can bounce a radio wave, a radar pulse, from the Earth to Venus, and some of it will reflect back like a little mirror, and it'll come back to your telescope or to your radio dish, and you'll detect how long. You'll measure the speed—sorry, you'll measure the time difference between the pulse that you sent out and how long it took to get back to you, and you'll divide that time by two.
- 3.27** And that time will then be used with the constant speed of light to measure to exquisite precision how far away Venus is. And we can do that and predict it. And we've been doing that for the last 80 years, or 60 or 80 years or so. So now comes the problem. Originally, it was very hard to find the distance even of the Earth's surface. Then, thousands of years later, we could measure the distance to Venus and then the Sun, and then apply Kepler's laws, and now radar astronomy. But then, going beyond that becomes quite difficult again.
- 3.28** So, again, the strategy is to make a very solid foundation for your ladder that's measuring distances, called the cosmic distance ladder, and then calibrate your future measurements against past measurements that you understand and know very well. So we need this ladder. Luckily, nature provides us with various ways to measure distances, some with great precision, some with less precision. But when all you have is that tool, that's about as good as we'll be able to do until new technology and new objects are invented or created or discovered that can play a role in the cosmic distance ladder.
- 3.29** So we start off with basic measurements on Earth. We go from the Earth to the Moon. From the Moon, we go to the Sun and to Venus. And then for the outer planets, we can also use things like radar and so forth. But we want to go way beyond the solar system. Remember, the object that's beyond the orbit of Pluto is only about a light day away

from us, Voyager 2. And we want to go to the nearest star, which is four light years away. We can't bounce radar off of it. So how do we measure its distance? Fortunately, there are many ways to do it.

- 3.30** So after we measure the inner solar system of thousands of minor bodies called asteroids, with radar bouncing energy off of it, measuring the time delay, it's very easy to do that. So the next rung on the cosmic distance ladder is called parallax. Parallax is the difference in perceived angular position as you move from one location to another location, just applying the laws of trigonometry, which is basically advanced geometry. It's predicated on what's called the baseline. The baseline is the astronomical unit.
- 3.31** It's a known separation. In this case, it's the distance between the Sun and the Earth that's known to meters in precision. And then using measurements made at one time and then six months exactly later, you can then construct a very accurate measurement of distance using pure geometry. And we can actually do such a measurement right now. So what's shown here is the Sun at the center of the Earth's nearly circular, but not quite perfectly circular, actually elliptical orbit. The Sun's position is fixed, even though it's actually moving in the galaxy. We'll ignore that for now.
- 3.32** The Earth is shown at two different locations, six months apart, or 180 degrees in phase around the circle. Circle is 360 degrees. You go around 180. That's half a year. Or in this case, it's around 2 astronomical units in distance. So we'll do this measurement with our tools, basically just our eyes. So imagine, instead of waiting six months, you're just going to use your eyes, and the separation, which you could measure very accurately with a ruler. So what I want everybody to do, as I told you, you all have two refracting telescopes in your head. They're separated by some distance between the center of each pupil.
- 3.33** I want you to take out your thumb and do what you've been dying to do. Cover up my mouth. Stop me from yapping, as my kids say. OK, so cover me up and then close one of your eyes. Close one of your eyes and cover me up. And then open your eye. Open the other eye and blink back and forth between your eye. My position should shift with respect to your thumb, right? You're not moving your thumb. You're moving basically your vantage point, from one eye, your left eye, to your right eye, say.
- 3.34** That causes me to shift with respect to the background stars that are at infinite distance. Let's assume they're at infinite distance. They're not, but they're at a very large distance, compared to the separation between you and me. Now you can measure the distance between your eye; that forms the base of a triangle. The length between you and me is what you're trying to calculate, the distance that I am away from you. This is the parallax method.
- 3.35** Once you measure that angle between your two telescopes at two different positions at the same time, or at different times, it really doesn't matter, then you can actually calculate, based on the shift, how far away I am. So that is the basis of parallax

measurement. You can actually measure it to a very exquisite precision as long as I'm not too, too far away from you. And we don't just do this on Earth now. We've actually launched satellites to measure increasingly precise distances to not just planets, but the stars, and even to find distances to galaxies. So the very first extra-galactic stepping stone outside on the cosmic distance ladder, very first rung.

- 3.36** So one parsec. One parsec is a measure of distance, and it's done in terms of the base of the triangle. Remember, the base of the triangle has a length of one astronomical unit, or 93 million miles, 140,000 kilometers or so. And to get a shift in angle, when you did back and forth, you're shifting tens of degrees. What this is saying is that, when you shift by 1 arc second, when that angle in here—should have drawn this a little bit more sharp— when this angle, denoted here θ , when this equals 1 arc second, which is equal to 1 degree divided by 3600, which is equal to 60 times 60.
- 3.37** So 60 seconds in 1 minute and 60 seconds in— 60 arc seconds in 1 arc minute of angle, the product of those two is how many arcseconds there are in 1 degree. So imagine you take a degree and, for reference, the Moon, the full Moon is 1800 arc seconds across. It's half of one degree across. So that big crater that Galileo showed you, that I've shown you a few times, that crater is actually only a few arc seconds across. That's a fraction of a fraction of a degree. And what this is saying is, if you measure an object that has a parallax, the angle that you measure is not tens of degrees. It's not one degree, it's not half a degree, it's not even $1/60$ of a degree.
- 3.38** When it's $1/60$ of $1/60$ of a degree, that's 1 arc second. That object is 3.26 light years away from you. OK? It's a weird unit. A parsec is 3.26 light years. It's not usual you have a unit and then you have another unit that's only three times bigger. It's like saying, oh, here's a kilogram, and then you have a par kilogram, and it's only, it's 6.6 pounds or something like that. So in this case, it's still this enormous distance of 3.26 light years. But guess what? There are no stars that are even that close to the Earth. The closest star is about four light years away. So it's more distant or the angle is smaller than $1/8$ second.
- 3.39** And yet, astronomers have been able to measure distances like this using powerful telescopes and accurate geometry, tools of geometry and trigonometry, that are tens or hundreds or even a thousand times this distance. Here's another schematic of it. You measure a distance to the Pleiades, that star cluster that Galileo claimed was made up of innumerable stars that he couldn't resolve with his telescope. Of course, it's not. That cluster of stars shifts by an amount. It's actually measured to about more than 1,000 light years away.
- 3.40** So the angle difference is minuscule. And yet we can measure it. And now we're able to measure— actually, back in the 1830s, incredible, astronomers were able to measure less than a third of an arc second variation over a timespan of six months apart. It's really amazing that astronomers were able to do this. This is about 200 years after the

invention of the telescope. And we're measuring things that are located literally dozens or more light years away from the Earth. It's an incredible tool that we have.

3.41 It was upgraded with a satellite called Hipparcos in the 1990s, and then later a satellite called Gaia can measure stars with an accuracy, not one arc second or a tenth of a— one millionth of an arc second. So it can go millions, if you like, millions of times farther away or more accurate than we could do, say, measuring the diameter of the Moon's biggest craters. It's just astounding. So it's been able to do this for a billion stars in our own galaxy, and that's foundational, and that's basically as good as you can do with this effect, because eventually the stars are not able to be separated or you can't see the distance shift, even at the greatest, highest magnification that this telescope is capable of, which is basically the resolution of its camera.

3.42 And so this really won't be duplicated or even bettered, but we really can't improve upon this. It's about as good as you could possibly do. So you might think, well, that's it. We can only measure things in our own galaxy, and we can measure a billion stars. But that's not good enough. We're actually going to go farther out into the galaxy using another property of geometry in our universe, which is the fact that light's intensity, as we measure it, falls off in what's called an inverse square fashion, which might sound bad, but it actually allows us to make a type of ruler based on the brightness of an object.

3.43 So when you look at starlight coming from the Sun or any object, there's a certain amount of energy being produced every second. That comes from nuclear fusion, which we'll talk about in subsequent lectures. The amount of energy produced per second is the luminosity of the object. These lights that surround us, the projector, a light bulb, a match, the butane torch that we used before, they all have some intrinsic production of energy per second. That is their luminosity.

3.44 For a light bulb, it might be 100 watts, and it's stamped on the light bulb. 100 watts means 100 joules of energy produced every second. That's intrinsic to the source. The Sun produces a lot more than that, about 40 trillion trillion times 10 watts. And that's producing, that energy is consistent day after day, year after year. Now, the light that gets in is what we detect in our telescopes, is what we detect. That's called the flux. How much intensity do we catch? How many of those raindrops are captured per second in our bucket?

3.45 Remember, the bigger the bucket, the more drops, the more sensitive the telescope is. It can see things that are farther away. The wider the aperture, the area increases as the diameter is squared, and the intensity of the source, as measured on Earth, changes depending on the distance we are away, squared as well. So each of these three candles, they're not combustible candles. These are just LEDs. They all have the same exact power. They all have the same batteries inside of them, producing the same amount of light. They're flickering to kind of make it more romantic, I guess, in here.

- 3.46** But what's really beautiful and attractive is that you can use the falloff in measured flux, knowing that the luminosity is constant, to measure the distance that you are away from it. The camera is capturing a fixed amount of light. It is a telescope, after all. And the telescope has a diameter. And that captures a certain number of photons every second. Each photon has some energy and is detected on the CCD chip inside of it. The intensity of the one closest to it appears much brighter, maybe even four times as bright, because of this falloff in flux.
- 3.47** The light produced by any candle gets distributed around an imaginary sphere. And that sphere's radius, as it grows, it grows in the area of which the light is dispersed by the square of the distance. The same amount of light is captured by a sphere surrounding it right here, as a sphere that fully surrounds it out here. The same number of photons are coming out. They have the same energy. But the intensity on each square meter of the sphere at a distance D is going down by one over D squared. D is what we want to measure.
- 3.48** So we can invert the equation of detected flux, written in terms of luminosity, which is stamped on the candle, which is produced by the three batteries inside of here, each one being the same. And as long as each one's the same, you can measure the distance in terms of this one, in terms of that one, in terms of that one. And if you happen to be able to do parallax, if you take out your fingers now, you could measure very accurately the parallax distance, and then confirm that each one is farther away by this inverse square law.
- 3.49** You're combining two rungs on the ladder. You're stepping on two rungs at the same time. And this allows you to make very exquisite measurements, because it's much easier to see something's intensity at a much larger distance than to measure the parallax angle. That's fundamentally limited by the Earth's orbit or by a satellite's orbit. But you can't have a huge, enormous orbit the length of the galaxy because eventually that would take too long for information to be transmitted across from one orbit to another orbit. So this then becomes a very fast, accurate way of determining distance.
- 3.50** This is a ruler. We have constructed a ruler based on observations only using a telescope or a camera. It's foundational. It's incredible. And we can use this technique to measure things not just in our solar system or in our galaxy, but in every galaxy that's visible over the entire history of the universe's existence, back to the beginning of time, almost. The one caveat, the one consideration that you must have—how could I be wrong? You might assume that each candle is the same, but somebody took out one of the batteries or dimmed the light a little bit. Those are called systematic error effects. We'll discuss those in subsequent lectures.
- 3.51** So this is the luminosity distance. It's foundational to the cosmic distance ladder. So you find the distance between, from the closest one to you using parallax. Then you can confirm the measurement of distance to the ones that are farther away. Then, as

long as you have an object that produces enough light at a distant galaxy, you can measure galaxies out to nearly the beginning of time itself.

3.52 So again, this is predicated on knowing the intrinsic luminosity. If you don't know the intrinsic luminosity, that's OK. That's OK. You can measure just the relative distances of things, which is sometimes just all you need. And then if you have another way of calibrating your cosmic distance ladder, you can then make an absolute distance measurement, which is what we actually want.

3.53 So again, we measure its flux. The luminosity is what's produced. We make assumptions or models about it. Here's an example of how we use flux on the Earth. This is a solar farm in the Mojave Desert, not far from where I live in San Diego. Captures about 1 kilowatt of solar energy at the surface of the Earth. And Mars is about twice as far, which means, according to the inverse square law, solar panels on Mars get at most one quarter of a kilowatt, or 250 watts. That's not even as much as some light bulbs that you can buy. A halogen light bulb would require more than 250 watts.

3.54 So it's very difficult to collect solar energy on other planets, especially as you go farther out. And yet we've been able to make satellite trips beyond one light day outside of our entire solar system. It's just astounding. So back to the solar system. The solar system is comprised of our Sun, which contains about 99.8% of the mass of our entire solar system.

3.55 Eight planets and many, many minor planets, which are like asteroids. There are meteoroids, and we'll talk about as well comets and other things called minor bodies, as we discuss, eventually discuss, how life on Earth could have been started. We are living on a planet that's associated so closely with its Sun that actually the Earth came about and is as old as our Sun. And that'll play an important role in discovering how old the universe is, as we will discuss in the next lectures.

3.56 So this is not to scale, but it captures an essential fact about all planetary systems, whether our solar system or exoplanets, as we'll discuss the exciting possibility based on the discovery of thousands of planets that have features almost identical to the Earth. Some are like Earth twins, that all planets move in elliptical orbits. An ellipse is basically a squished circle, or I should say a circle is a special case of an ellipse, where instead of having two different axes, a major and minor axis, where they're different, as in the case of an ellipse, with the Sun at one focus and the planet at another focus or another object, and the other focus being somewhere else in the solar system, that the circle just has the minor and major axes be equal.

3.57 Of course, there are many things we can't see, but we can sense using instruments that are specialized to detect things like the magnetic field of the Earth. It happens to be that the Earth has a magnetic field, and almost all the planets do. The Sun has a very powerful magnetic field. And we've learned about it through astronomical instruments

and in some cases spacecraft that have voyaged around these objects. The magnetic field of the Earth is quite crucial for our life to exist.

- 3.58** If we didn't have it, the Earth would be inundated by cosmic radiation, by solar wind, by particles and solar storms that would basically alter our DNA in a way I'll describe in Lecture 8, the final lecture, when we start talking about life. And it's unclear if these other planets, some of which look identical to the Earth, if they have or do not have such a protective bulletproof coating like our protective magnetosphere that surrounds the Earth. So too they might not have other planets like the massive planet Jupiter, which also acts like a bodyguard, a secret service agent protecting the Earth from life, destroying asteroids and collisions with other objects that we'll talk about, that have played a role in the Earth's history and formation of life as we know it here on Earth.
- 3.59** The Earth and all the planets have different relative orientations between the orbital direction that they will revolve around the Sun. The orbital motion where the Earth or any planet makes a complete— makes a periodic ellipsoidal orbit around it is called the period of the Sun. That phenomenon is called revolution. And in fact, that was the title of Copernicus' book, the Revolution of the Heavenly Spheres. The daily, in the case of the Earth, day is from the 24-hour rotation of the Earth on its axis. The Earth has an axis going from the South Pole, where I've been twice in my life in Antarctica, to the North Pole, where I've never been.
- 3.60** But that is tilted with respect to the horizontal plane, which is called the ecliptic, which is the plane that the planets primarily are rotating in. So this plane in which the planets and even the Moon and the Sun appear to orbit in is called the ecliptic. And the constellations, as perceived at great distances behind the Earth-Sun distance, for example, that's the zodiac. So the constellations are way out here. Parallax tells us that. And the path, fictitious path that they go through, is called, that they seem to go through, is called the zodiac. Those are the constellations that are far away.
- 3.61** The path that's physically or a fictitious plane as well, called the ecliptic, is the plane that more or less the Earth and all the planets orbit inside of. The Earth is tilted, by the way, by some 23 degrees with respect to the ecliptic. And because of that we have seasons. If it was perfectly parallel to this perpendicular axis to the ecliptic, we would never have tilt. And because of the tilt, the light from the Sun has to go through thicker or thinner amounts of atmosphere, depending on whether it's summer in the hemisphere you're in or winter in the hemisphere you're in, or during equinox when the two times during the year when the relative obliquity or perpendicularity of the Earth-Sun system doesn't make a difference.
- 3.62** All points on the Earth get the same amount, 12 hours of daylight and nighttime twice per year. And then in the ecliptic, the Sun could either be tilted in a way so it's closer to us, or the northern hemisphere. The northern hemisphere is tilted more towards the Sun and you get more solar radiation—it gets warmer— and the opposite happens in the winter. So the Earth and Moon are kind of like a pair of planets separated by

250,000 miles. That's nothing in the cosmic time, that's nothing in the cosmic distance ladder.

- 3.63** So because of that, we have these eclipses and these incredible things that only happen—of the 300-plus Moon-planet pairings in our solar system, the Earth-Moon pairing is the only one where you get a perfect solar eclipse. Tonight, as I said, we're going to have a lunar eclipse. And that'll be wonderful if we can see it, if it's not too cloudy or rainy. But that phenomenon is only visible in the solar eclipse configuration. That type of syzygy, when the Moon gets—pretend this is the Moon, this is actually Venus—but when the Moon obscures the Sun's rays from reaching the Earth, that's a solar eclipse.
- 3.64** When the Earth casts a shadow that obscures the Moon as seen from Earth, that's a lunar eclipse. And when the Sun is between the Moon and the Earth, that's called an apocalypse. No, that never happens. So there's no so-called dark side of the Moon, apologies to Pink Floyd. There is a far side of the Moon and that's because the Moon's orbit over time has locked itself by the effects of its effect on our tides, called gravitational tidal locking, that the Moon's year, which is the time it takes for it to revolve around the Earth, is the same as the Moon's day.
- 3.65** So if we had, imagine if we had a 365-day-long day, OK—imagine our day was 365 times 24 hours long—then we would always face the Sun as we went around it. And that's what happens with the Moon. We are far away from the Sun, so we don't get tidally locked to the Sun. But Mercury is. Mercury only shows one face to the Sun. Its gravitational field over time has become synchronized. So it's always facing the Sun, the same way the Moon's facing us all the time with just one face.
- 3.66** There are times when you can see some of the far side of the Moon as depicted here. This is obviously taken from a spacecraft, so it looks radically different. We don't see the same kind of lava flows and whatnot. And there are reasons for that as we'll discuss in Lecture 8. Here's a depiction of these different phenomena ranging from new moon to new moon. That's one lunar year, which is one month lunar cycle. So starting on the left, you go from a new moon, where the Sun is behind where the Earth is looking towards the Moon.
- 3.67** And its possibility during a new moon and only during a new moon to have a solar eclipse, either partial, total, or annular, when you have a ring around the Sun, a ring around the Moon. Then a full moon is the only time when you can have a lunar eclipse. Because the Earth could cast a shadow, as shown in the middle picture, from Sun could create an eclipse or a shadow to darken the Moon's surface, as will happen today. I didn't arrange that to happen, it just kind of happened to happen.
- 3.68** And then the other crescents that were so-called first and third quarter, and there are terms that refer to whether the appearance of the light side of the Moon is getting brighter over time. That's called waxing. And then eventually, like our energy levels in the late in the day, it starts to wane. So here's an eclipse again. The syzygy, my favorite

Scrabble word, won many games like that. So there's almost 300 planet-Moon systems in our solar system, and the only one where this type of exact perfect total solar eclipse can happen is the Earth.

- 3.69** What do you make of that? Make of it what you will. I witnessed two of these. Well, I've witnessed many solar eclipses, but only 2 total solar eclipses in my life. And this is just an astounding thing where people get the chance to visit it, if you have a chance. The next one in the US, I think, will be in Alaska in 2026, maybe slightly after, or maybe that's somewhere else. But eventually, the next one in the continental US is in the 2030s. So you've got some time to prepare. Start booking hotels, they fill up fast.
- 3.70** This is what lunar eclipses look like. There are many more chances for a lunar eclipse to happen, total or partial, than a solar eclipse. If you miss tonight's, no big deal. There'll be another one in a few months, in March of 2025. And there you're seeing kind of the atmosphere of the Earth, projected, the shadow of the Earth. Plus the atmosphere acts like a lens, like a lens surrounding the planet, like a fluid lens, that focuses the light of all the sunsets on Earth happening at that time. And all that gets focused onto the Moon's surface. So just as many of the sunsets that we see are reddish or orangish, that light, the Moon becomes very red and very darkened.
- 3.71** And throughout history, eclipses, both solar and lunar, have been omens of significant potential disasters. In fact, the word disaster and catastrophe contain the words astro in them, because people thought that stars influence their daily lives in existence for many thousands of years, and that a comet or an eclipse was actually a disaster. In fact, it's rumored that Christopher Columbus saved his life in the West Indies in, I believe, 1500 or so, when his crew was captured and he happened to bring along, as I said, astronomers are very important for telling time, marking seasons, and predicting for navigational purposes.
- 3.72** So on his ship, he had an astronomer with him who knew that there was predicted to be a total solar eclipse visible from the island. I think it was Jamaica where they had been kidnapped. And he said, "If you don't release us, our god is gonna eclipse your god and kill your god, the Sun." And then it happened, and they let him go. So astronomers saved Christopher Columbus's life.
- 3.73** The Moon is the dominant force that produces tides on the Earth's surface. Galileo, remember, had many brilliant ideas, but his least brilliant, in some sense, was that the tides that we experience here near the ocean, four times a day, two high tides, two low tides, he claimed that was caused by the rotation of the Earth around the Sun and kind of the sloshing of water on the Earth. But that's not what causes it. He was trying to prove a theory, which is basically—his hypothesis was correct. The Earth does go around the Sun, but his evidence was wrong.
- 3.74** So you can be right for the wrong reason. You can be wrong for the right reason. And that's part of the fun and interest of being a scientist, but also why you should always

be skeptical and don't just listen to someone because they're literally Einstein, who made at least seven catastrophic blunders. And it's too bad, because he could have had a very, very good career, that guy. He could have been famous. No, he turned out just fine.

- 3.75** So the tides are caused by the gravitational force and the fact that the Moon is orbiting around the Earth. And so it's pulling on the fluid slightly more by the inverse square law, pulling more on the side facing the Moon, than on the side farther away from it. That also kind of pulls the Earth a little closer to it because of Newton's first law, which we'll discuss in just a bit. And that causes this bulge to make the oceans highly exaggerated in the slide, but makes it sort of elliptical and makes an ellipsoid. And then, as the Earth rotates within that, you get the sloshing and you get this rising and falling of the ocean at any point on the Earth's surface where there's an ocean.
- 3.76** So that causes two high tides and two low tides because there's this ellipsoidal shape. Now, the Moon is also special because we've been there. It's the only other body in the solar system where humans have been, for now. Maybe we'll get to Mars someday. We're going back to the Moon. And one of my friends might be, very much likely to be, Jessica Meir, who's a PhD from Scripps Institution of Oceanography. It was part of my home institution, UC San Diego. She may be the first woman on the Moon in the next five or so years. So hopefully she'll be there. We haven't been to the Moon in over 50 years. It's crazy.
- 3.77** So I mentioned, when I was a kid, Pluto was a planet. And maybe for some of you, that was true too. What is a planet? Why isn't Pluto a planet anymore? It was decided in the early part of the 2000s, about 20 years ago, that a planet must meet the following three criteria in order to be considered a planet. And that was brought about because of the rapid discovery rate of finding so-called exoplanets. Were they actually planets? Were they part of the solar system? Were they outside of the solar system? Were they, even if they were bigger than, say, the Moon, do they count as planets?
- 3.78** The planet Mercury is not much bigger than our Moon, and yet it is a proper planet, so it's not purely based on size. So they have to meet these three criteria. They have to orbit the Sun. In our solar system, they have to orbit or in another solar system, they have to orbit their host star. There are planets, there are weird rogue planets that are flying through space, that aren't associated with actual stars that are at their center or focus of their elliptical orbit. The planet has to have enough gravity, as Aristotle predicted 2,400 years ago, to be spherical.
- 3.79** And lastly, they have to clear the neighborhood via the gravitational force, collected all the space schmutz and debris that surrounds that planet. And that's where Pluto fell short. It is spherical. It does orbit around the Sun in an elliptical orbit. It's a very strange orbit. It's highly inclined. It's also elliptical, but it has a lot of bodies near it that haven't been affected enough by its gravity. So it's kind of like being really famous or just Instagram famous. The gravity of a situation, or the gravity or the influence of a person

or of a planet, is really determined by its interactions with other people or other bodies.

- 3.80** OK, I'm not comparing these two types of stars. I'm not that famous. But the point being that, unless it does that, it's really not dominating gravitationally enough to be considered other than a minor planet, which is what Pluto is now considered. So planets are smaller than stars. Again, the Sun is 99.8% of the mass of our entire solar system. All the asteroids, all the meteoroids, all the planets, comets, minor bodies we'll talk about—99.8%, 0.1% of the solar system's mass is the planet Jupiter. And then all the other planets, comets, everything else, make up 0.1%. It's phenomenal how incredibly unequally distributed the mass of our solar system is.
- 3.81** So it's impossible, in fact, to have stars that are 1% of the Sun's mass, 2%. In fact, you have to be about at least 8% of the Sun's mass to consider yourself a star, which means that Jupiter couldn't be a star, even though it's made of gas. But it can't create the fusion processes that are necessary to be considered a star. There are objects that are kind of like failed stars that are much more massive than Jupiter. We're talking 100 times more massive than Jupiter, but not quite the 8 to 10% needed to qualify as an honest-to-goodness star.
- 3.82** They radiate heat. They're like super-Jupiters, many times bigger than Jupiter. They do produce, some of them can produce fusion, but not enough to really ignite and create the massive amount of luminosity that we ascribe to the Sun or other proper stars. So it'll eventually cool down, if it was fusing gas into heavier elements earlier on in its history. It can radiate light. But that light is typically far, far infrared light. Now, there's many more brown dwarfs or super-Jupiters than there are stars like our Sun in our galaxy.
- 3.83** So again, a star has to have a mass just in terms of the biggest planet in our solar system, which is highly unlike the Earth, Mercury, and Venus, and Mars, in that it's a gas giant. It's many, many hundreds of times the mass of our Earth. And it's primarily made of gas, methane, ammonia, other things. It may have a rocky or metallic core, but we haven't yet explored it enough to understand what its composition is. So that is going to be done over the coming years with satellites and various spacecraft.
- 3.84** A brown dwarf is many, many, sort of like 13 to 80, not quite the mass of the planet Jupiter. And then an ordinary planet like the other planets, like Jupiter itself, can be up to about 13 or so times the mass of Jupiter. So there are exoplanets that we'll discover that are giants, gas giants, and they're actually orbiting closer to their star than Mercury or Venus. And we'll describe those in a subsequent lecture when we discuss exoplanets.
- 3.85** So is there an upper limit? What's a lower limit? The upper limit's about 13 times. Otherwise, there's a lot of other chunks of rock, gas, ice, and dust that do orbit the Sun and other stars called comets. We call them comets or minor bodies. So again, for a

long time, when I was a kid, Pluto was a planet. It has been visited in the last 10 years by a satellite called New Horizons, which was the fastest-ever constructed spacecraft to ever leave the planet. It made it to Pluto in less than 10 years. It's an incredible feat when you consider how far away Pluto is.

- 3.86** Pluto's strange. It is spherical. You can see it from the picture. It has many moons that orbit it. It has a very long orbit. In fact, Pluto was discovered in the 1930s, which means that it has not yet completed one full year, Plutonian year, as seen from the Earth. So we will hopefully be alive to see that in the year 2180. Keep taking your vitamins. And so now we found things that are in our solar system and orbit beyond the realm of the planet Neptune, which is an honest-to-goodness planet, some 10 or so times the size of the Earth. Mass of the Earth, not the mass of Jupiter; it's much smaller than Jupiter.
- 3.87** So these are objects. And it's actually hypothesized that there is a ninth planet, called planet nine for now, that orbits beyond the orbit of Neptune, which is the boundary of the—that's the last planet in our solar system that we know so far. That doesn't mean it's the last planet that exists. There could be other objects clearing out their neighborhood, spherical in shape and orbiting the Sun. And those are called trans-Neptunian objects. So there are already objects that are 1100 times, there's almost 1100 objects or more that are beyond the orbit of Neptune. But they don't qualify as planets. They're called minor planets.
- 3.88** Effectively, they're spherical and they orbit the Sun, but they haven't cleared out their orbit. They haven't basically been the dominant celebrity. And they're part of the neighborhood of our solar system. There was one found in 2005. It's called Eris, and it's the largest object yet known. And it was shown by its orbit to have these properties. And even we're learning about its mass and its composition. Here's a bunch of these objects. Not all of them are spherical, but they're compared to the planet Pluto.
- 3.89** So you can see, Eris is thought to be even bigger than Pluto, which is another reason that Pluto was essentially demoted around the same time. So Eris is bigger and spherical than the planet Pluto. It also has a moon, potentially—Pluto has six or seven moons. And yet, if Eris wasn't considered a planet, that meant that Pluto couldn't be considered a planet. So those two things happened around the same time. So now we don't consider either one of them to be a planet. But that doesn't mean there aren't planets beyond the orbit of Neptune that we just haven't discovered yet.
- 3.90** In fact, we're looking for them with one of my telescopes that primarily is going to look for the fluctuations in the origin of space time itself. Again, see my cosmology class for more information about that. But the key point is that you can detect these things in a variety of different ways. And once you detect one that meets these three criteria, could be that we discover the first new honest-to-goodness new planet in 200 years. Because, remember, Pluto's no longer a planet, so that 1939 or so discovery doesn't count.

- 3.91** There are other objects. One's called Makemake, and it's a fun one to pronounce in addition to Eris. And they're called dwarf planets now, which is kind of funny because Pluto— there was a Disney character named after the planet Pluto, or around the time, or Pluto was named after the Disney character Pluto. I forget what came about. But now there's, you know, they call these things dwarfs. And of course, Snow White and the Seven Dwarfs is a famous Disney movie. So some of these things came about around the same time in the 1930s.
- 3.92** So we'll talk more about exoplanets and the potential for habitability of these exoplanets. What will be required for a life form to potentially exist there? So far, spoiler alert, we don't know of any for sure that exist and harbor life. But it's a very, very hot topic for a variety of reasons, because it may help to explain not only the origin of life on Earth, maybe perhaps coming from other planets, but perhaps the origin of life in the entire universe. And the existence of extraterrestrial life could imply the existence of extraterrestrial intelligent life, not just some mold or bacteria on Makemake or something like that, but on another planet with a civilization, perhaps that we could communicate with.
- 3.93** And some say that they're already here. So that's an exciting thing. We talked about spectroscopy in Lecture 2. We can use the Doppler shift, this change in frequency due to the gravitational pull of a planet on its host star. So even though we're tiny and Jupiter is tiny compared to the Sun, $1/1,000$ th the mass of the Sun, it's still enough to be detectable via my ingenious colleagues and friends in astronomy can detect the gravitational pull of Jupiter on the Sun. And it can not only do that for Jupiter on our Sun, but it can do that for a planet the size of Jupiter or even less, size of Neptune, on an exostar, on another star in another solar system many, many light years, dozens of light years away.
- 3.94** And then using the laws of Kepler, et cetera, we can predict what its orbit is and measure its mass. So we can measure these tiny fluctuations in the Earth's orbital speed— sorry, in the planet's effect on the Sun's dynamics. We can measure the Doppler shift of those spectral lines that we showed in Lecture 2, and how they shift back and forth as Jupiter orbits around every seven to nine years or so, or 11 years, I guess it is. And as that happens, as seen from Earth, the Sun will wobble a little bit more, depending on the direct orientation of Jupiter with respect to it. We can do that with many, many stars in our own galaxy.
- 3.95** And in that way, you can detect the existence of exoplanets and measure their mass and their—and measure their mass and their speed and their dynamics just as well. There's also an effect that's just like the transit of Venus that occurs for other stars with other exoplanets. When a star—say this is not the Sun anymore, say this is a very distant star, and say there's an exoplanet. Let's pretend that Neptune is the Earth. When this star transits its host, sorry, when this exoplanet transits the host star that it

orbits around, it blocks out a tiny fraction of the light, the flux that we receive from that star.

- 3.96** It's like a miniature transit or miniature eclipse. It is. It's called the transiting. It's called the transit method. And we can detect the size of the object, as well as its orbit when that happens, because it only happens for a short period of time. So astronomers have done this with satellites, called the Transiting Exoplanet Survey Satellite, called TESS. This is what it looks like. The light is somehow constant for a while. Then it goes and makes an eclipse.
- 3.97** So what's happening here is, there's actually, they call it two different eclipses. So there's a planet and a star. The first thing that happens is the planet, the red ball, goes in front of the star. OK, so here's the planet as seen from here. It goes in front and blocks off maybe 1% of the light or so. And then it goes out, it orbits away from it. And then you can see the combined light of the planet, sorry, the Sun, the star, and the exoplanet in reflection. There's some light getting reflected.
- 3.98** So we would see this, like a crescent, a crescent exoplanet, like the first phase of the Moon as we see it. It'll be half illuminated from the perspective of the Earth. So there's more light in this configuration than when the planet goes behind the star. So that's called a secondary eclipse. So now the Sun is eclipsing the exoplanet, or the star is eclipsing the exoplanet. So you get slightly diminished light, not nearly as much as the primary eclipse. So this only happens when you have a star that's being orbited by an exoplanet.
- 3.99** But it's very rare that these three bodies line up in my favorite word, syzygy, between what's established by the Earth exoplanet and exostar. It's very rare. But there's a lot of stars in the Milky Way galaxy, and there's about ten or hundred bodies that are the size of planets in almost every star that we've ever looked at in the hundred billion stars in our galaxy alone. So it's incredible that this occurs. It occurs very rarely, but with very large numbers, it happens enough that we can actually use this to measure the properties of the exoplanet.
- 3.100** And then the coolest thing that people are doing now that we'll talk about in Lecture 8 is, when this happens, imagine that this is—I'll rotate in Venus, OK? Now, this is an exoplanetary system. There's a star, there's the Earth looking at that star, and looking at the planet. When that primary eclipse happens, the transit occurs. This planet, if it's like Venus, has an atmosphere. Some of the light will go and get obscured by the planet, but some will make its way through the atmosphere, get focused like a lens to our telescopes.
- 3.101** That means we can do spectroscopy on the atmosphere of an exoplanet, which is enormously interesting, because you could see things on this planet. Do you know what the first technology was invented by humankind? It was agriculture. Agriculture changed the chemical composition of the atmosphere, putting out methane, changing

much more carbon dioxide, being absorbed by biological organisms. So, one of the signs of not only life but technology could be revealed by these transits that illuminate the atmosphere. You could measure the chemical compound.

- 3.102** Imagine if you're looking out in the atmosphere and you saw that there were, you know, basically propane and butane, and there were other molecules, freon, other molecules that you could determine, and you could rule out the fact that they were made naturally occurring. Of course, you have to rule out that there's processes that could also produce things. But the more complicated the molecule, freon is a pretty complicated molecule. Or, you know, there are many, many other gases and chemicals and so forth that you could sense.
- 3.103** So, if you were looking at a planet and you saw that, it would be a marker perhaps not only of life, but of technological life. And that intriguing possibility will be a big part of the subject of the end of this course and maybe a future course we'll see. Life on other planets. Again, you get two shots, two bites at the apple, if you will. The primary transit, when it's blocking most of the starlight that it's going to block; again, this is a fraction. It's hard to read the scale here. This is going from 100% to 98.3%. This is a planet that's about 70% the mass of Jupiter. And it's still only blocking about a percent or two of the light, based on the orientation of its distance from the star and our distance from it.
- 3.104** We measure it and we find that has a density of about one third of water. And that's interesting because the Earth has a density a little bit more times water because it's actually made of mostly rock, the core of the planet, and only a tiny drop of it is really the form of water. So if the density were much higher in the form of rock, we could measure its density. How do you measure its density? Well, you know, the size based on how long the eclipse took and the shape of this eclipse. That gives you the disk, the area of the planet. Assume it's a sphere. Now, you know the radius. You can compute the volume of the planet.
- 3.105** And if you can measure the mass of the planet using Newton's laws, then you can calculate the mass per unit volume, which is the density. Imagine you found a planet that had the exact same density as Earth does, in terms of rockiness and also water content. It would have to have a blend of those two properties. Very exciting. So in our next lecture, we're going to talk about other stars and how they illuminate the universe and what they teach us about the size, structure, and composition of our galaxy, which will then lead us onto an amazing discovery that galaxies are primarily not made of stars, but made of something much more mysterious called dark matter. Thanks a lot.

Lecture 4: The Life of Stars

At a glance — The life of stars.

- 4.1 Last lecture, we talked about what our solar system is based upon. The planets, the minor bodies, the asteroids, comets, et cetera. And we'll have much more to say about those objects as we proceed in the course. But the most important actor, the star of the show, if you will, is our Sun. Now, our Sun is far from the only star in our galaxy or in the observable universe. In fact, it's one of perhaps at least 100 billion other stars in our own galaxy alone.
- 4.2 And there may be as many as that same number of galaxies in the observable universe, meaning there might be 100 billion squared stars, some of which are just like our Sun, throughout the entire observable cosmos. And we'll talk about that in subsequent lectures. So, what is a star? Well, at the broadest definition, it's a giant ball of gas that's undergoing nuclear fusion. Nuclear fusion is one of the four fundamental forces of nature, including electromagnetism and gravity.
- 4.3 There are two types of nuclear processes that we'll talk about today. These processes result in the production of energy. Sometimes you get more energy out than you put into the system, and that then produces a self-sustaining nuclear reaction, in contrast to, say, a chemical reaction, combustion, or something similar. So a star is usually, typically a spherically-shaped ball of gas which has enough gravity to compress it to a point in which its fundamental nuclei, primarily at first, in the case of a star like our Sun, protons by themselves fused together to make hydrogen into helium.
- 4.4 Helium, of course, the name of the Greek god of the Sun, and was the first element, if you will, discovered not on Earth, but on the Sun, based on spectroscopy, which we discussed as one of the primary tools of astronomy in Lecture 2. Now, nuclear fusion doesn't occur everywhere in the star. It only occurs in the densest portion of the star's core, which is about 15% of the volume of the star. The other 85% is really the production of heat interacting with the other materials that have been produced in the star's history, including the hot plasma region that's on the outside of the star.
- 4.5 But the inside of the star can be at several millions of degrees above absolute zero, just an incredible process. Now, the actual recipe to create nuclear fusion is quite complicated and was only understood in the last 50 years or so. And this process is as follows. We take hydrogen. Where did the hydrogen come from? You couldn't go to Whole Foods to pick it up. The hydrogen was produced in the Big Bang. The hydrogen in this glass of water, as we'll talk about later in the lecture on cosmology, was produced in the very first few minutes of our universe's history, in what's called nucleosynthesis, the synthesis, production, of the very first nuclei. That'll be a subject in Lecture 7.

- 4.6 In the Sun, once the raw ingredients were produced some 14 billion years ago nearly, that material stayed around and was eventually attracted to each other by gravity. Helium nuclei would form and coagulate in regions that would eventually become protostars. When the protostar's density got high enough, gravity pulled together, purely the force of gravity, pulled together protons with enough force, if you will, that they overcame their electrostatic repulsion.
- 4.7 Protons, remember, represent one unit of positive charge. So, to get them beyond the point of electrostatic repulsion, which is produced by like charges, like magnetic fields, and like charges, they repel, opposite charges attract. To overcome that, you need a force much stronger than electromagnetic repulsion, and that's nuclear fusion. And that force is called the strong nuclear force, one of the two types of nuclear forces that we'll discuss in this course.
- 4.8 Once the protons come together, they fuse together. Now, how can two similarly charged objects, two protons, come together? It must be that there is an internal structure to the proton that has a type of charge, not electric charge, but a type of nuclear charge that can overcome with greater force the electrostatic repulsion of the two like charges. That is the so-called inner structure, the quark structure, of atomic nuclei that we'll discuss later.
- 4.9 Once that force is overcome, the protons and their subsequent quarks combine together, and what you get is astounding. One of the protons changes its flavor altogether in a type of alchemy that the ancient alchemists could barely dream of. Transmutating a proton into a neutron, a neutral, non-charged entity, subatomic particle. When that happens, to create what's called charge conservation, you have to balance the two positive charges. They have to be balanced. The nucleus that was a proton splits into a photon, a neutrino, and what's called a positron. That balances charge conservation and other things like momentum.
- 4.10 Energy is always conserved, even though mass may not be conserved. Once you have the conversion of one of the two protons to a neutron, the proton can fuse with the neutron and that creates an isotope which is a heavier version of the original element. Original element was hydrogen, now it's still hydrogen. The element is determined by the number of protons it has. It still has one proton, but it's twice as heavy because now it has a neutron attached to the proton. This is called deuterium. It was discovered in the 1930s by Harold Urey, after whom our chemistry department at UC San Diego is named.
- 4.11 That then, the deuterium nucleus, proton-neutron fused together, can then combine with another proton, again produced during the Big Bang, perhaps in our case of our Sun some 4 to 10 billion years earlier. And that proton combines with the deuterium nucleus to then create what's called helium-3, which is an isotope of helium. There are two isotopes of helium; there are three isotopes of hydrogen. So we encountered two

isotopes already. Helium-3 is now the one that then releases heat in the form of a photon.

- 4.12** And then, helium-3 can combine with another proton, or in fact, it can combine with two. Helium-3 nuclei can come together, producing in the end a helium-4 nucleus, which is the ordinary nucleus that we know and love from parades and helium balloons. At the end of the cycle, you're left—you started with four protons, and at the end, you're left with two protons and two neutrons. Two of the neutrons have changed their identity forever, fusing themselves with protons and then making into helium-4.
- 4.13** This process was only relatively recently understood. And as a byproduct, when I cook, the byproduct is always a mess and wasted ingredients, and sometimes a lot of heat and fire. But in the case of the solar fusion process, it's a tremendous amount of heat energy. And if we go through the details of that recipe, we'll find out exactly how much energy is liberated. That heat then goes to heat up the star and the light that we then get some portion of, some tiny fraction of, on Earth.
- 4.14** And yet, that tiny fraction, 1,000 watts of energy per square meter of area at the surface of the Earth, that's enough to do tremendous things, power huge amounts of our cities and so forth. And the source of all of our weather and every phenomena on Earth and climate and weather is responsible primarily to, driven by the Sun. So, how does this work to produce energy? Just combining two things together, you'd think that would take energy, but it actually gives off energy. And the reason it does that is because the neutron actually has slightly more energy than the proton.
- 4.15** And when these neutrons combine together and fuse together, we eventually get two protons, two neutrons combined in a helium nucleus. What's left over is this astonishingly small amount of what's called rest mass energy. So this amount, which is 0.0458 times 10 to the minus 27 kg per reaction. So a minus 10 to the 27 kilogram is a billionth of a billionth of a billionth of a kilogram, and then 4/100 of that amount of pure energy, via Einstein's $E = mc^2$. You can convert this amount of mass—that's what kilograms measure; it's a unit of mass—converting that into energy.
- 4.16** It still turns out to be a very small amount of energy, as this calculation shows. The amount of energy produced is about 4 trillionths of a joule. And to give you some notion of how small this is, so a light from a 100-watt light bulb is the equivalent of 100 joules of energy released every second. So this process, if it happens, it's a single fused helium nucleus from two protons and four protons in the original combination, only driven by gravity.
- 4.17** There's no pumps or gears or levers or nuclear reactors. It's purely gravity that's smushing these things together with enough force to cause them to fuse together. It's almost again, to me, as a physicist, the more I understand, the more I think it's almost miraculous that we're even here. And the fact that the neutron is slightly more massive than the proton, otherwise they're almost identical, except for their charge, obviously,

is another accident without which we wouldn't be here. They're perfectly identical. We would not be here.

4.18 So this releases a trillionth, $1/100$ trillionth of the amount of energy in say, 1 second, as a light bulb does. Imagine a trillion light bulbs, still be less energy. Sorry, it would need a trillion of these to make even less than 100th of the energy of a light bulb. So how does this work then, to produce the heat that we know and enjoy and go out and sunbathe in and grows our crops and drives our weather and powers our solar panels? Well, there must be a lot of masses. Well, there must be a lot of kilograms of the Sun, of protons on the Sun, that are being converted into it. And in fact, it's an astonishing number.

4.19 So this one amount of four protons is small, but when you convert the number of protons in a kilogram and multiply by that tiny number, you get an enormous number. You get something like 600 trillion joules. You went from fraction of a joule, but now you're converting it because you're multiplying by this enormous number, c , the speed of light, 300,000 kilometers per second. Square that number, that's an enormous number, and when you convert mass to energy. So when you convert per kilogram, it comes out to be 600 trillion joules of energy for every kilogram of protons that you put together.

4.20 Still, it's kind of hard to wrap your mind around it, right. So now you convert the 620 trillion joules of energy fusing 1 kg of hydrogen, and now imagine you had the same amount of hydrogen, the gas that was inside the Hindenburg that blew up; that would only release 140 million joules, which is still—it's a lot of energy, but it's a million, it's more than a million times smaller than the amount of 1 kg of fusion. It shows you that chemical reactions, as powerful as they are, they're nothing compared to nuclear reactions.

4.21 So, converting it—maybe now we can start to convert it to something that you could start to comprehend. So 620 trillion joules converted to gasoline is about the amount of energy, equivalent fuel you'd need to drive 300 million kilometers, which obviously no one has ever been able to do and would take quite a long amount of time to do that. Now, that's only for 1 kg. Now think about how many kilograms on the Sun right now are being converted from hydrogen to helium so that we can live, or at least have been doing it up until at least eight minutes ago. We don't know for sure if it's still going on right now. Hope so.

4.22 There's about 6 trillion times the energy released every second in the Sun than was used for the first atomic bombs ever made for scale. And there's enough hydrogen in this core of the Sun that we don't have to worry about the Sun running out of fuel, running out of gas, literally, for the next 10 billion years or so. So keep paying your taxes, keep paying your tuition. So don't panic. It's only about half, it's only about middle age. So like some of us, it's about halfway through their life expectancy.

- 4.23 Now, how do we know the life expectancy of the Sun? In fact, how do we know how old the Sun is? Because you go to a doctor, the doctor's going to ask you, "How old are you?" Going to measure your vital signs, your temperature. It's going to measure your pressure, your blood pressure, and that's to make predictions about your life expectancy. Now, again, we can't do experiments on the Sun. I can't say, well, what if we replaced some of the quarks with antiquarks and some of the neutrinos? We can't do any experiment, but thank goodness there's 100 billion other stars in our galaxy alone, and 100 billion other galaxies. So we can do a lot of statistical ensemble types of experiments.
- 4.24 The theory of formation of our solar system involved a preexisting star. And that preexisting star is long dead. And it was made purely out of the hydrogen that was made during the Big Bang, after its life came to an end, about 100 million or 200 million years after it formed, which was many billions of years after the galaxy itself formed, and after the Big Bang. That star became massive enough to explode and produce what's called a supernova. And that supernova spread out a lot of its energy and material into our neighborhood in what would become our solar system in the galaxy.
- 4.25 Because the star also didn't burn up all of its hydrogen, a lot of the hydrogen in that primordial star, called Generation 2, for reasons I'm not going to discuss—our Sun is called Population 3, the first generation of stars, Population 1, by astronomical nomenclature, and the generation that probably produced our Sun was called Population 2. That's just a nomenclature; it's not that important. But when that star died, it produced and released into our galactic neighborhood all the raw materials, not just of our Sun, but of you. Your materials came from that same explosion that existed, that came about some 4 or 5 billion years ago.
- 4.26 Because there's no source of matter that's just coming into existence. So all the matter had to come from preexisting sources, including the matter that made up our Earth. That means we can use the Earth as a cosmic chronometer. And that cosmic chronometer tells us the age of the Sun as well. And then once we see the age of other stars and how long they live, and compared to the age of the Sun, we can predict how much time is left before our life insurance policy comes due.
- 4.27 Now, for a long time, it was thought that the Earth itself was only 5,500 years old. In fact, before the 18th century, in the 1700s, the chronology of the Bible was accepted as the true age of our Earth and the concomitant age of the solar system as well—because, after all, the Sun was created on the fourth day of creation, according to the Bible. Now, Isaac Newton was actually a very religious person. Actually, he considered his scientific accomplishments and his alchemical experiments to be paling in comparison to his religious devotions. He was a very deeply committed Christian, and he viewed them as trivial compared to what he tried to accomplish as a theologian. He was a deeply religious person.

- 4.28 So again, in the 17th century, 1600s, before Newton's time, this Archbishop Ussher, he wrote that the beginning of time fell on the first day of creation. So in the Hebrew calendar, the first day of the week is Sunday and the last day of the week is the Sabbath, which is Saturday. So the creation came on day one, which is a Sunday night. So Hebrew days start at night and the biblical time, they start at night and it's a lunar calendar, but that's irrelevant for this. He calculated, going back from various calculations, that the beginning of creation was actually the year 4004 BC, or BCE as we say now.
- 4.29 Now, according to biblical scholars, of which I am not, the creation story does not refer to the creation of the universe as we think about it, because after all, the Sun again was, and the Earth, the Sun, and the Moon, created on the fourth day. It refers to the days after which the first people, Adam and Eve, were created. According to, again—this is biblical or Torah scholars, again, I'm not one, but so, there could be some great amount of time between the creation of the first people, Adam and Eve, and the biblical narrative of the creation of the world or day one. Again, not here. This is not a theology class, so take that what you mean for what you will.
- 4.30 But it's interesting to note that it doesn't necessarily have to correspond. To some people it does. Some people say it's literally 24-hour days, but that doesn't make so much sense when you consider, again, the day is defined by one solar rotation as seen from the Earth. The Sun wasn't created till day 4. So this was very accurately dated 23rd of October, 4004 BCE. Now this held for a long time. Remember, it was almost heretical to speak against the biblical narrative. The Catholic Church was incredibly powerful during the late Renaissance.
- 4.31 And even to talk about other planets and stars, as you know, from our first encounter with people like Giordano Bruno and later Galileo, didn't end well for the scientists sometimes. But later on, scientists in the so-called Enlightenment, which I always loved, they called themselves the Enlightenment— it'd be like I called myself, "I'm a member of the genius generation"—but they called themselves the Enlightenment. And during that period, the practice of science became more regularized, where it had a standardized pattern of hypothesis, followed by testing, iteration, and replication, publication, which we now see in modern form.
- 4.32 And during that time, science like geology and other sciences, paleontology, even atmospheric science, chemistry and things like that, converged to give some hint that there were cycles to time that could be used to create an Earth-based chronometer, a way of measuring the Earth's age. So they would look at the fossil behavior, of fossil patterns on Earth, and the structure of sedimentary layers; and basing that on the pattern of floods and tides and so forth, they could construct how many tidal periods, how many years had gone by, by counting the different layers.
- 4.33 And we can still do this today. In San Diego, there are cliffs of sandstone. You can count layers upon layers. You see different fossils, creatures and so forth. And in particular, in

Dover, in England, where the Royal Society was based, they were able to measure all these different types of fossils based on tiny little shells. And then they could do dating on those by counting numbers of layers. And they found there was no way it could be consistent with 5,000 years. So they started to predict that it was over 100 million years old, just based on very crude calculations of the thickness of a single layer. Say that's a year, and then you have 100 meters-worth of sediment and different, 1 million cycles-worth of sediment.

- 4.34 You can see that in the Grand Canyon, if you've ever been there. It's completely obvious that there's been different geological epochs over, literally, millions of years. So it couldn't be thousands of years old. But it really wasn't until the advent of called radioactive or carbon dating that it was possible to date things accurately using chronometers, not in the form of watches or counting layers of seashells, which are, you know, they have some degree of precision, but not much.
- 4.35 But now you're dealing with the realm of the nucleus, which is very accurately measured, very stable atomic clocks, that basically function as ways to keep the time, based on when these structures were formed and how much of them exist to this day. And then using the law of radioactive decay to calculate how much time has elapsed since their formation. That's the basis of radioactive dating. And you're more familiar with it in the form of carbon dating, where the supposition is that an organism, it could be a plant, tree, dinosaur, whatever, was ingesting organic material, either carbon dioxide in the case of plants, or eating plants in the case of animals, dinosaurs and whatnot.
- 4.36 And when it ingests that material, it has a certain amount, a ratio, of radioactive carbon-14, which does decay and eventually become stable carbon-12, which is the predominant source of carbon. And by measuring the ratio with the assumption that the animal stopped eating stuff when it died, or the organism stopped metabolizing carbon-14, then the ratio of carbon-14 and the ingestion stopped on its death day, which is good enough for calculating things that are thousands or millions of years old. And then you measure how much is left, and when you use what's called the half-life.
- 4.37 So, some nuclei are unstable. Depending on how many neutrons to protons a nucleus has, that determines its stability. This was discovered by a professor at my university, Maria Goeppert Mayer, in the 1940s and 50s as part of the Manhattan Project, and she later won the Nobel Prize for her work. And the stability of atomic nuclei is a very strong function of the ratio of protons to neutrons. So uranium-238, which you may have heard of or even seen in the famous movie Oppenheimer and other features, has an atomic number which is the sum of its protons and neutrons.
- 4.38 And it has a lot more neutrons, about 50% more neutrons than protons. And that means that it's relatively unstable. It will decay eventually into lead-206, which is more stable. And the ratio, again, of these two things to one another, it decays via the weak nuclear force, which is a weaker form of the subatomic forces. It's a random process.

We can't predict when any given neutron will decay into a proton and electron and a neutrino, but it can be predictable. On ensembles where you have a chunk of material, it might have a trillion trillion nuclei.

- 4.39 So you can't predict when one of them will, but you can predict when half of them will be gone, on average. Just like in this room, there's trillions of air molecules moving around in every cubic centimeter, and I can't predict the velocity of any given one of them. But when I have a trillion of them, trillion trillion of them, I can predict very accurately their average behavior, how fast they're going to be moving on average.
- 4.40 And on average, their velocity is zero, right? There's as many moving to the left as to the right, except out of my mouth, where there's a lot of hot air coming out, right? But on average, you can't predict—you know exactly accurately how their average velocity will be zero. And what you calculate is the square of their average value. And that tells you a gauge of their temperature. That random motion, when squared, is the quantity that we associate with the thermal or heat capacity or temperature of a gas, for example.
- 4.41 So, nuclear decay works in this fashion, as Maria Goeppert Mayer and others worked out. You have some atom, and it can move down in terms of number of protons and total number of atoms, atomic nucleons, protons plus neutrons, and it can do various different things. It can lower its charge, and it can shift to becoming more positive by the emission of an electron. That's called beta decay. That shifts the Z number, the number of positive charges, to the right. When it produces a neutron, it doesn't change its charge, so it just moves in a single column.
- 4.42 When it produces a proton, it reduces the number or emits a proton, it reduces its atomic, its proton number, or its charge, Z. And it can also emit a positron, which also makes it more negative and doesn't change the, changes the isotope, but doesn't necessarily change the parent atom, the periodic table on which it exists. So after one half life, which can be measured in milliseconds for some unstable nuclei that aren't very useful for measuring billions-of-year-old things, to literally hundreds or thousands or millions of years, in the case of some very, very stable and long-lived nuclei.
- 4.43 So, after one half-life, half the amount of the radioactively decaying material is left. As I said, if you know that uranium-238 eventually decays to lead-206 via many, many decay processes, but eventually it'll stabilize and it won't decay anymore, all you have to do is measure the ratio of the amount of lead to uranium-238. There'll always be some uranium-238 there. Eventually, there'll be a lot of lead. And I'll tell you how many half-lives have elapsed. If it's 12.5%, then you know that three half lives have elapsed because two to the third power is one-eighth, and so one-eighth of it will be gone after 12, after three half lives.
- 4.44 So in the case of uranium, the half-life is 4.5 billion years. So that's extremely stable. So you can measure things that have elapsed for billions of years. So, in looking at objects

on Earth, special types of objects, as we'll talk about, we know that the Earth has experienced at least 4.5 billion years-worth of decay, meaning it was formed about 4.5 billion years ago. So there are objects that geologists have found in the last hundred years called zircons, which are some of the oldest objects, oldest geological specimens, on the Earth.

- 4.45 And some of these have ages dated because they have uranium within them, and they separate out with lead being separated out from one another. The uranium is more dense than the lead, so it will sink during the molten process, it'll sink to the bottom, and so it'll be physically separated from the less dense lead. And then by measuring the profile of how much has decayed, as a vertical slice through these objects, they can actually find the time since the rock itself coagulated and solidified from being magma to being a rock.
- 4.46 Now that's not a huge brick of uranium. It has other metals mixed in it. Predominantly, there is a lot of lead in it, but there are other metals in there that aren't important for this process. So we date the age of the solar system using this property, and in fact, I will give you in an upcoming lecture a sample of the early solar system that's as old as the Earth and the Sun itself. And it's a meteorite, and it's also very highly metallic, and it has radioactive components in it. Don't worry, it's not dangerous. Only a few people have passed away since my last—no, nobody's passed away since I've given these out.
- 4.47 And the half-life again is very long for some of the cobalt, the radioactive cobalt that's in it. But again, it doesn't pose a danger. Can't even detect it on a Geiger counter. So I will give you a sample or a fragment of something that is as old as the solar system, and you at home can also get it at briankeating.com/PA for Peterson Academy, and I will send it to you as well. So the Sun is about middle-aged. How do we know that? Because we look at other objects.
- 4.48 So Galileo is my hero, and that's one of the first sketches ever made of what are called sunspots. And sunspots are extremely important. Remember, we're trying to determine the property of our Sun. Use that to understand properties of other stars, use those properties to understand galaxies, use the galaxies' properties to understand the universe as a whole, using techniques like spectroscopy, photography, luminosity distance, parallax. These are all ways to survey, understand the composition, the density, and the destiny of our universe, as we'll see in later lectures.
- 4.49 Galileo saw these sunspots; he didn't know what they were. Nowadays, we know there's an intimate relationship between them and the magnetic field of the Sun and its temperature. The Sun's surface is nowhere near as hot as the Sun's core. The Sun's core is millions of degrees Kelvin, and that's the temperature associated with the nuclear fusion reaction. Now, a photon that's produced at the core of the Sun doesn't just zip right out because it's surrounded by this huge amount. 85% of the Sun is not undergoing nuclear fusion, but it's still extremely hot plasma, glowing gas in the state of matter that physicists call plasma.

- 4.50 Plasma is relatively hard for photons to get through. So it actually takes about, on average, 100 years for a photon produced at the core of the Sun to make its way out, bouncing around all the plasma and eventually getting out and coming in a direct straight line to us on Earth. So it takes eight minutes for light to get from the surface of the Sun to us. It took a million years, sorry, it took 100 years for the light particle, the photon itself, to get out of the core of just the Sun itself.
- 4.51 The Sun's only about, well, the Sun is about 1 million kilometers across, quite big, but it would only take a fraction of a few seconds for it to get out. And yet it takes eight minutes for it to get here, plus that whole century for it to get out of the core. Now, it turns out that the Sunspots that Galileo saw, and you can see with a filter in front of your telescope—Galileo was so brilliant. He didn't have solar filters, but he was able to project the light of the Sun onto a piece of paper after focusing it from his telescope.
- 4.52 And remember, this is in an age when they didn't even have clocks. There wasn't even like a grandfather clock or a wind-up watch. Nothing like that existed for 100 years or more. And yet he was able to do extremely precise timekeeping using other tools and techniques. He also measured, day after day, the properties of these sunspots and came up with the first estimate of the rotation speed of the Sun. I mean, how brilliant and cool is that? So he measured that the Sun takes about 30 days for a sunspot to move across the entire disk of the Sun.
- 4.53 He didn't think it was like a planet or something. I probably would have thought, "Oh, it's cool planets, aliens, whatever." He knew, no, it's something going on on the Sun's surface. And he used it to measure the rotation, the day. The solar day, if you will, is 28 Earth days. Pretty cool. Later we'd learn, later we'd learn that the temperature of the surface of the Sun is about 5,000 Kelvin. It's colder, if you will, than the Bunsen burner or the blowtorch that I used earlier. That's hotter, and that's why it glows blue, where the Sun glows quite lower at a temperature about 5,000 degrees Kelvin.
- 4.54 And the difference of 5,000 degrees and 4,500 degrees is enough to make a visible difference that we see as sunspots. So a sunspot, that region of the Sun, is cooler for a variety of reasons. It also has a magnetic field, but that's not necessarily related to its appearance. But it's like a shadow. It's like you have a BIC lighter in front of a blowtorch. You would see, basically in contrast, the BIC lighter looks like it's basically freezing, compared to the blowtorch, much hotter. So these sunspots are regions where the Sun is some, almost 1000 degrees Fahrenheit colder than the surrounding material that's not present with sunspots.
- 4.55 So, what will happen? Someday it will happen. Someday, I'm afraid, we won't be able to have cosmology courses, astronomy courses, and study the Sun. That'll take 10 billion years or so. When it runs out of hydrogen, it starts making helium. Once it stops making helium, it will eventually make what's called a white dwarf, which is a star of very, very dense properties, smaller than the size of a city, but with a mass comparable

to a fraction of the Sun's mass. So it's incredibly massive, much more massive than a planet, in a much smaller region than a city on Earth. That's called a white dwarf.

- 4.56** They play a role in later discussions, and they do play a big role in cosmology, because they allow us to understand the acceleration of our universe's expansion. Now, a star that's much more massive than the Sun has a lot more fuel, lives fast, and dies young. It burns out much faster, and its end is cataclysmic. What happens when a star, much more massive than the Sun, it makes helium, then eventually it makes even heavier elements—carbon, nitrogen, oxygen, silicon. Eventually, it tries to fuse two nuclei together to make iron.
- 4.57** And at that point, the end of its life comes because, when you fuse iron, it takes more energy to make iron than the iron gives off when it fuses together from the two parent nuclei. And that happens, the heat is no longer present to prevent gravity from sucking in the star's outer core. When all the material, now it's basically these massive layers of carbon, nitrogen, oxygen, they come colliding together, they collapse in what's called a core collapse supernova. That blasts out the material at the core of the supernova and everything else, the carbon, nitrogen, silicon.
- 4.58** And thank goodness for that, because the iron in the hemoglobin molecule in your blood came from the supernova that exploded 4, 5, 6 billion years ago. It was a much more massive star. We don't know about it. It gave its life so we could be here to discuss it. But it's kind of beautiful. As Carl Sagan said, "You're made of star stuff." Where else would you get it from? Where else could it possibly come from, other than from other stars? This was also discovered at UC San Diego by my late great colleagues, Margaret Burbidge, her husband, Geoff Burbidge, Willie Fowler, and a great cosmologist named Fred Hoyle.
- 4.59** I talk about the formation of this nebula called the Crab Nebula, which was once thought to be one type of supernova; but now, thanks to the James Webb space telescope, this image on the right, was now found to be a different type of supernova, exactly like the one I described, an iron core collapse supernova. Now this supernova that made the Crab Nebula in the constellation Taurus is actually one of the oldest, if not the oldest, astronomical phenomena ever observed, besides the Sun and the Moon and the planets.
- 4.60** It was witnessed by Paleolithic and early astronomers in prehistoric America, in New Mexico, at a place called Chaco Canyon. And they made a depiction of it with the hand. And there's a Moon and there's a star that could be seen during the daytime. What they were depicting was that there was a new star that came about. Imagine seeing a star as bright as a tiny fraction of the Moon, visible day and night? That was the process that was liberating so much energy in this object that we now call the Crab Nebula.
- 4.61** So before, it was a red giant, then it collapsed, and then made this incredibly dense object called a core collapse supernova. And the material that gets blasted out then

became the material from which our solar system is made. And the meteorites that I'll be passing out, or you can get at my website, [briankeating.com /PA](http://briankeating.com/PA), those objects are primarily made of the last thing the star made, iron. That's why they're so cool and metallic and they're magnetic as well, as you'll be able to do experiments.

- 4.62 So now there's many, many other stars. There's 100 billion other stars in our galaxy alone. And then we see stars in other galaxies, too. What are they made up out of? We can tell a lot by going back to our old friend, spectroscopy. Spectroscopy tells us, based on the color, the intensity, the luminosity of the stars, how massive they are. And they tend to follow an evolutionary track like biological evolution. They start in one branch of this so-called main sequence diagram, and they move up the slope— sorry, they move down from hot and blue to large, cool, and red.
- 4.63 And sometimes they blow up depending on their mass. The mass has to be several times the mass of the Sun. And so most of them lie on this graph. And there are exceptions. There are brown dwarfs, so we've talked about those. There are white dwarfs, we already talked about those. The Sun is an ordinary, what's called a main-sequence star. And there's an equivalent large number of them, probably about 20 billion other stars on the main sequence, more or less like our Sun in our galaxy.
- 4.64 But that means there's about 80 billion that are not like our Sun, either brown dwarfs or objects like white dwarfs that can last for billions of years, more than our Sun will ever live for. Another gift of spectroscopy is the measurement of velocities and dynamics of stars. So you can learn about the environment that a star is in by how it's moving, what gravitational forces are affecting it, what other processes are affecting it, are impacts and so forth, and other objects are affected as well.
- 4.65 So the original model of the solar system and the universe was by Ptolemy and Aristotle. It was an Earth-centered cosmology. And to accommodate these weird phenomena that were witnessed, like retrograde motion of planets like Mercury, Venus, and Mars, it was only possible to accommodate those in an Earth-centered model if each planet had extra or ancillary orbits associated with it that aren't physical. So he was basically trying to teach to the test.
- 4.66 And he was basically trying to figure out a way that he could keep the Earth at the center of the solar system. But it was so complicated that it was eventually too big to succeed, and it had to eventually give way to the heliocentric system. So this is the Ptolemaic, with adding on epicycles, additional orbital-type motion. That was eventually done away with by the great Johannes Kepler.
- 4.67 Now, Kepler conjectured these laws, that planets move in elliptical orbits, that their orbits' period is related to their orbital distance and the parameters of their ellipses. But he didn't prove that that was true. It took Isaac Newton to prove, via the law of universal gravitation, that gravity was a force that affected the motion of objects. And you could relate the dynamics of the orbit to its orbital parameters. In other words,

how fast it was going; you could relate that to the size and distance from the star to the planet. And we'll talk about that.

- 4.68** So I mentioned that even for minor planets, dwarf planets like Pluto, we have yet as a species to witness a complete orbit of the planet of the minor planet, dwarf planet, Pluto. It simply takes too long and it was only discovered less than 100 years ago. So distant planets move slower. They also move on ellipses, but they move slower. They also typically will have a nice closed orbit moving at, in different, moving along elliptical paths. In close-in situations, you can have open orbits that don't actually converge and make complete ellipse-type behavior. Those are pretty rare. We don't really consider those.
- 4.69** But they're a consequence of the law of motion that Newton has conjectured. And we'll talk about the law of motion as it comes to gravity. We'll prove this in a later lecture. But suffice it to say that the velocity of a star planet around a more massive star or planet behaves in this mathematical form, that the farther you get away, if you quadruple the distance of a planet compared to another planet, that faraway planet will have half the velocity of the close-in planet. So the proof of that, as I said, will come later. But this is a consequence of Kepler's laws, as derived from Newtonian law of universal gravitation.
- 4.70** So in low Earth orbit, where the space station is and SpaceX rockets go to, and even SpaceX Starlink satellites are in low Earth orbit, they need to travel at enormous velocities because they're so close. They travel at 18,000 miles per hour. And if you take into account the fact that the Earth's circumference is about 24,000 miles around, that means it takes 90 minutes for a satellite to go completely around the Earth. If you've ever seen a Starlink at night, you'll see it moving quite fast. It's moving at 18,000 miles per hour, to stay in that orbit.
- 4.71** We can actually calculate the Earth's rotational speed. The Earth's circumference is roughly 24,000 miles. The day is about 24 hours. That means at the equator the Earth is rotating, if you will, relative to a fixed observer, about 1,000 miles per hour. Obviously, we don't feel that. But that was a big source of confusion for early physicists, including Galileo. If we're moving at this incredible rate, how can we not fly off? How can we not feel it? Well, that was a great puzzle for many years. But effectively, it's because we're attached to that motion, and it's impossible to distinguish relative velocity.
- 4.72** Two observers in relative velocity cannot say which one of the two of them is moving. If you've ever been on a train and there's another train next to you, and that train starts to move, it feels like you're moving, or sometimes the other way around. That's because as long as it's at a relatively constant velocity, no one of you can say truly if you're at rest or if they're at rest. So we don't experience that because we're in relative motion at the exact same rotational rate, and we're attached to the Earth.

- 4.73 But that was a big mystery and kind of a disproof in a lot of ways in people's minds that the Earth, no, it's not in motion. The Earth has to be fixed, as the Bible said. Now, the Sun also moves. So the Earth rotates on its axis. It revolves around the Sun once per year, but also, the Sun is moving. As I said, the Sun's rotation of its surface is about 28 days. But the Sun is moving around the galaxy's center about 220 kilometers per second. Again, we don't notice that because we're attached to the Earth. The Earth is orbiting around the Sun. And that velocity, if you convert it to miles per hour, is enormous.
- 4.74 But we don't sense it, we don't mention, we don't notice it. And it will take some several billion years for the Earth and the Sun system to make a complete orbit. A year, a galactic year, will take some billion years or so. That means in the history of the universe, we've only experienced 13 years or so of galactic time. But it's not really anything particularly important. So gravity makes the Earth orbit around the Sun. It makes the Moon orbit around the Earth. And what else does it do? It also is responsible for the behavior of other galaxies.
- 4.75 And again, we use the Doppler shift to determine velocity. We can't take a radar gun and measure the velocity of the Andromeda galaxy. It's too far away. We start off by measuring its position. For local things, we can measure parallax using geometry, the blinking eye technique. And then we can even do parallax to measure velocities locally, as well as Doppler shifts. But then, once you get beyond our galaxy, you really only can use the luminosity method to determine distance. And you can only use the Doppler shift to determine velocity.
- 4.76 So again, remind you, Doppler has this great idea that things moving towards you will be shifted in frequency, of a constant frequency will be shifted towards the blue. That means if I have a laser and I point it and I move at a good fraction of the speed of light, it will become, go from red to blue in the direction I'm going towards, and it will get more and more red in the direction I'm moving away from.
- 4.77 You can use this to measure the orbital dynamics, say, of exoplanets, as depicted here, but also of stars and eventually of galaxies. Talked about redshift, the physical compression or elongation of wavelength, as an object's source is in motion relative to an observer. We can measure the absorption features and physically calculate on a photograph or a CCD how many millimeters or so and how many wavelengths equivalent is the light source shifting and dividing by the frequency and wavelength it is at rest in our laboratory, where we're not moving with any relative motion.
- 4.78 We calculate the shift, red or blue. That is physically known as delta. The delta is the difference between the observed wavelength and the emitted wavelength, which was emitted in the same exact frequency or wavelength as it would be in a stationary lab. That turns out to be, for low velocities, exactly the ratio of the source's velocity relative to the speed of light. Now, even the Sun, moving at 220 km per second, that's a tiny fraction of the speed of light, which is 300,000 kilometers per second.

- 4.79 So this law holds true for sources that are moving much faster than this. We have to take into account Einstein's theory of special relativity, which will be part of the subject matter that we'll cover in my course next time, which is on black holes, general relativity, wormholes, and time warps. But that's a later course. Of course, an object can have motion that's not radial, towards or away from the observer. There's no reason to assume that all stars or all planets are moving exactly along a radius separating us and them.
- 4.80 In fact, many of them are moving perpendicular to that direction. That's called transverse velocity. And the vector sum of the transverse velocity plus the radial velocity is the total velocity. So the sum of those two things is equal to their actual velocity through space, not relative to us, which is an abstract and fundamentally and fundamentally coordinate dependent. Depends on where we are, how we're looking, and how we're moving as well.
- 4.81 So the proper motion is the total motion through space of this object. So here's an object that has a radial velocity in red. It's moving away from us. That causes a red shift, but it's moving perpendicular. It doesn't cause a blue shift, but it's indicated in blue here. The proper motion in astronomy jargon is denoted by the Greek letter mu, and it can be measured for certain objects. And there are certain objects nearby to us that we can measure quite accurately. Because we can measure the radial distance via parallax, and we can actually see, is it moving also tangentially as it moves radially.
- 4.82 There are stars in our galaxy which, on average, the total value of the velocity is zero. In other words, there's as many stars moving towards us as away from us in our galaxy. Just like there's as many air molecules moving towards you as away from you. It's sort of random. Now, that is not true for galaxies. There's 99.9999999% of galaxies are moving away from us. And that's an astounding feature that was only known about 100 years ago, discovered only 100 years ago. We'll talk about that when we talk about cosmology.
- 4.83 So the farthest spacecraft, Voyager 1 and Voyager 2, are moving at about 17 kilometers per second. And we can actually measure them because they're sending radio waves to us. We know exactly the radio frequency that they're emitting at, because it was built using certain radio transmission technology. And we know exactly what that was. And there are some stars that are moving slightly faster than that, but not much more.
- 4.84 There's a star that's only about 6 light years away, or 1.8 parsecs. It's called Barnard's Star, named after the astronomer who discovered it, and it's moving about one degree, which is about twice the diameter of your pinky fingernail held at the length of your arm. It's about twice the distance. It's about twice the angular diameter of the full Moon. And it's moving that fast every 350 years. And it's also moving in some combination of radial velocity and transverse velocity. That's the fastest star that's moving in this way, at some hundreds of kilometers per second, but again, just a random star.

- 4.85 It happens to do that. Now, why is it doing that? There's speculation of what could be causing it to move so fast compared to something else. But usually these things are formed, and the gravitational environment causes them to have the peculiar properties that they have. But it's not necessarily something fundamental about the universe. There's got to be somebody, as I always say, somebody graduates last from Harvard, right? So there's going to be some star that has the highest proper motion. It happens to be one that we can measure accurately because it's relatively close to us.
- 4.86 But now we think about the Milky Way galaxy, which will be the subject of the next lecture. How stars agglomerate into structures of hundreds of billions of stars called galaxies. We'll discuss what the structure of the galaxy looks like. We've covered the structure of the solar system, the planets, what defines a planet. We've discovered the structure of stars, how they form, how they die. The next subject that we'll turn to soon will be the structure of galaxies, and beyond the galaxies, we'll come to the structure of the universe.
- 4.87 And these structures can be traced primarily through the emission or absorption of light. Again, we can't go out and measure with some tape measure. So we have to use resourceful tools that have developed by astronomers over the centuries. And we'll do that to discover astonishing features of our galaxy. When we look at the stars' motion, of our Sun and other stars in our galactic neighborhood, we see something that we would expect. They also move in elliptical orbits. That's a consequence of Newton's law of gravity.
- 4.88 But we see other things as well that can't be described, that can't be understood by the mere presence of stars and the mass of the stars as we know them and estimate them to be. There's something else at play. So if there was some way to determine the center of the galaxy, some measuring tape, or some way to measure to the distance to the center of the galaxy, we could measure where we are relative to the center of the galaxy, and ask the question, "Are we the center of the galaxy?"
- 4.89 We know we're not the center of the solar system. Maybe we can assuage our cosmic ego and say, "Oh, we're the center of the galaxy." If not, maybe we're the center of the universe. No, we'll see we're not. Maybe we're the center of the multiverse. That's a question we address in my cosmology class in Peterson Academy. Our anatomy lesson for the day concludes by looking at these unique and very, very important structures called globular clusters, globular meaning globe-shaped, ball-shaped, spherical shape.
- 4.90 These are objects of about a million stars. So something less than 10,000 times lower mass than our galaxy. Still huge. Imagine, it's like a mini galaxy with a million stars in it. Still, these are tiny compared to the galaxy itself. These form in a way that, I like to imagine, are like bees flying around a hive. And they're trying to get to the hive, and they're attracted to the hive. They are distributed spherically, symmetrically, around the center of the Milky Way galaxy. That wasn't always known.

- 4.91 It wasn't always known if the galaxy was the entirety of the universe until the discovery of the properties, the structure, and the shape of those globular clusters was understood. And next time we'll find out. 100 years ago this year, the debates that were raging about the structure of the galaxy, trying to address this fundamental question. Is the Milky Way galaxy all there is? Is it the entirety of the universe? And if not, what do you make of other objects that seem to look like the Milky Way? Seem to be galaxies themselves. And their behaviors, their position, their great distance.
- 4.92 And most importantly, their uniformly, seemingly uniform motion, moving away from us. That will lead to the astonishing conclusion that we live in a dynamic universe that had a beginning. And that is a subject that we'll turn to in the upcoming lectures. Thank you.

Lecture 5: Galaxies and Gravity

At a glance — Galaxies and gravity.

- 5.1 Welcome back. We've done a tremendous amount of exploration already in this course today. We've covered our own planet and its size. It's our nearest neighbor, the Earth's Moon. We've even gone out towards the Sun and the outer planets and gone beyond them to the farthest reaches that human technology has explored in the outer parts of the solar system.
- 5.2 But to be honest, we've only scratched the surface. In fact, even our galaxy alone, once we consider it in the later part of today, only comprises about the same as a bathtub out of the entire Pacific Ocean. We have much to explore, and the tour will be even more fascinating as we encounter mysterious types of matter and energy that don't exist anywhere on Earth. We talked last time about how Galileo Galilei, the first scientist, really, to use technology and tools of modern science to advance new hypotheses about the universe, how he observed the Milky Way Galaxy through his small perspective tube, which he called the telescope.
- 5.3 And that device allowed him to see stars you couldn't see with the naked eye. The naked eye stars that you could see are shown here as open stars, and the ones that you could only see with the telescope are shown as filled ones. The modern view of a galaxy is depicted here. A galaxy is a vast conglomeration of almost uncountable stars. In fact, it is impossible to count the number of stars in the Milky Way galaxy, for example. Then when you consider there's probably an equal number of other galaxies, your mind really, truly gets blown.
- 5.4 We'll discover very soon how we come to know that there are even other galaxies in just a moment. But take note of the structure of what we now perceive as the Milky Way galaxy. We see a luminous band. In fact, the word galaxy comes from the Greek galaktos, which means Milky Way. Actually, we call the Milky Way galaxy, is actually the Milky Way, Milky Way. It's redundant. But going out from where we are, located about two-thirds of the way out in the outskirts of the galaxy in this disk of stars, reveals a very different neighborhood of our solar region compared to the various other parts of our galaxy and other galaxies, as we'll describe.
- 5.5 Now, if you could somehow take a giant selfie, the biggest selfie in history, perhaps, which is impossible because you'd have to get some million light years away to take such an image, this is what our galaxy would look like from above. There's no up and down in space, but just for convenience, I'll use that terminology. The Sun is located about half to two-thirds of the way out on the disk. We're in the disk. So when we look along the disk and perpendicular to the disk, we see very different types of behaviors of the stars, different concentrations of the amount of stars that we see.

- 5.6 So, fundamentally, a galaxy is now known to be a gravitationally bound collection of stars, planets, gas, and dust. Those are really the main ingredients. And the many different types of elements on the periodic table of the elements are found within these different compositions that are described here. There'll be other types of matter as well, completely unfamiliar to most of us, unless we happen to work with neutrinos on a daily basis. That will be called dark matter. There's not very much dark energy on the scale of our galaxy, but it plays a tremendous role in cosmology, which is the subject of my other course so far here at Peterson Academy.
- 5.7 The Sun is held in its orbit around the central bulging mass of the galaxy by the same gravitational force interplaying with centrifugal force that keeps planets in orbit around the Sun. It's exactly the same. Except instead of the matter being provided primarily by the luminous objects, the stars that we see, in the case of a planet orbiting a Sun, we actually know that the Sun's mass and the total amount of stellar mass that we see is incapable of providing the types of dynamical behavior of orbits that are observed for our Sun around the galactic center.
- 5.8 Now, how did we come to know this? This has been, essentially, a battle of philosophical prejudice that led us to, again, think that we were the center of the universe, which was considered to be the Milky Way galaxy. The discoveries we're talking about in this lecture are less than 100 years old. Some are very recent. Gravity is one of the four fundamental forces of nature or physics, the other three being electromagnetism, and there's two types of nuclear forces.
- 5.9 We encountered the strong nuclear force, when we talked about stellar fusion in the proton-proton chain that makes up the helium from basic protons in the Sun's center. We discussed weak nuclear processes, the weak nuclear force, in the context of radioactive decay that we use to assess the age of the Earth, which in turn can tell us about the age of the solar system and the Sun itself, because they formed at this exact same time.
- 5.10 These types of electromagnetic forces are very familiar to us. They're actually responsible for electromagnetic radiation, all the forms of light and radiation that we've talked about so far, from radio to optical and beyond. These are very familiar to you, as is gravity. Gravity is also very familiar. The lesser known forces are actually more important only on the scale of the subatomic level, either for nuclear fusion, nuclear fission, or radioactive decay.
- 5.11 We'll encounter this weak nuclear force in the context of dark matter, because it turns out that, while we don't really know what the dark matter is per se, we know how much of it there is. Because of the existence of neutrinos, we know at least one type of particle that behaves like dark matter. I'll explain what that means in just a bit. Remember, the strong nuclear force has to be able to overcome the electrostatic repulsion that like charges naturally have. Charges that are the same repel each other.

Putting two protons together requires some other force that's much stronger than the force of electric repulsion.

- 5.12** That force is the weak nuclear force, and it operates at the level that is beneath the level of the proton itself, at the so-called quark level. It only interacts over very short distances, but it is the strongest possible force, many, many orders of magnitude stronger than the electric force, which is many, many times stronger than the gravitational force. So gravity is the weakest force. And in physics, the strength of a force is inversely proportional to the distance over which it acts.
- 5.13** So gravity can act over the scale of our solar system, whereas you can't put an electric charge on Neptune and another electric charge on the Earth and have any kind of interaction between them. But actually, the effects of, say, Jupiter do affect the Earth, as we'll see in later lectures. So there are systems dominated by gravity, like the formation of the planets; and the gravitational force that eventually causes stars to collapse and create supernova are actually driven at first by gravity and, eventually, then by nuclear fusion.
- 5.14** A big core constituent of our galaxy is the enormous, giant black hole at the center of the Milky Way. It's not what's causing us to orbit around the Milky Way, per se. It plays a tiny role in that behavior. But these objects, black holes, wormholes, time warps, and so forth, they're enigmatic. They're not just in science fiction movies. They're also extremely important for understanding of—they're also extremely important for understanding of gravity, and eventually, whether or not we can have a quantum mechanical theory of gravity, which has eluded us so far. But stay tuned for my upcoming course on black holes and wormholes and time warps for more about that.
- 5.15** The core principle of gravity is that all masses fall at the same rate, that there is no way that heavier objects will fall faster than lighter ones. This was a realization that Galileo had, that overthrew the received wisdom he got from good old Aristotle. But again, Aristotle made great breakthrough, like the fact that whales are mammals and women have fewer teeth. Now, I always joke about that because he was brilliant. He did construct the laws of logic and many mathematical principles, but he didn't do these simple tests that would later be done by Newton and Galileo.
- 5.16** So Isaac Newton played a huge role in determining and discovering three laws of motion that bear his name. The first law of motion was exactly in contrast to what Aristotle believed. Aristotle thought that the nature of matter was to be at rest, and things had to be moved by some motion, some mode of force, to get them to move. In actuality, it's the opposite. Things stay in motion unless they're acted upon by a force. A force then leads to Isaac Newton's second law of motion, which is that to change the speed or velocity of an object, you must apply a force.
- 5.17** Because a change in velocity, by definition, it requires an acceleration. Velocity is a speed plus a direction. The wind is blowing three meters per second in that direction. A

speed could just be so many miles per hour, and a direction could be pointing north, south, east, or west, for example. But when you change either the direction or the speed, that requires acceleration, and that requires a force. The force relationship to acceleration is shown here, F equals MA .

- 5.18** Lastly, Isaac Newton's third law of motion is the principle that, as I sometimes said, for every action, there's an equal and opposite reaction. But really, it means that for every force that's applied, there's a counteracting force in order for those forces to be balanced. For example, when you push on the ice, the Earth is being pushed back a tiny bit, and in reality, you're pushing forward, and those forces are balancing such that you will then have an acceleration. That action is effectively a new acceleration.
- 5.19** Later on, Einstein would come to revolutionize the concept of gravity. Where Newton left off, he didn't explain why forces act in the way that they do, and what the true nature of gravity was, and how fast gravity could interact with particles of matter. That was left to Einstein. Newton assumed that space was static. It wasn't changing with time. Flat, meaning that all triangles within it would add up their interior angles to 180 degrees. We talk a lot about this in the context of my cosmology class, for those of you that are interested. We'll talk about it even more in the context of the gravitational mind-warping and space-warping effects of black holes in a later class.
- 5.20** Remember that Kepler described the motion of planets as behaving in elliptical orbits, but he couldn't prove that that was so mathematically. He got the right relationship between the functions of time and space that planets exhibit, but he did not know the reason why, mathematically. But Newton did provide the rationale behind the equations, which is extremely important. He did so by postulating the so-called law of universal gravitation. Universal meaning, this law holds throughout the universe.
- 5.21** It actually doesn't hold on very, very small scales or very, very large masses. We have to modify it, and that's what Einstein did, but that's, again, for cosmology and for my black holes course. The mathematical relationship is one of an inverse square behavior that we encountered when we talked about luminosity and measuring luminosity distances. The strength of gravity is like the strength or intensity of flux. The amount of gravitational force flux, if you will, decreases as the inverse square of the distance from the two massive objects.
- 5.22** If one massive object, the bigger one, capital M , sub G , has a mass, and it's acting on and equally being acted upon by a lower-case mass, lower-case m , they will interact via this behavior. Then, if you take Newton's second law, F equals MA , equate the acceleration times the mass of the smaller object in the gravitational force field of the larger object, F capital G , you'll see that the lower-case M s cancel off. Well, they will cancel off. You get that a mass of any composition or any amount of value, from a cinder block to a pebble, when dropped, will always fall at the same rate.

- 5.23** Allegedly, Galileo did this experiment at the Leaning Tower of Pisa, about 400 years ago, and observed this fact to be true. And then later on, about, I believe it was in 1970, astronauts went to the Moon and did this experiment because there is a consideration and a complication when you do it on Earth, that we have an atmosphere. So if you take a piece of paper with a large surface area and drop a pebble, they won't fall at the same rate in an atmosphere.
- 5.24** When the astronauts landed on the Moon, they did this experiment. You can see it on YouTube. And they dropped the feather and a hammer, and they both fell at the same rate, hit the ground at the exact same time, or hit the lunar surface at the same time. Now, remember, Kepler postulated these laws of planetary motion, but he could not prove them. He could compare to data that he observed, and Tycho Brahe observed. He could calculate and describe the orbits, but he couldn't derive why they had this behavior.
- 5.25** But it took the laws of Isaac Newton from the 16-1700s to actually prove using the law of universal gravitation. For an object like a planet with the mass of the Sun, knowing the behavior of these objects, he could then prove the three laws of motion. The first law of motion is that planets move in elliptical orbits. The Sun is at one focus, and another focus is essentially empty. Remember, a circle, a perfect circle, is an example of an ellipse with both axes equal, the major and minor axes.
- 5.26** Newton's law of universal gravitation applies everywhere in the universe, at least on large enough scales. For example, to get to the Moon, we didn't need Einstein's laws of general relativity. We only needed the laws of Isaac Newton, and those were discovered and discussed hundreds of years before we actually needed them to go to places like the Moon or even the outer solar system. All the satellites that have visited all the distant objects in the universe, from Voyager 2, which is now one light day away from Earth, an incredible distance, they all got to where they were purely by the gravitational effects, purely by the physical effects of Newton's laws.
- 5.27** You can use Newton's laws to measure things about other objects. For example, if you know the force, if you can somehow measure the force, or if you can measure the radius, or if you can measure one of the two masses, you can then invert this equation and use it to make a scale where you measure the mass of something else, for example, or if you know two of the masses, you measure the equivalent radius, the distance of a planet from the star. It becomes incredibly powerful.
- 5.28** When you want something to achieve an orbit, like a SpaceX satellite, something like that, you need to launch it into an orbit where its velocity is equally matched to the velocity at which the acceleration due to the Earth's gravitational field is causing it to fall. If you just put a satellite up in space with no velocity, it'll drop it down, it'll eventually crash into the Earth. But if you want an object to orbit around the Earth at a constant speed, say in low Earth orbit, you need to give it a tremendous velocity.

18,000 miles per hour will cause it to orbit in low Earth orbit, about 200 miles above the Earth's surface.

- 5.29** You can get a little bit of a boost into orbit by launching satellites near the equator. That's why they're launched primarily from Florida, at least in the US, and California, and Texas now. That's much closer to the equator. It's much harder, there's less equivalent helpful motion in the direction of orbit from, say, close to the North Pole. It would require much more energy to do that. That's why rockets are launched near to the equator than farther from it.
- 5.30** Now, if we equate the gravitational force to the centrifugal force, we can then solve for, as I said just a moment ago, we can solve for the mass of the object. We can weigh the Sun effectively by measuring, say, the radius of the Earth's orbit, which is the astronomical unit. We got that from a variety of different ways in previous lectures, when we talked about parallax and radar distance measurements. We can measure its velocity as well for a variety of ways that go back to the 1700s, as I said. Around the Sun, we can measure that orbit. Then capital G is a constant that can be measured in a laboratory here or can be measured in a variety of circumstances.
- 5.31** Doing these things together gives us the mass of the Sun. We don't have to go there, put on a scale, count how many protons it has. Now we know its mass and we know its luminosity. Several of you wondered yesterday, how can you assume that all these objects that look the same have the same properties? Well, if you measure their masses and you measure their luminosities, you can pretty much assume, measure their radii, you can measure their densities, and you can say exactly what they're made up of. They should have the same properties in our galaxy as in any other galaxy.
- 5.32** We're not talking primarily about orbits of satellites and even planets. Again, we want to discover what is the shape, structure, and composition of our galaxy, which we now know is not the whole universe. But to do that, we had to prove, how big is our galaxy? How large in extent, width, height, etc., is it? And also, how many stars are within it? Then we could start asking questions of the other objects we see in the universe through our telescope. How far away are they? How big are they? What are they made up of? And what are the dynamics of them?
- 5.33** We're not so much concerned with orbits of satellites, as cool as that is. At this point, we're considering the orbit of our Sun around the Milky Way, as if it's a gravitating object, as if it's a solar system, if you will. In fact, it has many of these properties of a solar system. It has the structure where there's a lot of mass, maybe not all the mass of the galaxy, as it is in our solar system. Remember, our solar system, 99.8% of the mass of our solar system is in the Sun. In other words, the only object that's producing tremendous amounts of light and heat is also responsible for the preponderance of the mass of our solar system.

- 5.34 That's not true in our galaxy, not by a long shot, as we'll see. But to do that, to prove that, we want to be able to measure the mass of our galaxy, including the effective mass of our Sun, which is the galactic center. Then we'll show that that mass is far too insignificant to account for the velocity of the Earth around it. The conclusion will be, there must be some mass that we're missing. Obviously, that mass is dark matter. But that was only realized in the last few decades.
- 5.35 We're not really concerned with orbits and getting to planets as we are of the orbiting mass—as we are of the central mass that's causing the orbital dynamics to behave in the way that they do. We can use Newton's equations as a type of scale to weigh the objects that's producing mass. Now, the key insight is that it's only the mass within your radius that causes your orbital velocity to be what it is. There could be a lot of mass outside of where you are, but it's really the mass inboard or inward of where your location is that determines your orbital behavior.
- 5.36 There are effects from other objects. Jupiter does affect the Earth's orbit, and the Moon is affected not only by the Earth but also by the Sun. We actually have tides in the ocean. Partially, the tides are caused in a small fraction by the Sun, not just the Moon. We're going to do the analogous thing. We're going to ask the question, are there things that are orbiting around the Milky Way, like a planet orbits around a star, that are farther away from the Milky Way's center than we are, so they will be attracted by all the mass of our galaxy, effectively all the mass of our galaxy? To do that, we're going to then discover the size and shape and dynamical behavior of the Sun and every other Sun in our galaxy.
- 5.37 Once again, keep driving this home. It will force upon us the startling realization that the matter, the stuff that we're made of, protons, neutrons, croutons, my favorite part—no, there are no crouton particles—but the mass that we're made up of is not the dominant form of matter in the universe. It's an astounding realization that physicists and astronomers have only had for the past few decades.
- 5.38 In a previous lecture, I talked about these stellar cities that have about a million inhabitants. They're called globular clusters because they're globe-shaped, ball-shaped. They have many, many, perhaps 2-3 million stars within them. Some of them are very old, very old structures. They're nothing like anything we have encountered in any of the stars we've seen before. When you see them through a telescope, they're quite beautiful. But they're also quite useful because they are objects that orbit well beyond the orbit of the Earth's and the Sun's position in our galaxy.
- 5.39 They're like bees swarming around a hive. The hive is not filled with honey. It's filled mostly with dark matter, as we'll see. These objects change our view of the entire cosmos. Exactly 100 years ago, when a great debate broke out, rivaling any presidential debate you've ever seen, between two astronomers, one who thought the Milky Way galaxy was the whole universe and one who thought it was just an island amidst an

archipelago of many, many millions, billions, hundreds of billions of galaxies, as we'll see.

- 5.40** The structure of the Milky Way intrigued astronomers for generations. Was it the whole universe? Was it just a fragment the whole universe? To be honest, we really didn't have enough data to understand it in between the time that Galileo first used the telescope and mistakenly thought that all the glowing material he could see was actually the effect of innumerable stars crowded together. When in fact, we know there's a whole bunch of other stuff in the cosmos—gas, dust, nebulae, rocks, planets, asteroids, comets, minor bodies, dwarf planets, all sorts of things that make up the glow.
- 5.41** But the most interesting things are the stars. The question is, is the Milky Way all there is, or are there worlds, universes beyond what we could see? The word galaxy comes from the Greek, *galaktikos kuklos*—my Greek is always rusty—or milky circles. Again, we're saying the Milky Way galaxy, it's really saying Milky Way, Milky Way, or Milky Way, Milky Circle. A good old friend, our friend Aristotle, this interesting character, thought that the Milky Way was actually in the Earth's atmosphere. It was a really gross underestimate.
- 5.42** Later, William Herschel and his sister, Caroline Herschel—remember, the first woman scientist, professional scientist in history, was an astronomer. The first scientist, male scientist, human scientist, was an astronomer, Galileo. The first woman, I mean, it's incredible how much astronomy has given in terms of our cultural understanding of how we relate to the cosmos, because it's for everybody. As I said, it's your universe, and we want to understand it because we're going to live here, hopefully for a long time.
- 5.43** William and Caroline explored the Milky Way galaxy, making the very first crude map, and this is it. He put us at the center of the galaxy, which is this incredible sentiment of the human condition, that we have this cosmic ego. Yes, we know by 1785 and even by the late 1500s, and certainly by the 1700s, when Bradley proved that we're moving around the Sun and not the other way around, we knew we weren't the center of the solar system. Maybe we could salvage our cosmic ego if the Sun were the center of the galaxy.
- 5.44** The reason why this appears to be the case is actually an illusion caused by dust. You can get a sense of this. If you've ever been on a plane and you've sat inside the plane looking out the window and you're inside of a cloud, the cloud could be shaped like a pancake. It could be like a cylinder, you don't know, the perfect sphere. Physicists love to assume everything is a sphere. But you can't tell because you're only seeing a very small, apparently spherically centered shell of light that surrounds you. It's like being in this room, but it looks like a sphere around you. That's because you're only seeing the last time the photons scattered from a water droplet to get to your eye.

- 5.45 All these clouds, pilots talk about being inside of a ping-pong ball. That's the way it feels when you're in the cockpit. Everything just looks like it's surrounding you like a glowing orb. Well, the Milky Way is not filled with water vapor, but it's filled with these tiny grains of dust and rock. We'll get to that when you discuss meteorites in the final lecture. You'll actually get samples of these types of dust that fill up the Milky Way galaxy and every other galaxy that we know about.
- 5.46 When you're inside a very strange and irregularly shaped galaxy like the Milky Way, it doesn't matter. You'll perceive yourself as being inside of this ping-pong bowl, not made of water vapor droplets but of dust. It was caused by this illusion that he came to make the map like this. Here's another version of it. The darkest dot in the center, so to speak, the center, is where he thought the Sun was. It took 10 years for William and Caroline to make this map. It was a huge step forward in humanity's understanding. But again, falling into the same trap that we are somehow central to the universe's workings.
- 5.47 For many years after this, people still thought the galaxy was the entirety of the universe. Again, we went from, it's something in our atmosphere, according to Aristotle, to now it's this singular object that fills up the whole universe, and everything in the universe is what we see as the Milky Way. I like to think of the Milky Way as the superstar. And obviously, that's also human-centric, anthropomorphic in some sense. But it has these outliers. It has this gang of paparazzi and onlookers and hangers-on and a retinue of followers in an entourage. And some of those are actual, honest-to-goodness other galaxies.
- 5.48 Small galaxies called satellite galaxies, but they'll play a hugely important role, as we'll see in just a moment. But the other thing it has, and those satellite galaxies, I should say, are distributed anisotropically; they're not uniformly distributed around the Milky Way. You could not mistake them for being any way centered on the Milky Way itself. Even though they're gravitationally bound to the Milky Way, they're not centered on it. But globular clusters are centered on the Milky Way. They're in a spherical halo that surrounds the center of the Milky Way but extends beyond the visible disk by a good amount.
- 5.49 In 1920, amidst a giant presidential debate season, 1920 was a presidential election year here in the United States, there was a debate held by two competing forces. There were two astronomers, Harlow Shapley and Heber Curtis, two names you don't really hear so much anymore, Harlow and Heber. They debated whether or not the nebula that could be seen through a telescope, or even in this case, with the naked eye, were in our galaxy or external to it. They look similar. On the left is the Great Nebula in Orion, and on the right was the Great Spiral Nebula in Andromeda.
- 5.50 Are they distant? Are they near? Are they within our galaxy? Are they far removed from it? Shapley thought that the Milky Way was the whole universe, and Curtis believed that the spiral nebula were objects very different from the non-spiral nebula, but also

galaxies unto themselves. And this debate was held in Washington, DC, of all places, at the National Academies of Sciences. And they used as their tools, or one of the prime reasons for the argument that was made by Curtis, was to look at these objects called globular clusters. Balls of stars, globes of stars, about a million or so in total.

5.51 These globular clusters buzz and whirl about the center of the galaxy, far removed from the disk. So they're not part of the disk. There's none of them inside of the disk of the Milky Way. So they can't be confused as objects in the plane of the galaxy that our solar system finds itself in. And to take a step back, there are other types of clusters that aren't as dense and compact. This is the Pleiades, which has maybe 1,000 or 2,000 members in it. Galileo sketched it with the open stars and the closed stars. These are very, very different than the globular clusters. You don't mistake them. Any astronomer can tell the difference between these.

5.52 And then as we go farther out, from being able to use the local distance, either radar measurements, we can use parallax techniques, measure distant objects. Then we can use objects called standard candles. The standard candles have a range of luminosities. These are the scientific tools. This is the experimental setup that would play an incredible role in our understanding of the shape of the Milky Way galaxy, but also what it's made up of. Remember, we're trying to find the mass of the galactic center, analogous to trying to weigh the Sun using the Newtonian law of universal gravitation, combined with the orbital properties of centrifugal motion.

5.53 Those then lead you to invert the equation, to measure mass of the central gravitating source. That's our goal. To do that, we have to make measurements of the radius of different objects. In the next lecture, we'll discuss how the globular clusters' distances were measured and then find that they're being attracted by a sinister, much darker force, a darker mass, if you will, than we could ever have conceived of 100 years ago today. Thank you.

Lecture 6: Distance and Dark Matter

At a glance — Distance and dark matter.

- 6.1 Welcome back. If you remember back in Lecture 2, we talked about the importance of measuring distance based on a variety of different tools. We eventually discussed the cosmic distance ladder, starting with local measurements of the Earth's surface between two cities in Greece and Egypt, and then eventually used that to leapfrog our way up to the Moon and then to the other planets and the Sun.
- 6.2 Now, if we know that the Sun is the center focus of the elliptical orbits of all the planets that go around it, then you could find the distance to the Sun from any planet by measuring all the planets and looking at their common point of, their common focal point. Now, for a galaxy, you could do that, too, if only there were objects, like the planets, that you knew for sure went around the center of the galaxy. Turns out there are, and these objects are called globular clusters. These contain about a million stars in a very compact region, just a few dozen or so light years across.
- 6.3 These objects can then be used as these beacons that trace the central location or give away the central location, but not only that. Not only do they tell us the distance to the center, they tell us a good estimate of what the mass of the central gravitating object is. That's a huge accomplishment if it can be done. The globular clusters are these beacons that are like these bees buzzing around the dark matter, honeycomb-filled hive, that we call the galactic center. The galactic center also has a black hole in it, but that doesn't cause—even though it has a mass of about 6 million times the mass of our Sun, it's nowhere near enough to cause the Sun to orbit around it, as we'll see.
- 6.4 So locally, back to the cosmic distance ladder, we made the observation that there's as many stars moving towards us as away from us. That means, according to Doppler, there should be as many blue-shifted stars as red-shifted stars. But as you go farther and farther away, you can calculate velocities and calculate not the average velocity, but the square root of the average velocity squared, which is an unusual way of thinking about it, but it's a better characteristic for the net or average behavior of a typical globular cluster.
- 6.5 Because if you just average them together, the velocities will add to zero. There's as many globular clusters above the galactic plane as below. If you just take their velocities, which is a speed and a direction, average them together, you get zero. But if you take the square of them, then the negative signs will cancel off, and you'll get a very good estimate for the average velocity once you take the square root of the sum of these squares. That's how it's done.
- 6.6 We talked in a previous lecture about a very unusual type of ruler, of distance-measuring tool, called luminosity distance. That was shown with these candles here. The closest one is about half the distance to the camera, to the audience, as the one

that's farthest away. Therefore, it will have four times the apparent flux for a constant aperture, for a given aperture, either your eye, the camera's lens, etc. Will have four times the intensity for something that's twice as close. That's the basis of the inverse square law as applied to stars using their luminosity to measure their distances.

- 6.7** Now, it turns out, there are actually other stars that are more useful than just knowing the typical luminosity of a star of average intensity like our Sun. These are stars that vary. They vary in their intensity as a function of time, kind of like these candles do, but not quite. These candles are more or less twinkling randomly. But there are stars whose brightness varies in this not quite sinusoidal but periodic, it looks like a sawtooth pattern. It goes up to a maximum, decays linearly, then dwells at the trough for a little bit, then rises very sharply, and then comes down.
- 6.8** Well, it turns out these stars are breathing. They're expanding, they're contracting. The nuclear processes within them are causing them to be more or less luminous, depending on how compact or diffuse they are. So their brightness depends on their luminosity, and their period of oscillation between maximum and minimum are very well matched to, for example, an astronomer's schedule, because they vary on scales of a few days, perhaps a month, but not—they could have existed in ways that they take a billion years to vary, and then we'd never notice them.
- 6.9** But thank goodness, an astronomer named Henrietta Swan Leavitt— again, we don't have many people named Swan anymore. I think we should bring it back. She was one of the pioneers in astronomical history. She developed a very, very accurate tool called the period luminosity relationship. Now it's known as Leavitt's law, and that extended the ability of astronomers to measure distance. Because if you think about it, measuring intensity and either making a sketch or taking a photograph or using a CCD camera's exposure, is a lot more difficult than just counting.
- 6.10** How many times per month did the star go from bright to dark or dim? It doesn't go all the way off, but it gets brighter and it gets dimmer. That Cepheid period luminosity relationship is much, much easier for an astronomer to then take an observation of the number of cycles per month, for example, and look up on this law, on this plot that she made over 100 years ago, and use that and look up. Well, that star has a luminosity of so many watts. And now we measure so many watts per square meter in our telescope, we know exactly how far away we are. It was a revolution for astronomers.
- 6.11** It wasn't appreciated much at first. And then after she died, it became crucial to our understanding of the expansion of the universe that we're going to encounter in the next lecture. I told you about the luminosity distance relationship. But just to remind you, it's predicated on the fact that a telescope like this one, or your eye, or the camera's lens, has a given aperture, some diameter. That aperture lets in so many square meters; it lets in so much light. The wider it is, the larger the area, the more photons will get in, and the more sensitive the detector will be.

- 6.12** Also, the better the resolution and separation between the closest stars that can be resolved together can be obtained. Bigger is always better, which is why, as a 12-year-old, getting my first telescope, slightly smaller than this, I immediately wanted a telescope like this and kept trying to implore my parents to get me one, and they told me to get a job. I've never had a real job, I've always been an astronomer, basically since that day. I did work many jobs as a kid to afford my addiction to aperture size.
- 6.13** Once you measure the flux, you know the area, you can just measure it. You know then, if you can know the luminosity, you have an exact measurement of the distance. Now, obviously, the period luminosity distance relationship is not perfect, but there's the job of the scientist to categorize and classify not his or her brilliance and knowledge, but his or her ignorance. That's what separates a good scientist from a charlatan who's trying to make some point for some reason, I don't know what. But in that case, we know the luminosity relationship to within some precision and accuracy.
- 6.14** Then the more of these objects we can measure, not only in our galaxy, not only in the satellite galaxies I mentioned earlier, that'll be very important in just a moment, but in every other galaxy we can see across the observable universe. This is the longest ruler that we know about, effectively. We can use it to measure distances out to great, great lengths. Here's an object. This is a Cepheid variable measured by the Hubble space telescope, wide-field planetary camera number 2. That's what all those acronyms stand for.
- 6.15** You can see it's going, between May 4th and May 31st, it's undergoing a complete oscillation period. So bright to minimum, back to bright. So one full period. That's just one of many. What are these objects? Well, it wasn't even known, and it wasn't even that important to know them 100 years ago, what they were made of. Actually, we don't have a good understanding, according to the leading astronomer, one of the leading astronomers of our time, Wendy Freedman at the University of Chicago, who was responsible for using the Hubble space telescope for the purpose it was designed, which is to measure the Hubble constant, which we'll talk about next lecture.
- 6.16** She ran that project. She's one of the most eminent scientists. I've interviewed her several times on my YouTube channel. She claims that we don't actually understand, even to this day, that intrinsic nuclear physics processes that are causing the star to pulsate the way it does. But that doesn't really matter. I mean, you might not understand how this five-watt light bulb is made, but if I tell you it's five watts, then you can do all sorts of things with it. Sometimes we don't have to know to find something incredibly powerful as a tool to do the research that we want to do.
- 6.17** They do vary. Some of them vary by a factor of 100 in luminosity, from the brightest to the dimmest. They're given in terms of the Sun's luminosity here. That's what that L with that circle and a dot means. That circle and a dot in astronomy represents the Sun. In terms of solar luminosity, all of them are many hundreds of times brighter than our Sun. They're supergiants, and our Sun is what's called a subdwarf or a dwarf star.

There's a huge variation between these objects. Their periods range from a day and a half to two months.

- 6.18** But again, if you're a graduate student working in my research group, you're much more glad that these don't take seven or eight years to undergo one oscillation because that would be longer than your PhD, or hopefully it is. They can be used in a way that's kind of fascinating, at least it is to me, that you don't actually have to know their intrinsic luminosity, if you give up some of your desire to measure distance absolutely. In other words, you could measure them relatively to one another.
- 6.19** If I don't tell you what their actual luminosity of each one of these candles is, but you know that they're all the same, you can still use them to measure the relative distance between this one, that one, and that one. Then, if some other tool, say the parallax method, where you blink your eyes back and forth and you measure the distance between your eyes, that could then give you the distance to the closest one. Then you can measure absolute and relative distances.
- 6.20** What Henrietta Swan Leavitt did that was so phenomenally important was to recognize that there were these dwarf satellite galaxies that were very close to the Milky Way's edge. Whether they were in it or slightly beyond it was actually not that important because what they would then be used to do was to measure every other object in terms of these objects that are just at the boundary between the Milky Way and its environments outside of it.
- 6.21** It turns out that she found many of these cepheids, a dozen or two dozen of them, in one of these two satellite galaxies. They're called the Magellanic Clouds. Can only see them from the southern hemisphere. It looks like a piece of the Milky Way with incredible brightness and density of stars broke off and rebelled against the Milky Way and is leaving it. It looks like a bright band. Again, Milky Way, nebula, cloud, they all have the same connotation. They're quite beautiful, but you can only see them from the southern hemisphere.
- 6.22** The smaller one, called the Small Magellanic Cloud, is about half or a quarter the size of the larger one. But she discovered more of them in the Small Magellanic Cloud than the large one for reasons that aren't that important. But she realized that every single one she saw was closer to the ones around it than it was to us here in the Milky Way galaxy. In other words, this is a population of stars that were very close in distance to one another. Exactly how far away they were was unknown as yet, but they were at least much closer to one another than they were to us. And that was important.
- 6.23** Because that meant if she could see two cepheids that had the same luminosity, they could be used to get a double check on the distance to the Magellanic Cloud. And then once she got measurements to the Magellanic Cloud, she could then measure any other galaxy that happened to have cepheid variable pulsating stars. The hint that's really beneficial for us is that all of them do. Every single galaxy we can see, for some

reason, has many, many, sometimes hundreds or thousands of these objects. In particular, the Milky Way's biggest neighbor, Andromeda, will have some in it, too. That will play a crucial role in settling the Curtis-Shapley debate.

- 6.24** We can infer relative distances. Here now, I'm showing the Large Magellanic Cloud and the Andromeda Galaxy. Both of them, fortuitously, have cepheid variables in them, many of them. Let's say there's a class where you measure two different cepheids, one in the Large Magellanic Cloud, one in the Andromeda Galaxy, or nebula. Remember, at this point, they thought it was a spiral nebula in the Andromeda constellation in our galaxy. They have the same period, so they have the same luminosity.
- 6.25** We don't know what that luminosity is, but we know that they have the same period, therefore, they have the same luminosity unknown to us. But that's not important for measuring the relative distance ratio of Large Magellanic Cloud to Andromeda. It was in October 1923, the great astronomer, Edwin Hubble, took a photographic image of the Andromeda galaxy, the nearest neighbor, which, again, was known at that time as the Spiral Nebula in Andromeda. They didn't presume that it was another galaxy.
- 6.26** This plate, which I've held, is 101 years old right now. You can see the incredible excitement, at least for an astronomer. I could notice it immediately. You see in the upper right, these letters V-A-R are written. What's harder to see is that, above the letter V, written in Sharpie or whatever, is the letter N with two lines. The two lines are pointing to the star that was originally thought to be a nova. A nova is either a supernova, an exploding star. It's a one-off event. It brightens for some reason. It could be exploding, it could be undergoing some cataclysmic process within it. It brightens, and then it dims, and it's dimmed forever.
- 6.27** So he first observed it in early October and thought it was just a nova. He saw it changed its brightness over a couple of days. And then he came back to it at the end of October, and he realized, no, it's a cepheid variable star. It was the first one discovered that could be conclusively proven to be a cepheid variable in the Andromeda nebula. And so he was very excited. He puts an exclamation point there, crosses off the nebula. He doesn't erase it and undo history. But he discovered this cepheid variable, immediately knew, "I can take Leavitt's law, apply it to the cepheid that I just discovered," counting the days, and he kept monitoring it and monitoring the brightness, and confirmed it was actually a variable star.
- 6.28** Then he used Leavitt's law to measure the distance. This is what he found. He found the equivalent star in the Large or Small Magellanic Cloud, doesn't really matter, with the exact same period, had a much, much brighter amount of luminosity. In fact, it had 230 times the luminosity of the exact same periodicity star in the Andromeda nebula. Well, that means he measured a flux 230 times bigger, and the distance ratio when measuring luminosity is proportional to the inverse square. The flux is proportional to the inverse square of the distance. So he knew that this thing was at least the square root of 230 times farther away, or about 15 times farther away.

- 6.29** If the larger small Magellanic clouds are at the boundary of the Milky Way, and the Milky Way's distance across could be measured using parallax, using the stellar method of blinking your eyes or measuring the same star, six months apart, he knew immediately that this object could not be in the Milky Way galaxy. That meant it had to be external to the Milky Way, and it had to be a galaxy in its own right. This was the first conclusive proof of extra-galactic astronomy, of an object not in our solar system, not in our local galactic neighborhood, not in our galaxy at all, 15 times farther away than our galaxy itself.
- 6.30** What's amazing about good scientists is that, once they discover a tool, they use it to do as much science as they can until they realize they have to invent a new tool. Hubble used the tools of Leavitt and others, and then he classified all these nebulae. He started to look at them and say, "You thought this was a spiral nebula in the constellation Ursa Minor? No, it's actually a galaxy, and it can start to measure the galaxy's distance." And he did that for dozens and hundreds of galaxies. It became an industry for him.
- 6.31** And then later, building on the work of another astronomer who doesn't get as much recognition as Hubble does, and Hubble deserves every bit of recognition. But like Henrietta Swan Leavitt, there was another astronomer, also with a very cool name that you don't hear anymore, Vesto Slipher. Astronomy is full of cool names. Good old Vesto would provide information that would lead, when combined with Leavitt's law, to the inescapable conclusion that the universe is expanding, as we'll discuss in the next lecture.
- 6.32** What Hubble did is classify the different types of galaxies he saw into two broad characteristics. One characteristic of a galaxy is to have a spiral or roughly swirling-type shape, as Andromeda has. Another type of galaxy is basically a blob. It's not a globular cluster. They're way too far away. They're not in our galaxy. They're actually other galaxies. They're essentially structureless ellipsoids or spherical balls of stars. Some of them have way more mass than our Sun, and some of them have enormous amounts of energy being emitted along huge jets that radiate out from the central black hole.
- 6.33** These are powered by enormous engines, millions and tens of millions of times the mass of our Sun in the form of a black hole, creating mass onto it and shooting it in opposite polar directions out of the galactic center. We'll cover that in my course on black holes and wormholes and time warps. He made what was called the Hubble tuning fork relationship between spiral galaxies of different types. There's different flavors, different taxonomies, different phylogenies, I guess you'd call them if you're Bret and Heather Weinstein-Heying in their course here.
- 6.34** They have different evolutionary paths, just like when they teach their course on evolutionary inference here. There's an evolutionary synthesis that occurs because spiral galaxies can be combined together by the force of gravity, attracted by the dark matter within them to form clusters of galaxies, but also galactic collisions. This galaxy that we see as a spiral galaxy here, M51, the Whirlpool galaxy, is actually two galaxies,

and they're combining, perhaps they're going to crash together, perhaps they already did crash together, but eventually it becomes this Vitamix Ninja blender, and all the stars get mixed together.

- 6.35** Given enough time, they'll merge together and all the stars will have fully randomized and equilibrium values, and you'll make the spherical or ball-shaped or elliptical orbits. These objects are much older in some sense because they're the byproducts of collisions of younger or preceding. In some cases, they can be older or represent the past collisions of even older spiral galaxies that came together to merge, to create this jumbled mess. Actually, not so messy because they are distributed like a ball, but they're the result, these elliptical galaxies, ranging from zero ellipticity, E0, up to fairly elliptical, E7, with pronounced ellipsoidal shape.
- 6.36** We know now that almost every galaxy we see has a redshift, which will lead us in Lecture 7 to the conclusion that the universe is expanding. But the galaxies that are not redshifted or blueshifted, they are being attracted to us or we are being attracted to them, in particular, the Andromeda galaxy. The Milky Way galaxy and the Andromeda galaxy are set to collide and merge. Eventually, there will be an elliptical galaxy in this region. If the universe lasts that long, we'll talk about the future of the universe in the next lecture.
- 6.37** But what will happen when they collide is a subject of much debate and can be answered not by observations right now. We have to simulate and predict, based on the laws of physics and the galactic dynamics, their composition, and the bulk composition of the universe itself in terms of its matter and energy. As well, we can predict what it may look like. To get some indication of what it may look like, we can look at contemporary, right now, although these collisions occurred many millions or billions of years ago; in fact, we're just seeing them now.
- 6.38** We can look in the universe and see different collisions and after-effects of the combination of spiral and spiral galaxies or elliptical galaxies with spirals, etc. And these are called galaxy clusters. There are many of them that are extremely rich. This one shown here. The back is representing the Coma cluster, which is one of the most important clusters in cosmology because it was one of the first to really exhibit the manifest signs of dark matter. We won't talk about the way that Fritz Zwicky at Caltech in the 1930s came up with the name dark matter, and also the concept based on the observation of the Coma cluster.
- 6.39** Because, to a perhaps nonspecialist audience, although you guys are becoming more and more specialized, the evidence from the rotation properties of individual spiral galaxies is much more convincing. And that was provided by Vera Rubin, who worked for some time at my home institution, UC San Diego. In fact, she learned the tools and technology of how to make so-called rotation curves that we'll turn to next, from my late colleague, Margaret Burbidge, who I knew for many years until she passed away in 2020.

- 6.40** Vera Rubin was one of the pioneering astronomers of the 20th century. She and her colleague, Kent Ford, noticed the following behavior, that if you were to look at the orbits of stars in a galaxy and trace them as they orbit around their galactic center, and you were to assume that these stars are orbiting around the center of a galaxy are just like planets orbiting around a central star, you got very different behaviors. The stars behave very different than planets orbiting around the stars, in other words.
- 6.41** The planets orbiting around a star obey Kepler's laws. The velocity falls off as one over the square root of the distance of the planet to the star that is causing the gravitating force. But galaxies, this doesn't happen. You see the dashed curve in this image. The dashed curve shows, as a function of distance from the center of a galaxy in terms of light years, how it should obey, how the velocity should decrease with distance, as one over the square root of the distance. That's how planets in Kepler's laws behave. That, in other words, is what happens when the central mass is also the central luminous source.
- 6.42** If you apply that same logic and say, well, the central luminous source of the galaxy is the bright part of the galaxy we can see, it must also have 99.8% of the mass of the galaxy. No, you're way off. In fact, for a large distance, tens of thousands of light years from the center of the galaxy's brightness, the velocity not only doesn't decrease, it increases. As if there's some enormous mass, much more illuminant, much more massive than the luminous matter would suggest it to be. That extends way beyond the visible light radius of the galaxy. That's what she conjectured and measured for many, many galaxies.
- 6.43** These so-called rotation curves indicate beyond a shadow of a doubt that there's some unseen matter that is responsible for the high speed of the stars as they orbit around the visible light center of the galaxy. Something else must be at play. We now call that dark matter. There's many, many different pieces of evidence, ranging from Fritz Zwicky's original measurements of the Coma cluster that I cover in my cosmology class, to the measurements via Rubin, to measurements of the cosmic microwave background radiation that I study, and others.
- 6.44** Let's look at this. How should it behave? Applying Newton's laws in a centrally gravitating context, you derive Kepler's law, this fall-off with radius as 1 over the square root of the distance from the central massive core. That should be the circle of velocity. We don't see that. We see an increase, and then eventually a plateau, and then eventually it does decrease, but you almost can't see it because there's no stars out there whose orbits can be measured. If you could, they're moving so slowly, the effect is minuscule.
- 6.45** Looking at our galaxy, we see the same exact behavior. It can be measured with optical telescopes, radio telescopes; and as I said, many, many other different types of techniques lead to this inevitable conclusion that the luminous matter is far from what we see. In fact, it's almost irrelevant in some comparisons. If we look at the edge of the

Milky Way galaxy, there is approximately a mass equivalent to 200 billion times the mass of our Sun. But if you actually count up how many stars you can see and their luminosity of each one is putting out, you don't get 200 billion solar luminosities, as you would expect.

- 6.46** If the galaxy were really made up of 200 billion masses of the Sun, you would expect, well, maybe there's 200 billion Suns' worth of light, but there's way less than that. There's only 10% as much luminosity as you'd expect, purely based on mass. That means there's 9 times or 10 times as much unseen matter, mass, as the light-producing stuff you can see. It's also because most of the stars in the Milky Way are less luminous than the Sun. Many, many of them are not nearly as bright as our Sun, even though our Sun is what's called a dwarf star, and it's not as luminous as these giant stars or other stars that we've talked about.
- 6.47** So this is the view of our galaxy. They're bright stars that agglomerate in the disk, and you can see them. And thank god they're there because we wouldn't be here if those stars didn't exist. But actually, there's this huge, perhaps spherical halo that surrounds each and every galaxy of the type of the Milky Way galaxy. So you have this halo. It's hard to see, but you have this halo of invisible matter. It's really what it should be called, not dark matter. It's really matter that doesn't interact with light. It doesn't produce light; it doesn't reflect light; it doesn't absorb light; it doesn't emit light.
- 6.48** It's very different from any matter that we're used to. Even our bodies in the dark can be detected, night vision goggles. It's giving off infrared radiation. We just can't see it, but that doesn't mean that it's anything mysterious. If you had an equivalent dark matter person, they wouldn't give off any heat, any light. They would have mass. If they went on a scale, they would deflect the scale the same amount. We know this exists because of its gravitational effects. We see it in innumerable different situations. I talk about that in my cosmology class.
- 6.49** But to give you one insight into what these objects are not. They're not just chunks of cold matter, black holes, even. They can't be, because of work done by my friends and colleagues, on the constraints of ordinary objects which can be measured due to their effects. It's called gravitational lensing. Mass, according to Einstein, warps space. When you have warped space, light moves on curved paths. In fact, around a massive object like the Sun or any other star, light can be bent so much that the actual positions of stars way farther away in the galaxy will seem to shift.
- 6.50** A phenomenon called gravitational lensing. It honestly acts as if it's a lens, like a telescope, distorting, translating, and moving the position of light trajectories of photons. Shown here is the behavior of such an object. These objects go by the name MACHO, not because they're toxic masculinity or anything, but because they are massive compact halo objects. Remember, the dark matter that exists in our galaxy is not concentrated at the central core. It's not concentrated on the galactic disk. It's in

this halo that surrounds at 10 times the distance where the Sun is from the galactic center. So it's called the galactic halo.

- 6.51** That means if there were particles of chunks of rock, icebergs, black holes that made up this missing matter, this 90% of the missing matter in our galaxy alone, that they would act as if they're tiny little lenses. As they passed in front of a star, they would amplify the light of the star, just like if you put this telescope and looked at the Sun. Please don't do that. But it amplifies, it concentrates the light of the Sun. Then, if the Sun were to move out of position, it would go down. You get this behavior in the so-called light curve.
- 6.52** For a long time, it's shown on the lower right; it's flat, it's just constant. Then an object happens to do a transit or syzygy, my favorite word, between the observer on Earth and the distant star in another, in the Large Magellanic Cloud. It has to be outside the halo of our galaxy, which the Large Magellanic Cloud is. And so astronomers measured many, many of these events where they saw a one-time brightening of the star, a dimming, and it never repeated. So it wasn't a nova. It wasn't bright enough to be a nova. And it didn't have the properties of an exploding star, of a supernova.
- 6.53** It was very characteristic of a mass truly passing in front of another star in a more distant dwarf galaxy called the Large Magellanic Cloud, causing that light to rise and then fall. One time only. And so there are these MACHOs, but they calculate how many there are, and they're nowhere near the amount necessary to account for that gap of 90% between the luminous matter that we see in stars and even in the planetary systems around every single star.
- 6.54** Remember, it's almost negligible. 0.2% of the mass of our solar system is mass other than the Sun. These objects almost contribute nothing. They have discovered these objects. They could be black holes in some cases. They're probably chunks of planets like brown dwarves, maybe, perhaps, but they're not nearly enough to compensate for what we know. We see the same behavior, not only in individual galaxies like the Milky Way and Andromeda, but when galaxies come together, imagine you have all these little tiny disks, CDs or DVDs, and then each one is surrounded by this much bigger beach ball of dark matter, literally the same ratio of a beach ball to a DVD, if you still remember what those are.
- 6.55** And then you have a whole bunch of those DVDs surrounded by beach balls. And then they're coming together and they're not hard shell like a beach ball. So the dark matter halos start to overlap of the individual galaxies when they're put in a cluster. You can start to think about the three-dimensional topology of such a combination of disks and beach balls, but then the beach balls can be combined together. And when you start to look at it, you get this weird pattern of dark matter filaments, punctuated by tiny disks of luminous matter, where the galaxies can trace the much more dense cores of dark matter halos that surround them.

- 6.56** In fact, simulations of the universe exactly exhibit this behavior and represent what we see in terms of clusters, the velocities of galaxies in them, the number of redshifts, blueshifts, the rotation properties of galaxies, the formation of galaxies and the stars within them, exactly mimic it only when you include non-luminous or invisible or dark matter. That's another clue that dark matter is real. But what is dark matter? That's a huge mystery. Only about 20% of the missing matter, the invisible matter, could possibly be in the form of MACHOs. Chunks of honest-to-goodness matter, protons, neutrons, croutons. That's all it could be. 20% at most, it's probably much less than that.
- 6.57** There must be some other type of matter that doesn't interact with light. Remember, it has to be dark, can't emit, absorb, scatter light whatsoever, which means it can't interact electromagnetically. Light is related to the electromagnetic force. But it could interact with the strong force only, it could interact with the weak force only, and it could interact with gravity only. Those are your possibilities. I want to give you a quick rundown of what I call particle physics for astronomers.
- 6.58** There are basically two different types of particles that we know about and we encounter and are familiar to us. Light is called a boson. It's a type of subatomic particle or elementary particle, has no mass, and you can have an infinite number of photons in one place at one time. The other type of matter is the matter that we're made up of, protons, neutrons, and electrons, and those are called fermions. For abstract reasons that I won't get into in this course, you can only have—each fermion has to be in its own particular state. They're not very extroverted, so they like to be completely isolated in terms of their quantum mechanical behavior.
- 6.59** But photons don't behave that way, and neither do neutrinos, even though they are fermions, for very, very arcane reasons that I won't get into. So one possibility for dark matter would be a new type of particle, as I said, that only interacts via gravity or the strong force or the weak force, except it can't only interact via the strong force because the strong force interacts with electromagnetism. You can actually break apart, as we saw. As we've seen, when you have nuclear fusion, for example, one of the decay products is a particle of light called a photon.
- 6.60** It couldn't be pure, strongly interacting material, but it could be weakly interacting material because weak particles don't interact with light. If you had the only purely weakly interacting particles that we know about, to a large extent, are neutrinos. Neutrinos are the byproducts typically of nuclear reactions, of radioactive decay. They've been detected, but they're among the most elusive particles to ever be detected. In fact, they were only detected a few decades ago, and we don't know their exact masses yet. We know the masses of all the other 17 elementary particles, but these three different types of neutrinos, we don't know their exact masses. We have limits on them, we don't know their exact mass.

- 6.61 That's one of the goals of the projects that I work on. Many, many of them can pass through us and are passing through us unimpeded because they don't interact with the other types of fermions that we're made of. They don't interact with electrons, protons, or neutrons. So they pass right through the Earth. In fact, trillions of them go through each and every one of us every single day, but we don't notice them. They're very, very low mass, but they have mass. So they could be dark matter. They are dark matter.
- 6.62 In fact, they're the only form of dark matter that's a true particle that we know for sure exists. The catch is, in the spectrum of different masses of particles, remember, this has to make up 90% of the mass of our galaxy, more or less. You'd like each one to have a significant amount of mass. Physicists measure mass in terms of the equivalent energy of the particle, converting $E = mc^2$, taking the equivalent mass and representing it as an energy divided by the speed of light squared.
- 6.63 The electron in those weird units has a mass, what's called rest mass energy, of about a half a million electron volts. That's how we call it. The neutrinos have a mass that's less than 0.1 electron volt. We're talking 5 million times lower than the lightest elementary particle, which is the electron. There better be a tremendous number of these neutrinos to make up for the missing matter in our galaxy alone, or they're interesting, but not really that important. And that's the state that we find ourselves in.
- 6.64 Now, these, in contradiction to MACHOs, are called WIMPs. That's the quirky humor of these silly astronomers that have names like Vesto and Swan and all sorts of things. WIMP stands for weakly interacting massive particle. So it can interact with gravity because it has mass, and it doesn't interact with electromagnetism because it's only weak interactions that affect it. These WIMPs could be like neutrinos, but they'd have to be much more massive or much more abundant.
- 6.65 We know, for example, how many neutrinos there are in the universe. If you go into interstellar space, intergalactic space, far from any nuclear reactors, a.k.a. stars, and you go deep between, say, the distance between, hundreds of times the distance between us and Andromeda, for example, and you had a little box, and you asked, "What's inside of that box?" So the majority of things inside that box would be photons. There'd be about 400 photons from the Big Bang inside of every cubic centimeter of every part of the entire observable universe. Imagine how many photons there are. It's like 10 to the 80th photons in the universe.
- 6.66 Then slightly less than that, there'd be about 100 neutrinos, also from the Big Bang, and the formation of the elements we'll talk about next lecture. They'd be inside that box, too. Then the rest would be dark matter and a lot of dark energy inside of there. But we know even less about dark energy than we do about dark matter, as I discuss in my cosmology course at Peterson Academy. Now, galaxies are vast repositories of dark matter, all the more so when you have clusters of hundreds or thousands of galaxies in the Coma cluster, the Virgo cluster.

- 6.67** And then clusters can cluster together to make superclusters. And there are even structures far bigger than that, that range in size over billions of light years, just incomprehensible distances, bound together by one thing, gravity, otherwise completely unrelated to one another. What's causing the gravitational force? Primarily dark matter. Dark matter has this long-range effect because gravity is weak, but the weakness is its strength, because the weakness allows it to have interactions over the entirety of the observable universe.
- 6.68** Now, my colleagues are very clever, and they want to detect these particles. They know how hard it is to do because perhaps they only interact via the weak interaction. Perhaps they only interact via gravity. And imagine trying to get enough of these very, very light but numerous particles in one place. How could you possibly do that? Well, some of my colleagues try to do that here on Earth.
- 6.69** In fact, at UC San Diego, my colleague, Kaixuan Ni, is building a detector to try to see if he can capture the rare but important interaction of dark matter particles coming into his laboratory, which is actually in the Gran Sasso mountains of Italy, and interacting with a very, very cold gas called xenon, which doesn't interact with any other molecules because it's what's called a noble gas. When it's cooled, it liquefies, and it can actually be a very sensitive detector of a recoil of a WIMP particle with the nucleus of an ordinary particle, like a proton or a neutron, and the quarks within it; they will interact if it's a weak force interaction that is mediated by these dark matter particles.
- 6.70** But if it's only by gravity, he's out of luck. There's only one tool that could be used in that case. In the case where the dark matter only interacts via gravity, every other technique would be out of luck. You would not be able to detect it using these cold tanks of liquid xenon. You would not be able to do it with these cold chunks of germanium and silicon that some of my colleagues use. Remember, what you'd have to do is collect enough of these objects, these dark matter particles, that they could be weighed, put on a scale of some kind.
- 6.71** Well, it would be very hard and, quite frankly, impossible to do that in the laboratory. You just could not get enough of them together to make enough of gravitational force that could deflect a scale. But thankfully, nature has done a lot of the hard work for us, because a cluster of galaxies is exactly that. It's basically a collection of dark matter in its most compact form, in its most gravitating form. And so we can use the property and the structure of galaxies and clusters of galaxies as traps, as collectors of dark matter, if you think about it.
- 6.72** Now, how do we use them? We use the propagation of light going through and near those clusters. Where does that light come from? The light has to come from beyond the realm of the galaxies. When we talk in astronomy about how challenging astronomy is, it is challenging. We cannot do experiments. We cannot do honest-to-goodness variables and change the density or the composition or the interaction of dark matter with ordinary matter. We can't do that. We can't produce dark matter,

although some are trying to do that in different experiments, but we can't produce dark matter and test how it interacts with ordinary matter.

- 6.73** That would be a proper experiment. That's how particle physicists measure things like the Higgs boson. But we can take these clusters of galaxies, using them as gravitational lenses, deflecting the light of even more distant objects into our telescopes. Now, the telescope compensates in some way for the fact that we can't do experiments. Because a telescope is a time machine, which my particle physics friends don't have access to. My biology friends can't go back and actually see some dinosaur or turn into a bird or whatever. Again, take Bret and Heather's course.
- 6.74** But when you think about it, having a telescope as your time machine allows you to see a state of the universe that's much more primitive in an earlier configuration. And because of that, you just have to wonder, what is beyond the galaxies that we see? Because what is beyond them in distance means it is coming to us from an earlier epoch. We will see in the following lecture that the measurement by Hubble, thanks to Leavitt and thanks to this man Slipher, and many other scientists, theoreticians, like Lemaître, Freedman, and many, many others, that there was an inescapable conclusion that the universe was expanding.
- 6.75** It started in a more dense and high-temperature state and began expanding and cooling. There will be consequences that we can see using our time machines, our telescopes. We will see fossils every bit as important as Indiana Jones might reveal in the latest movie, artifacts that traveled through time and through space, that reveal the conditions of their formation. We humans and astronomers who are also human, we use telescopes because our eyes are our primary organ for interacting with the world. Our eyes are telescopes, as we demonstrated earlier in the course.
- 6.76** That means that, of particular interest to us, is to look for the oldest light in the universe. That will be the light that's farthest away from us and be useful for illuminating these clusters of galaxies that contain these vast repositories of dark matter. We will then use that light, called the cosmic microwave background, to make predictions about stuff that occurred even earlier than its formation itself. That will bring us to other artifacts, which are actually much more familiar to you, because those are atoms that have traveled since the beginning of time.
- 6.77** The hydrogen that makes up the stars, the first stars that ever formed, were formed in the first three minutes after the Big Bang. But that hydrogen means that unless you're making some in some crazy nuclear reactor, that means the hydrogen in your body also came from the Big Bang. We talked earlier about Carl Sagan's famous statement that "we're all made of star stuff." As I said, the meteorites that we're going to encounter soon have the same type of iron, the same ratio of radioactive iron to normal iron, as is found in the hemoglobin molecules in our blood cells. It's astounding. It is true.

6.78 But most of the universe is not iron or blood. Most of it's water. Most of our bodies are water. So you're really not star stuff, you're Big Bang stuff. And that will be what we turn to at the end of next lecture. Thanks very much.

Lecture 7: Beginning to End

At a glance — The universe from beginning to end.

- 7.1 Following the discovery of the extent of our Milky Way galaxy and the observation that there were other galaxies equally magnificent, large, and unique as our own galaxy, there were natural questions that came about. What were the relationships between those galaxies and our own galaxy? Were they attracted to ours? Were they formerly fragments of our galaxy? Were they moving in some predictable or characteristic way? That realization and those questions eventually led to the startling revelation that, not only was our galaxy not the whole universe, the whole universe is dynamic. It's changing.
- 7.2 It will change in such dramatic ways that, in the very near future, on a cosmic time scale at least, we won't even be able to recognize it. That's the subject of this lecture, which has to do with the Big Bang theory, the expansion of the universe, its deep past, and what its deep past tells us about the far future. We start with a question that has been asked for about 400 years, which is seemingly a very innocent and easy-to-answer question. Why is the night sky dark? Well, it's dark because the Sun goes down.
- 7.3 But actually, in the context of the cosmological models that prevailed for literally thousands of years, this seemingly trivial fact is a deep paradox that should not be the case. It should not be observable that the universe ever is dark, let alone that our night sky is dark for half of our lives and half of every day. The puzzle dates back 400 years or more, and it's illustrated by this woodcut by a philosopher in the 16th century named Thomas Digges. In this woodcut, you see, in a very hard-to-read font, he's proposing an infinite universe filled with innumerable stars.
- 7.4 It seems innocuous. It seems something that would have comported with most expectations of early antiquity, and even up until the recent 20th century, as we'll see later on. But this conjecture, this hypothesis that the universe is infinite and filled with innumerable or infinite numbers of stars, is fundamentally incompatible with this seemingly innocuous observation that the night sky is dark. How can that be?
- 7.5 Well, in a universe like ours, where we inhabit a galaxy, the galaxy itself is finite. The galaxy is composed of a finite, the large number of stars. It also has dark matter and other components that aren't contributing to the possibility of it being bright day or night. But as we observe our galaxy, every single star we can see with the naked eye is in our galaxy. At night, we can see those stars precisely because it's dark. But again, that is built on the understanding that we are in a finite galaxy of large but finite extent, with a finite, the large number of stars within it.
- 7.6 What if the structure of the universe is such that it is infinite in size and has an infinite number of stars or galaxies within it? How could it be in that case that we'd ever find ourselves in a situation where the sky, day or night, isn't as bright as the Sun's surface

itself? In which case, we would not be here to ask such questions because it would be way too toasty for us to even exist. Our mere existence and the existence of the dark night sky are refutations of the scientific hypothesis.

- 7.7** Merely by doing, not using a telescope, not using a particle accelerator, we can test and prove and falsify the hypothesis that the universe is infinite and filled with infinite numbers of stars. We'll show how that takes place right now. Olbers, who the paradox that we'll discuss is named after, in 1823, conjectured that the night sky shouldn't be dark in an infinite universe filled with infinite numbers of stars. The paradox can be seen when we look out, say, in a city even, the night sky is far darker than even the darkest countryside or certainly far, far darker than the cityscape in Manhattan.
- 7.8** Now, to make the analogy a little bit more clear, imagine you go to a forest, and in that forest, there are an infinite number of trees, and the forest is infinite in extent and length and width. Then somewhere, obviously, it can't be on the surface of the Earth. The Earth is finite. But imagine an infinite forest. Everywhere you look, every direction you perceive, you will see a tree. You will see the branches of a tree. The denser the tree, even with a very low density, very far-separated trees, as long as the forest is infinite and there's an infinite number of trees, no matter where you look, whatever direction you look, your line of sight will terminate on a tree. It's inescapable in an infinite forest with infinite number of trees.
- 7.9** That starts to reveal the paradox that, even in our case, with an infinite number of stars, even if they're as far separated out and dispersed as in our Milky Way, where the nearest star is four light years away from us, it's a pretty vacant place. Even if you had that level of stellar density, one star for every cubic light year or four cubic light years or even more, you would still see, everywhere you look, in an infinite universe like that, you would still see everywhere you look would look like a star. I can prove it to you.
- 7.10** It's in contrast to the so-called Planetarium model of the universe, where the stars are glued onto this dome that we talked about, the superdome model of cosmology, the biblical model of cosmology, if you like. So if there were a finite number of stars, then even the Planetarium model of the night sky would make sense. But you must agree, if there's an infinite number of stars, there's no way to place an infinite number of stars in such a way that none of them are completely overlapping. And in that way, the whole dome of the Planetarium would be uniformly illuminated.
- 7.11** This, by the way, is one of my great thrills as a scientist. I got to run the Hayden Planetarium, home of Neil deGrasse Tyson, my kid's favorite astronomer, and was able to do a Planetarium presentation of my own, and even got to show the image that I showed you on the first day of class, the first lecture, from Galileo, his hand sketches of the Moon that led him, and led me, paradoxically and similarly, on a path to my career today to be an astronomer. So that was a full-circle treat to pay homage to the great Galileo in the modern-day temple of the stars in the Hayden Planetarium in New York City.

- 7.12** Now, here's an image taken by the James Webb space telescope of a globular cluster, the center of one of the globular clusters, that was used to triangulate the central position of the Milky Way and eventually settle the great debate in favor of the context of Heber Curtis, that the universe was not the Milky Way galaxy. There's a large number of stars here. They almost seem to overlap, but they don't. Each one of these stars might be separated by many, many parsecs, for example, or they could be even closer. This is the most dense aggregation of stars that we know about in a galaxy or in the universe, even.
- 7.13** Let's assume it's true and ask, what conclusion are we led to? How bright would the daytime sky or the nighttime sky be in an infinite universe, filled, as Digges shows here, with an infinite number of stars? Let's take a density of stars, which is finite. It's going to have an infinite number of stars if you have a finite density and infinite extent, infinite volume. Let's look around the Sun's neighborhood. The closest star is about four light years away. In radius, that means that, inside a sphere, there's about one star like the Sun in a sphere of radius for four light years.
- 7.14** The volume of such a neighborhood is about, let's call it for ease of use, it's about 10 parsecs cubed. And there's one star in such a neighborhood, so there's one star, that means there's 0.1 stars per cubic parsec. Let's say each star has the luminosity of our Sun. The Sun is average, but it's actually above average in some ways in terms of luminosity. It's not really important. We can change all these numbers. It won't make a difference. Let's say the luminosity is this huge number of 40 trillion trillion watts or 400 trillion trillion watts.
- 7.15** Let's put ourselves at the Sun, for simplicity, obviously, we're not on the Sun, but we're at the center of this fictitious shell. The shell has some thickness, and it has an area, corresponding to the area of a sphere with that radius, which is 4π times the radius of that square. We're at the center of it because that's the way we perceive it. We can make this sphere as big as we like and set the density of stars in that shell, which has thickness times area as its volume, and set that equal to 0.1 stars or 0.1 solar luminosities per cubic mega parsec.
- 7.16** Then all we have to do is calculate the area of that sphere, which I said is $4\pi r^2$, and you could start the radius at any point, some distance away from where you're doing the observation, and you can make the thickness small or large; it really doesn't matter. Then the number of stars in that volume, that shell volume, is its volume times the density or stars per cubic meter, or stars per cubic, in this case, parsec. The total number is volume times that density.
- 7.17** Now, let's look at a single star. Let's look at the Sun, for example. If we count up the flux from one star, and then we multiply it by the total number of stars calculated by the volume times the density, the total flux will be that density times the luminosity of the star, times the thickness of the star. The flux also has the quantity $4\pi r^2$, which is also the surface area, A , of that spherical shell. So the volume contains the area times

the thickness, but the flux contains one over the area, so the area terms cancel out. And the total flux of each one of these little fictitious shells that we're making, of thickness, t , will be just the luminosity of the average star times the density of stars per cubic parsec. That's the total flux of each shell.

7.18 Now we just have to count up all the shells there are in an infinite universe. Do you see where this is going to go? You have a finite number, which is the total luminosity. You have the thickness of the shells, which is small but still finite. You're going to integrate or sum up all the contributions, starting at your toes or going all the way out to infinity. You're going to add up an infinite number of shells with a tiny, tiny amount of luminosity per shell. But since you have an infinite number of them, no matter how tiny, as long as it's above zero, the amount of flux per shell is, with an infinite number of shells, infinity times any number is infinity.

7.19 Actually, the flux, say, of such a shell, as we measure it, is quite small. It's 40 nanowatts per square meter, which is incomprehensibly small. But since there is an infinity multiplying this tiny number, the answer will be infinite. So infinity times any number is infinity. That means that the night sky's flux or the daytime flux should be infinite. Now, we don't see that. So then one of our assumptions in this paradox, or more than one of our assumptions, must be wrong. We only assume two different things: that the universe is infinite and it has an infinite number of stars.

7.20 Now, infinite means infinite in size. It could also mean infinite in time, infinitely old. When I say size, I mean size in space and time. That could be wrong. The infinite number of stars could certainly be wrong. Then we wouldn't see anything other than what we do see, which is a dark night sky. But the paradox would be wrong or it would be disproven that we live in either an infinite-sized universe or an infinite number of star universes. So no matter where we look, our line of sight, just like in a forest that's infinite, ends on a star's surface, if you have an infinite number of stars.

7.21 Now you can say, well, what if they're galaxies? Well, galaxies just makes it worse because each galaxy, in the same rough area as a star sometimes, might have 10 million stars within it. So you should see anywhere you look a glowing, bright, spherical shell that surrounds you, brighter even than this room. Something in our assumption pattern must be wrong. We must have made something incorrect about the universe. Either it's infinitely large, or it's infinitely large in space, or it's infinitely old, or the combination of them, or there aren't an infinite number of stars.

7.22 All three of those postulates, here is space and time combined together, but all three of them, when you include the infinite number of stars, put severe constraints on the structure and composition of our universe that we can then test to construct the actual model of our universe, how our universe is actually structured. The speed of light is the fastest possible speed any objects or information that we know about. It's incredibly fast. But the universe is incredibly large. Even if it is finite, it's incredibly large.

- 7.23** Now, if the universe has a finite speed limit for any information, including information from stars to enter telescopes, that means there could be regions of a universe that's finite but large, that haven't had enough time to reach our telescopes to indicate that they exist or to see the light from those stars, for example. That's one possibility that would take place when the universe is assumed to not have an infinite age, which, again, is part of an infinite size in the sense of space and time as one coherent whole.
- 7.24** Olbers' paradox and the woodcut that we saw before shows those stars out there in the distance. They're at a constant density. That's all fine for this argument. But again, it has to extend to infinity. But it's also static. They're not changing. The stars are not moving, and they can move left and right, but they're not moving coherently as a whole, i.e., they're not expanding away from each other and in particular from us. If they're expanding, different properties will take place in our observations of these objects. In particular, galaxies and stars will change in their position. Relative to us, they'll be Doppler-shifted.
- 7.25** The Doppler shift can have one of two different characteristics on bulk scales. It can either be that they're expanding away from us or that the universe is contracting. That was the assumption until the beginning of the 20th century. Not only did Albert Einstein, one of the greatest intellects of all time, not only did he believe the universe was static, which is impossible to reconcile with Olbers' paradox as well, but he believed that the universe had to have some mysterious substance within it to counteract the gravitational force of all the matter that he certainly knew existed because he is made of matter and the Milky Way is made of matter.
- 7.26** Slowly but surely, the universe was predicted to not be static, to not be infinite in size, to not be infinite in time or age as well, and certainly not to have an infinite number of stars. This was the heart of the Big Bang model, precipitated by this paradox from the 1820s, that the universe couldn't be infinite. Now, it's one thing to say what the universe is not, but it's another thing to say what it actually is and what it is doing. That is the goal of a scientist, to predict not only the way things used to be, but the way things will be in the future.
- 7.27** We discussed in our context several times the powerful tool of spectroscopy, in particular the Doppler shift, that is indicative of the relative motion between source of light and receiver of light or detector of light. If the universe is expanding or contracting, we should see characteristics of that behavior. Namely, we should see either all galaxies have their light being blueshifted if they're coming towards us or redshifted if they're moving away from us. The light that we talk about shifting, from blueward to redward, is indicative of expansion and motion away from us.
- 7.28** That is done by looking at the tell-tale chemical fingerprint of the atoms inside the galaxies. The hypothesis is, the galaxies are made of the same material that our galaxy is made of. Namely, it's made of stars. Stars are made of hydrogen, converting it to helium, primarily. Then they're made of dust and rocks and planets and comets and all

sorts of other stuff that we'll get to in the next lecture. But this universal fingerprint can be used as a tell-tale speedometer, and not only measure the actual speed, but the velocity. Remember, velocity is a directional vector. It determines the direction and a magnitude of the speed.

- 7.29** And in our galaxy alone, we see equal numbers of stars being redshifted and blueshifted. That's because the stars are in relatively random motion. They're not all expanding away from us. But we don't see that in the universe as a whole. We look at galaxies as tracers of the large-scale structure of space time. We're going to do some experiments now. Take out your diffraction goggles. I'll just remind you of these beautiful spectral patterns. Does everyone have their diffraction grating glasses? I'm going to put on two glasses, safety goggles and this. I haven't done this with this set of chemicals.
- 7.30** We have here a set of different chemical vials, different samples, different element, and they'll produce different spectrum depending on the atomic transitions that will take place inside the molecules that are inside the test tube now. The first one I've put up there is this blue substance, quite beautiful, called copper sulfate, CuSO_4 . It also has a lot of water, H_2O , in it. When we heat it up, it should start to glow, and it should produce some particular chemical signature. We may or may not be able to see it with these diffraction gratings, but with a true spectrometer, we would be able to see this quite accurately and be able to reveal not only that it has water in it, but it has copper, sulfur, et cetera within it.
- 7.31** Can we dim the lights? Because I'm sure this will be a little hard to see. See the color that's coming out of that? I can see a little bit of the spectral lines from here. It's going to get hot soon. This is one characteristic of, in particular, copper, in this case, primarily copper. We'll have another sample over here. This one is a potassium [hexocyanoferrate], which has iron in it, the same iron that you have in your blood. This one's more orange or yellow, as I see it. See a little bit of the spectral color. With the diffraction grating, it's hard to see, but you'll see little tiny dips or black lines. Then let's see other fun ones we can do.
- 7.32** Sodium carbonate. Baking soda, basically. This goes in there. This is pretty intense. You see any of the spectral lines appearing from this one? It has carbon in it. It's potentially produced from organic substances. Then the last one— when I was a kid, they didn't have these very safe samples. You got radium and uranium-238, things that could kill you, needed a license to use nowadays. The tell-tale chemical fingerprint reveals at a great distance what these objects are made of. In particular, it can reveal the presence of the most common element in the universe, which is hydrogen.
- 7.33** Hydrogen is the best steady. It's the simplest element there is. We can use it as a very, very accurate ruler, a very, very accurate speedometer, because this will tell us the relative velocity, not necessarily the distance. We'll see how we can measure the distance as well as the velocity, and we find a universal, literally universal relationship,

between distance and velocity. As we've discussed many times, the Doppler shift allows us to measure velocity, relative radial velocity. It doesn't allow us to measure complete vector velocity, but that's often not as important as measuring the radial velocity.

7.34 What astronomers were shocked to discover in the 1900s was that every single galaxy, with a handful, literally 10 galaxies of exceptions, every other one of the 100 billion galaxies we can see is moving away from us. Now, there's obviously something special about us, and maybe, and just maybe, it meant astronomers could be confident that finally, finally, we're the center of the universe again. No one's going to disprove us, not Copernicus, not Herschel. Nobody's going to disprove us.

7.35 It's basically the supposition that if you were in a region of the city and you heard all these ambulances, and every single ambulance you heard had its siren redshifted or lengthened in wavelength such that the siren appeared as if all the sirens were going away from you at a high speed, well, then you would assume a couple of things. You would assume that something happened and you are the center of it, and everything is now fleeing the center of probably a horrific accident. That would be the supposition of everything moving away from you.

7.36 Now, is that true of our galaxy in the context of every other galaxy? Now, if it was true for every galaxy in our observable universe that you could see and you could tell that every galaxy also sees every other galaxy moving away from it, then there wouldn't be something special about us. It would be a feature of the universe itself, meaning the universe itself is expanding, or your city is expanding. That could also happen. Stationary ambulances, but somehow we're stretching like playdough, stretching out the streets of the city really fast, and all the ambulances are moving away not only from you, but from everyone else. You would hear the same exact thing.

7.37 And that was the conclusion reached by Edwin Hubble. He took the measurements and theoretical models of Henrietta Leavitt. He used those to measure the luminosity distance to every galaxy he could find that had a cepheid variable in it. The cepheid variables have a period luminosity relationship. The period is proportional to how far away they were. He could determine, she could determine, how far away any galaxy was, as long as it had a cepheid. But good news, almost every galaxy has not just one, but many cepheids in it.

7.38 Then, Vesto Slipher had done measurements of galaxies, the few dozen closest galaxies to the Milky Way, and saw that—measured their spectra, and saw in particular that all of their spectra, with a handful of exceptions, the Andromeda galaxy being one of the only ones, almost all of them had a redshift. And this led Hubble to make the so-called Hubble law, which says the following. The redshift of a galaxy, which is symbolized by the letter z in astronomy, that quantity is the relative shift in wavelength towards the red of any galaxy relative to the wavelength that that light would be at rest in the laboratory on Earth.

- 7.39 That redshift is proportional to the distance to that galaxy. A galaxy at twice the distance, using Leavitt's law, also had a redshift twice as big as a galaxy half as close to us with a distance measured by a cepheid in that galaxy. If you plot the relationship, you get this characteristic behavior. Distance and redshift are linearly proportional to each other. A line is characterized by just one number, the slope of that line. The slope of the line tells you how much redshift is that galaxy moving at. Remember, the redshift is directly proportional to the velocity of that galaxy, the velocity being a direction always radially away from us, and the amount being proportional to the distance. This is known as Hubble's law.
- 7.40 The slope of that line is given as the symbol H_0 , or H-naught sometimes, and that's called the Hubble constant. It parameterizes how much the universe is expanding at what rate of velocity our galaxy is moving away from the Milky Way and away from us, as a function of distance. It has relatively strange units that we'll explain. The currently best accepted measurements by the community is about 68 to 72 kilometers per second per megaparsec.
- 7.41 That means, if there was a galaxy at one mega parsec, one million parsecs away from the Earth, which is actually quite close, it's closer than the Andromeda galaxy—remember, that's about three million light years, so it's a little bit closer than 3.26 million light years; that's a mega parsec—that that galaxy will be moving 78 kilometers every second away from us. A galaxy 100 times farther away will be moving at 6,800 kilometers per second, and so on and so on. You could use this measurement.
- 7.42 It's relatively easy to measure the spectrum of a galaxy. Just focus a telescope on it, put a diffraction grating or a prism or what have you, make a spectrometer. That means you could measure distance now by measuring one number, if this law is true, which it does appear to be holding over the observable universe. So Hubble made this conjecture. It's a law of astronomy. It's a law of cosmology. I talk about it a lot in my Peterson Academy cosmology class. It says the velocity is proportional to the distance. The proportionality constant is this constant of nature, a universal constant, H_0 , H-naught.
- 7.43 It's the same for any galaxy. If we were to go to Andromeda, we would also see every other galaxy, except the Milky Way, moving away from us. There'd be a handful of satellites of—Andromeda has some satellites. It ranges, it has some uncertainty. Different groups measure different things, but very high precision in each case. We write this, we'll say 70 kilometers per second per megaparsec. Again, that means a galaxy at one megaparsec will have a velocity of this amount because of the expansion of the universe.
- 7.44 It's far easier to do this measurement of a spectrum of a redshift. It could take a second or two to make a measurement of a galaxy. Compared to what it is on Earth, you get the redshift, you look up and multiply by 71. It's the easiest possible thing, much easier than using the period-luminosity distance relationship, much easier than using the

parallax relationship. It's almost like it's a gift from the universe. It makes it easy as pie to measure distances out to the limits of our telescopic vision.

- 7.45** I already said this, you measure the velocity. Sorry, you measure the redshift; this gives you the velocity. You convert the velocity into a distance using Hubble's constant. That's it, much easier than any of the laws we've discussed today. It's predicated on the fact that this behavior only applies in a universe that's expanding. It actually would appear in a universe that's contracting, but we don't seem to live in such a universe, obviously, so that's not relevant.
- 7.46** The actual velocity of the galaxy in units that we measure is just the speed of light times the redshift, exactly as Doppler would have predicted for sound in the 18th century. So what astronomers do is they go out and they measure the position, the brightness, and the redshift of every single galaxy they can see. And every single galaxy, once you get beyond a few dozen or 100 megaparsecs, is moving away at a very large velocity. Remember, this gets very large, very fast.
- 7.47** And it actually could be faster than the speed of light. There can be galaxies here that are expanding away from the Milky Way at this moment, faster than the speed of light. There's no way for us to communicate. It doesn't violate the laws of physics. It doesn't somehow invalidate what Einstein said about the fundamental speed limit of the universe. It just means when that light set out from a galaxy that's a redshift greater than one, the light was emitted, and we were much closer to that galaxy than we are now.
- 7.48** Now that galaxy is stretched beyond us and the space between us is such that the distance times this redshift, times Hubble's constant, is resulting in a velocity that's greater than the speed of light. We can't do anything. We can't send signals using that. It's as if you're on the surface of a balloon and stretching faster than the speed of sound. You could emit sound, but it would only be receivable if you admitted it at a time when you were close enough, such that by the time you've stretched out and we observe it, the light was detected or the sound was detected, but it'd also be very highly redshifted, very low frequency.
- 7.49** Mathematically, if we look at the units of the Hubble constant, the units are kilometers per second per megaparsec. A megaparsec and a kilometer are units of length. You're dividing a velocity, which is a length per unit time, by a length. You get something that has units of inverse time, one over time. It's like an inverse—it's essentially like a frequency. Now, it's nothing mysterious or strange about it. What it tells you is that, if the universe is expanding at this rate, you can invert the rate and measure, knowing the distance between the galaxies, to determine the time at which they were last in physical contact.
- 7.50** It's not actually what happened. It wasn't that the galaxies were all touching together. They weren't formed as galaxies. They were in a much different state that we'll talk

about later on in this lecture. They were composed primarily of hydrogen and helium and not composed of beautiful spirals with planets and all sorts of other items and objects within them. So Hubble's constant, when you take it with— Hubble's constant is related to the inverse time, and the time is the time between which the universe has expanded to be its current size, meaning that this is essentially related to the age of the universe since the universe first expanded.

7.51 One divided by H_0 gives you the age of the universe. If you compute that in seconds, it seems incomprehensibly long, but you realize that there's basically 31 million seconds in a year, and you can convert 31 million into years and seconds—sorry, you can convert years into seconds. Actually, this number comes out to be about 14 billion years. That's extremely close to the actual observed, measured age of the universe.

7.52 Age of the universe since what? That should be a question. Was the beginning of the universe the beginning of time? Was the beginning of the universe beginning of space? Were there preexisting universes? Are there other universes? Those are excellent questions, and they're addressed in my cosmology class. So luckily for you, you have a full subscription to Peterson Academy, and you'll be able to learn all the answers to those questions.

7.53 But for now, we suffice it to say, in this tour through cosmology for astronomers, that the expansion of the universe became one of the pillars of the Big Bang model. It suggested that if you ran the movie of our universe backwards, you would find a time at which all the matter in the universe that we see around, centered on us, was effectively in one point, in one place. It was called a primeval atom, a primitive form of the universe's existence. It wasn't actually an atom, but that's the way people thought about it in the early 1930s.

7.54 This is just seven years after Hubble discovered that the Milky Way was not the whole universe. The progress was exponentially fast. So this ushered in the notion that there could have been a beginning to time. There could have been a beginning of all matter and space and energy. And that's the heart of the Big Bang concept. The inverse of this weird unit H_0 is a time in seconds. Convert seconds to years, you get 14 billion years. That's the approximate age of the universe that we cosmologists accept to be true.

7.55 How does this address that fundamental question, the paradox of the night sky's darkness? Shouldn't be dark if the universe is infinite. Infinite in what sense? Infinite in space and infinite in time. That's true. It would be impossible to reconcile an infinite universe in time and space, with infinite stars with a dark night sky. We showed that. But if the universe is finite in size, finite in size in space-time, meaning it has a finite age, as we now believe it does, our observable universe, there hasn't been enough time, even if there were an infinite number of stars, or there would eventually be an infinite number of stars, there's not been enough time for that light to get to us to this point, even given the vast speed of light, because the universe is actually physically expanding.

- 7.56** So objects are moving out of the ability to show themselves to us. And so even if the universe did have an infinite number of objects in it—stars, galaxies, what have you—we would still not be able to see them because the universe is a finite age. So, that violates the conjecture of an infinite universe. So we've now found a way out of it and shown that it's not paradoxical that we have a dark night sky because we don't live in an infinite universe, at least in an infinite-in-time sense.
- 7.57** So, as I discuss in the cosmology class, it's not entirely the case that the Big Bang was the beginning of time, but it was the beginning moment for the structure and evolution of our observable universe. We can observe objects in the universe out to a distance in space, due to the expansion of the universe, the speed of light times the age of the universe, times a quantity of approximately the value of three. We can see things out to about 45 billion light years in any direction. We'll talk about that in just a moment. What are we actually seeing? What was around back then to see? And what does that tell us about the universe at close to the beginning of time?
- 7.58** Not as close as we would like. We'd like to go back to time equals zero. But ask yourself a question, as we do in cosmology, whether or not time actually passes at the beginning of time. What is time? It's a change in state of something. Well, if something only came into existence, then how could something exist before it to make a change with respect to? These are philosophical questions, but they're actually questions that physicists and cosmologists are trying to answer.
- 7.59** In particular, there are a whole numbers of cosmologists who believe there was a universe that existed before our universe, and the raw materials were provided from it to create our universe in the same way that a star existed in our galaxy that collapsed, exploded in a supernova, to create the materials that would be raw materials used to construct our solar system. What's interesting is, if you go to Wikipedia and you type in science, the image that comes up to represent all of science, from anatomy to zoology, is an image that depicts the history of our universe according to the Big Bang model.
- 7.60** In that sense, the Big Bang model, in just 80 years, went from being a laughingstock, not even believed by Einstein and mocked by many people. In fact, the name Big Bang is a pejorative given by a British astronomer named Fred Hoyle, who didn't believe in it, and believed in a static, eternal universe. It went from being a laughingstock to being the representative, at least to Wikipedia, of what science is. Hypotheses, testing, evidence, replication, reproduction. It has all those features, and it's brilliant and it's beautiful in many ways.
- 7.61** But we always have to ask, if this hypothesis is true, how could we prove it to be wrong? How could we, assuming the hypothesis that there was a beginning to our observable universe, what would be the things that we could see from it, such that if we don't see those things, we have less confidence or maybe no confidence in the Big Bang model? And fortunately, we've passed all these tests with the Big Bang model. I

shouldn't say fortunate, because actually, if you know everything about a science, it's dead. There's nothing more to know about it.

7.62 We've dissected every toad that we have access to. Once you dissect it, learn its anatomy and its molecular biology and its evolutionary history, that particular subject is closed off. But there are a lot of great mysteries in our understanding of the universe's origin. In particular, we don't know if there was a Big Bang, if it was a so-called singularity, if there are other universes. That's the subject of the research that I do with my team at UC San Diego and around the world.

7.63 What would you see if this hypothesis is true? What would be the relics, the fossils that travel through space and time to our telescopes and our detectors and our laboratories to this very day? Well, the best explanation for why the universe began expanding, rather than contracting, and why the universe started to have such an incredibly expansive, explosive beginning, is a theory that originated in a branch of science called particle astrophysics. The laws of particles in subatomic world applied to cosmological or astrophysical situations.

7.64 This is because the age of the universe, when the spark would have ignited it to begin expanding, like sparking this burner, that spark occurred over a fantastically small period of time, about a trillionth of a trillionth of a billionth of a second after the Big Bang, after time equals zero, if such a thing happened. That then caused, before that was pure energy, what's called a quantum field, but because this time scale is so short, if you multiply this time scale, 10 to the minus 34 seconds, by the speed of light even, the fastest thing that could possibly travel, the fastest speed there is, $300,000$ kilometers per second, you still get a microscopic number that's about the size of a proton or subatomic scale.

7.65 And so that's why the laws of particle physics come into play in cosmology in this realm. It was conjectured over 44 years ago that this was the spark that ignited the expansion, caused us to expand. That's called the theory of cosmic inflation. My first book called *Losing the Nobel Prize* is about this theory and how we try to test it with telescopes at the South Pole that my colleagues and I built and traveled to. We don't know if it's true. We're trying to test it. There'll be relics left over from it, as I discuss in the cosmology lecture; I welcome you to check that course out as well.

7.66 Now, more tangibly, also discussed in the cosmology class, but I can't resist because it's so much fun to talk about, are other types of fossils that travel through space and time. In particular, as I said just a few moments ago, although Carl Sagan said we're made of star stuff, we're actually mostly Big Bang stuff. We're made of hydrogen in the water molecules that make up 70% of our body, 70% of the molecules in some sense are related, have hydrogen within them. And that hydrogen isn't created in some laboratory somewhere. It was created, not in a star because you can't fuse something into hydrogen, but you can, during the Big Bang, turn energy into quarks, and quarks then into protons and neutrons.

- 7.67** Protons and neutrons then make the isotopes. There's three isotopes of hydrogen, and they can only be fused just like in the Sun at extremely high temperatures. So we can look for the ratio of those different isotopes of hydrogen called ordinary hydrogen, deuterium, which is a proton plus a neutron, and then tritium, which is two neutrons plus one proton. They all have the same chemical properties. They have different weights, densities, etc. Otherwise, you can drink.
- 7.68** I have a glass of deuterium oxide. Not here, I have it in my lab. I've made ice cubes out of it. They behave exactly as you'd expect. They're more dense than ordinary water. So a deuterium ice cube, you can float an ordinary ice cube on top of a deuterium ice cube, do all sorts of crazy things, make an ice cube that sinks to the bottom of the glass and tastes just the same. I haven't had many side effects since I've had it to drink. But even in this glass of water, ordinary water, tap water, there's a certain number of deuterium nuclei in the center of the water molecules.
- 7.69** That then can be used as a test to measure, what does science predict during the first moments of the universe's history when it was tens of millions of degrees Kelvin? What should be the ratio of those different isotopes? Because what happens is, the deuterium is the proton plus a neutron, the blue and orange colors. They get combined to make tritium with another neutron, and then the tritium combines with another proton to make helium nuclei. You should have a certain amount of helium, a certain amount of hydrogen, and of the hydrogen, there should be some tritium, some deuterium, and some protium, which is the name we call ordinary hydrogen.
- 7.70** We find that exact ratio. In fact, in the first few minutes after the universe's history, almost all the helium that would later be found in other clusters of galaxies and galaxies themselves, all of it was made, and all of the deuterium, and all of the hydrogen in our bodies. So we are fossils of the Big Bang, and we can test that. After the first few minutes, so in a time period shorter than one-tenth of the TV show, *The Big Bang Theory*, the universe stopped making nuclei, at least the lightest nuclei on the periodic table—hydrogen, helium, lithium, some beryllium, and their isotopes. That was it.
- 7.71** Then, for literally 400,000 years, nothing happened. There were protons, there were electrons, there were nuclei of helium, and there were some lithium and beryllium, very small amounts of those. We've measured all those back to very, very high redshift, very high distances. So eventually, we get atoms forming. And all the atoms that form to very good approximation are hydrogen atoms. We start with hydrogen nuclei and electrons, but the universe is too hot. Any hydrogen that forms an atom gets broken apart, obliterated, snapped into protons and electrons in what's called a plasma, plasma of deuterium and a plasma of tritium.
- 7.72** But at 400,000 years, the universe is expanded, and when things expand, they cool off. Once it cools off enough, the hydrogen can form out of protons combining with electrons. That process produces what's called the cosmic microwave background

radiation, which is, as I said earlier, how the bread gets buttered in the Keating household, because this is what my colleagues and I study with telescopes that see not light, but see microwaves. Microwaves were originally in the form of ultraviolet light, very high-energy light, that got redshifted by a factor of a thousand.

- 7.73** The universe expanded by a factor of 1,000 since this picture, the atoms formed in this picture to produce this signal. Again, covered in much more detail in my cosmology lecture. The universe went from being completely opaque in a plasma state. You cannot see through a plasma. You can't see through the center of the Sun. Can't even see through the flame once I make a flame here. A flame casts a shadow, so it's opaque. You can't see through it. But once the atom's formed, now it's no longer a plasma. Now it's a gas of neutral atoms. And that's the way it stayed for literally hundreds of millions of years.
- 7.74** So starting at 10 to the minus 34 seconds, going to when inflation occurred, potentially, going to the formation of the first atoms in the first three minutes after the origin of our observable universe to 400,000 years, and then nothing happens for millions of years, almost a billion years, half a billion years, until the very first galaxies form. We see those galaxies, we measure their properties, we see their dynamics, we can see their composition in some cases, we see they're mostly made of hydrogen. Those hydrogen-rich galaxies had hydrogen-rich stars, which eventually fused hydrogen into helium, helium into silicon, carbon, nitrogen, oxygen.
- 7.75** Eventually iron, and some of them blew up, causing supernova explosions, producing the raw materials of the next generation of stars and solar systems like our own. Now, as we will be turning to, we now find ourselves in a situation where we can take classes and courses about this subject, contemplating the type of matter that was formed and its history and how that matter could have spread and formed from inanimate matter, from just chemicals, hydrogen, into more complex elements like iron.
- 7.76** Then how that iron could then be in the blood of a creature that could then think about models of how the universe began after hundreds or millions of years of evolution of our minds and our thought processes, and, most importantly, of science itself. What will happen in a deep future, not to depress you, but eventually the Sun, as I said, is about middle-aged. It'll start to run out. That'll be when the age of the universe is about 19 billion years old. The Sun will swell and eat up the planets, including ours. So hopefully we'll have to figure it some solution to that, or otherwise it's the ultimate bad hair day.
- 7.77** And we see what happens to other stars. And so we know this process very well. We know when these supernova take place. I showed you a picture of the very first and earliest supernova observations, Chaco Canyon, in the United States, a thousand years ago, witnessed by Chinese astronomers, continuously monitored to this very day, even by telescopes that I use. So the Earth won't exist. It'll be swallowed up. And then eventually, it'll blast off the outer layers of its envelope, of the stellar envelope, and

become a white dwarf. And that will be very quiescent, very boring, for a thousand billion years, a trillion years.

- 7.78** The galaxies will find themselves basically these wastelands of white dwarves, stellar corpses, and black holes. And the stars will eventually burn out, and all that will be left is heat, very cold heat. The cosmic microwave background is currently about three degrees above absolute zero. This will cool off to just a fraction of a millikelvin, milli, one-thousandth of a degree above absolute zero. There will be black holes, and some say that black holes can last forever. Some say that if there was a universe preceding ours, it had black holes that still exist in our universe. That's a fascinating topic that we will discuss in the black holes course that I teach.
- 7.79** But some say, in this length of years, 100 trillion, 10 to the 27th years, there'll basically be nothing left of our galaxy or any galaxy except for one relatively massive black hole that contains the equivalent of all the mass energy that ever existed of that galaxy and all the galaxies that it swallowed it up along its 10 to the 27th-year-long history. A great scientist, Stephen Hawking, made a startling discovery that black holes aren't actually completely black, that they actually do emit some tiny amount of heat that's really infinitesimal. It's almost impossible to measure how small this is in Kelvin, fractions of a billionth of a billionth of a Kelvin. Just incredibly incomprehensible, as I say.
- 7.80** That has to do with the properties of quantum mechanics surrounding black holes. We won't talk about that so much in this class. Again, I will talk about this at length in the black holes class, so stay tuned for that, a little advertisement. So the black hole evaporates over time. And so eventually, there'll be nothing left. Evaporates into what? Not into water. It evaporates into photons. It carries away its mass energy, converts that into energy, pure energy in the form of photons. But that energy is incredibly low in extent.
- 7.81** That will last for 10 to the 100-plus years. Again, keep paying your taxes out there. The black holes eventually will emit these particle-antiparticle pairs and Hawking radiation, dissipate, and there'll be nothing left in the universe. Perhaps, though, according to some, the universe will then, at that point, respawn into another universe via an even more bizarre feature of quantum cosmology that I will discuss again at some length in my lecture on black holes, because that is the only possibility for hope in a universe of this enormous age that's truly incomprehensible.
- 7.82** But we come an extremely long way in this course, from the first astronomers to use naked eyes and even just their minds connected to their brains, their minds, their brains connected to their eyes. Then to telescopes, spectrometers, measuring things like velocities, positions, parallax, then later on measuring distances out in the universe. Then, of course, the question naturally comes—are we the only such creatures that can contemplate the universe's structure, composition, evolution? Are we alone? Or are there many, many civilizations teeming throughout our galaxy and other galaxies as well?

7.83 Some say either possibility is terrifying. Either we're alone or being watched maybe by many, many civilizations, or maybe those other civilizations haven't had time to get to us yet. And who knows what they'll do. We'll start to explore that in the last and final section of the course, in Lecture 8. We'll encounter what it means to be alive, how life could have originated on Earth and in the universe, and whether or not there are other life forms distributed throughout the universe. If so, where are they? And if not, what do we do about it? Thank you very much.

Lecture 8: Beyond Earth

At a glance — Beyond Earth — the final lecture.

- 8.1** Welcome back to the final lecture of this whirlwind tour of our universe. I hope you've enjoyed it, and I hope you've had a lot of questions along the way, including the possibility that we might not be alone in the universe. You may have wondered even how life got started on the Earth. Well, let your fears be rest assured that we are going to cover all that and more on the final lecture of our encounter with your universe in our astronomy introduction course.
- 8.2** You may have noticed that we didn't really cover comets or meteorites in this class, which is surprising because those are some of the few phenomena that are visible to the naked eye. In fact, meteors and meteor showers are visible almost every night of the year, depending on where you are and what type of a city you are, what type of a countryside you might be in. We discussed, way back when, in the beginning of the class, how there are so many interesting characters with interesting names in this course and in the history of astronomy.
- 8.3** One of the most important and most interesting and most fascinating names was Tycho Brahe, who was the first, as we said, to use the principles of parallax to measure, to set a constraint on the properties of comets, to think that, before his work, most astronomers thought that comets were in our atmosphere. After his work, he showed that he couldn't measure the exact distance to a comet, but he could measure the fact that they were relatively much farther away than our Moon, so they couldn't be in our atmosphere.
- 8.4** Remember, even Aristotle thought the Milky Way galaxy was somehow in our atmosphere, and that was a view that prevailed for thousands of years. Comets play an intimate role in the existence of life on Earth, and presumably in the possibility, although not necessarily a high probability yet, of life on other planets, as we'll discuss. Comets were thought for many, many thousands of years to be omens. Literally, the word disaster, catastrophe, these words contain the word astro or aster within them, meaning that a bad star, a disaster, a catastrophe, something catastrophic, literally, would be caused not by heavenly events, but by the stars.
- 8.5** Things like comets and meteors even, and certainly eclipses, were thought to be portents of very negative, nasty things. It's funny, they never thought of really lucky things happening when a comet or something would appear. And throughout history, we've associated comets with tracers, basically laboratories, time capsules, of what our solar system was like in its earliest state. And so too, meteors, which we'll discuss very momentarily, also play an enormous role in the history of astronomy, but also potentially of import on the search for life and the origin of life on Earth.

- 8.6** Meteor showers can be visible many times per year. And on my website, briankeating.com/PA, you'll get information about how you can observe meteor showers, what the different properties of meteors are, what their colors can tell you about their composition, as we've talked about. We can essentially look at them by eye. We don't need a spectrometer to determine what they're made of. We can also reveal aspects of the planet. We can reveal attributes of the parent body that created the meteors that we see in our atmosphere.
- 8.7** You can tell in images what's a meteor and what's a star. The stars are being tracked by the telescope mount. Most telescopes have a driven mount that tracks the rotation of the Earth and matches it exactly, such that the stars will remain these pinpoints, and anything moving with respect to those pinpoints, like a meteor, will leave these trails. The trails point back to the so-called radiant or origin of where the meteor shower comes from. That's typically named after a constellation. And that constellation will be the source direction to look for when these meteor showers arise.
- 8.8** There's essentially one per month I've listed on my website that you'll get access to, the four major ones, five major ones that you can see almost every single year. Sometimes you can't see them as well because the Moon is out, happens to be a full Moon during the day or the night at which the meteor shower peak is expected to occur. That's just the breaks. But sometimes you'll be treated to a show, which may have something like one or two meteor that you can see every minute for hours and hours. Sometimes there's thousands, but that hasn't happened in a very long time.
- 8.9** So what are meteors? What is the terminology? I should first say, meteors are the phenomenon where some body, typically a chunk of rock, ice, dust, or metal, makes its way through our atmosphere at tremendous speeds, many, many times faster than even a low Earth orbiting satellite. It could be 20,000, 50,000 miles per hour. At those speeds, hitting even our atmosphere at the upper levels of the atmosphere, it's like slamming into a brick wall, and these things explode, and they release a chemical explosion. They basically burn up in the atmosphere.
- 8.10** This causes a bright display of the fireball that you see on previous slides. They typically are broken into two classes, rocky, stony, rocky-type meteors and metal or metallic meteorites. It's actually much more common to find these metallic meteorites, but they're actually more rare in a sense. The stony meteorites that are found in some places are actually very hard to distinguish from ordinary rocks. They actually look like ordinary rocks. They give us a glimpse into the early solar system's formation because they're exactly as old as the Earth and the Sun system.
- 8.11** Actually, these fragments, which you'll each get, and those of you who join at briankeating.com/PA, your Peterson Academy, will also get your very own metallic meteorite. Now, a meteorite refers to this object having flown through the atmosphere and crashed somewhere on Earth or another planet or a body in the solar system. In fact, this is a fragment of a meteorite that originated on the Moon, on the Earth's

Moon. It got here on Earth and landed in northwest Africa after some meteor didn't shower. There's no atmosphere on the Moon, but an object, a fragment of an asteroid, hit the Moon surface, blasted off some of its material, making a crater maybe 20 million years ago.

- 8.12** That material floated around the Moon, eventually got captured by the Earth's gravity, much stronger, and fell to the Earth some maybe 10 million years ago. And some of it was found, the fragment of it was found, in northwestern Africa. That's where this sample comes from. You can come up later and touch it if you like. It's a real piece of the Moon. The astronauts brought back samples of the Moon. We can compare this chemical properties, as I'll send to you about the meteorites, but we've compared the chemical properties in our labs and so forth of the Moon rocks to the Moon rocks taken back to Earth from NASA astronauts.
- 8.13** It's actually a felony if I were to have possession of an astronaut-collected sample. But this was collected by Earth's gravity and the US Postal Service. So it's totally fine for me and you to interact with it. You won't get arrested like so many have in the past. As I said, they're named after—there's almost one a month—they're named after the constellation that they appear to originate from. There's nothing magical about the stars that are in that constellation. It has to do with the comet.
- 8.14** Typically, the meteor showers that produce the most active meteor showers that we can see are typically produced by a specific comet. You can see the associated comet for many of these is identified, the most famous of which being Halley's comet, which was last visible in 1985, 1986. Will next be visible 70 years after that, 2050 something. Hope to be around to watch that. That fragments of that comet, which is a dirty snowball, iceberg, floating around in space, broke off as it came close in its orbits, many millions of orbits over billions of years, orbiting the Sun.
- 8.15** Twice a year, the Earth plows through the trail, if you like, of the elliptical orbit that that comet is on. It happened to be tons, literally, of material from that comet still left that is also orbiting the same path as the comet itself orbits it. So twice a year, we intersect with those ellipse contact points, and that produces a meteor shower, which in October is called the Orionids. That's coming up. And in May, earlier in the year, six months earlier, five months earlier, it's called the Eta Aquarids. And there's many of them. As I said, there's basically one almost every month. You'll get information about that from my website.
- 8.16** Again, definitions. Meteoroids are basically like asteroids. They're chunks of rock floating and orbiting in space. When they enter the Earth's atmosphere, they become meteors. They burn up, they get obliterated and chemically dissociated in the Earth's atmosphere via heat. Meteorite is any fragment that survives its transit through the atmosphere without completely being obliterated to nothing or skipping off the atmosphere and going back out into space. That happens a lot, too.

- 8.17** Anything that lands on the Earth, whether rocky or metallic like these are, is a meteorite, rite meaning rock, and meteor meaning associated from a meteor, meteorite. Meteor itself means high in the air. So this was true, that they were atmospheric phenomena. The meteor itself is an atmospheric phenomenon. The meteoroid is a space phenomenon, and the comet is nothing to do with the atmosphere whatsoever. Typically, you can tell if a meteorite is a meteorite, and rather just some chunk of metal, like this chunk of iron over here, this basically shot put over here, you could tell by looking at its composition.
- 8.18** It will have a different ratio of different isotopes or alloys of metals that are found in this, which is human-made, versus this, which is a meteorite. Also, you'll see in these meteorites, if you have a large enough sample, or if you have a microscope and you can cut open one of yours that I give to you, you'll see this is a particularly beautiful one. This one is a combination of minerals that look like gemstones embedded inside of a metallic structure. But you also see in the metallic structure, they look like fingers or they're actually crystals. They're actually domains or crystals that are a unique characteristic of meteors. They form predominantly in zero gravity.
- 8.19** So this object cannot produce such patterns, these with [Widmanstätten] patterns, rather Widmanstätten patterns. It's a German word for this type of interesting structure that you can see; if you cut a meteor slice and then etch it with a type of acid, you'll see these crystals emerge, and that is only true of objects that form in zero gravity. We call these sometimes the poor scientists' space probe because you can actually go back in space and time and collect materials of the early solar system, and then compare them to different objects you find on Earth and on other objects, like the Moon.
- 8.20** I also have, I'm a connoisseur, aficionado of all things space, as you probably have gleaned. I also have a piece of Mars. I have a rock, a meteorite, that came, originated from Mars, and it cost more than all my other meteorites combined. My budget was really exceeded when I bought that little tiny slice, only one or two grams of the planet Mars, but it's really quite cool. The meteorites are named after the place that they land. So it's the constellation from which they originate is the meteor shower. When they're in the atmosphere, they have colors and properties.
- 8.21** And then they land. You name the landing site, the meteorite fall, the type of meteorite, for that place in which they landed. These are from, that you'll get, are from a fall in Argentina. Many tons of material landed in Argentina, probably about 7,000 years ago. And it was named after the landing site. These are called Campo de Cielo, which means field or camp of the stars or of the sky. I have many, many tens of kilograms of this material, then I brought much of it with me. Some of these, as I said, this meteorite was found in Africa. The Martian meteorite was found also in a different part of Africa. This one, I believe, is from China, where it has these beautiful mineral gem-like inclusions.

- 8.22** And so they all have different names based on where they were found. I mentioned the meteorite from Mars. There's also meteorites, sorry, rocks that are collected from the astronauts, compared to rocks that are found on the Earth's surface. The middle one is found in Antarctica, where I've been twice. And this is found in a region called the Allan Hills. And I'll talk about those in just a bit because they have an important role to play in our search for life in the universe as well.
- 8.23** And then there's a chunk of a meteorite, not from the Moon, not from Mars, but from the asteroid Vesta, which is one of the biggest meteorites in our solar system. So that's a chunk, a meteoroid, that was a fragment of an asteroid that became a meteorite after being a meteor, who knows how long ago. So it's a little confusing, but if you pay attention, you can keep track of it.
- 8.24** Very small particles make up what's called a power law. So you're much more likely to find small fragments of a much, much larger body than much larger findings or samples of a large body. It's called a power law, and there's something like hundreds of times more samples that are smaller than the sample that's 100 times bigger than a given sample. That's roughly the behavior that we see. The smallest objects are actually impossible to find. It's very hard to find these things.
- 8.25** Although your roof collects, actually, you could take a magnet and you could find tiny little fragments on everybody's roof in America, pretty much. You're sweeping out the gutter of leaves, you might find some tiny, but they're so small, you'd have to have a microscope to identify them. And then you'd have to look at their spectroscopy and maybe do some other chemical test to find out they're actually meteorites. So just take my meteorite as tuition payment and pay Peterson Academy. It's much easier.
- 8.26** They have different composition, either rocky, typically rocky, made of silicon and carbon, and other objects, those are called chondrites, rocky meteorites, and then there's metal ones that are interesting in many ways because they're highly magnetic. They're made of iron. Why are they made of iron? Because that's the last thing the supernova made before it exploded and could no longer keep its outer layers held out from its gravitational collapse, producing a core collapse supernova, an iron core collapse supernova.
- 8.27** So iron is very magnetic. And so we can take magnets and put them and attach them and see if the meteorite, and you'll be able to play with this with your meteorite. So here's a magnet. I can pick up this chunk here. These are even stronger rare Earth magnets, much more powerful. The Moon meteorite is magnetic. The Moon meteorite that I have here is not magnetic. It's a stony meteorite, so there's no magnetic behavior whatsoever on it. It's interesting, under the microscope or even to your eye, you can see colors. You're not used to thinking about the Moon as having different colors.
- 8.28** Some of this is proof that we can demonstrate that Apollo astronauts really did go to the Moon. We have samples of the Moon. They collected samples. They're not related

by collection agency. Now, NASA didn't collect this. You can compare, and they have the exact same chemical composition, even some of the same colors that are useful in distinguishing this from an ordinary rock. So, like this massive steel or iron ball over here. And then, as I said, each one of you will get one of these. And then they all act as magnets to one another, and you can play around with them. It's quite fun. And that's another way you know you're not getting just some random piece of stainless steel or something. I wouldn't do that to you. Not again.

- 8.29** Here are two falls. One is in Namibia. The natives that live around there made it into this monument. The Hayden Planetarium, where I got to do that display one time a few years back for my first book's launch, is quite spectacular. It's many tons. It's like 25 tons. Iron is incredibly dense, incredibly heavy. Remember, this is the byproduct of a nuclear explosion, of fusion process that took place 5 billion years ago. At least these ones are, or this one may have been made much more recently.
- 8.30** Now, what would happen if a meteor of this size, compared to, say, oh, this fellow over here, a dinosaur, were to have an impact of something many, many dozens of feet across? Well, it would cause some level of destruction. As shown here, we look on the list of sizes versus destructive capability versus the frequency of how often these things are impacting the Earth. We'll describe how infrequently or frequently these occur. I know I've given you a lot of anxiety. I've talked about the end of the universe, and I'm going to talk about the destruction of the Earth. But actually, some of these have to do with the creation of the Earth and the creation of life on Earth itself, potentially, and possibly the creation of other life forms in our universe.
- 8.31** So something like this size, if this is like a giant stegosaurus or something like this, this is many, many, let's call it 50 meters across. It's much, much smaller than that, even in scale to this guy. But if that were to impact, it would take out something the size of New York City. And there's a meteor crater in Arizona, not far from California, where I live, that I fly over on occasion. Coming out here, we flew near it. It's called Meteor Crater, Arizona. I'll show you a picture of it. That crater is essentially the size of most of Manhattan. So if that meteor had hit Manhattan, it would have destroyed the whole area.
- 8.32** There was an enormous devastating event that occurred in Siberia, which is very low population, and it caused a huge amount of devastation, but almost no lives were lost. That was over 100 years ago. Here's that crater, Barringer Crater, sometimes called Meteor Crater. And when these extremely large objects, like the 50-meter-diameter progenitor of that object of this crater, hit the ground, it was completely vaporized. There are some tiny fragments of it still around, and you can find those around the crater if you've ever been.
- 8.33** I've never been there on the ground, but it's the best preserved meteor crater in the world. And I always say it's really cool. This meteor hit the Earth right next to the gift shop. I mean, how lucky? They were lucky. Now, if it was near New York City, god

forbid, where I come from, that's how big the crater would be. Take out all of Lower Manhattan. Couldn't it hit in New Jersey? I'm sorry, I'm a New Yorker. I can't resist.

- 8.34** So they will find sometimes fragments of the meteor that produced it. But much more interesting are objects like this. This is in Antarctica. And you might think of Antarctica as a pretty boring place, but it's actually incredibly interesting. It's the least populated continent on Earth. It has some of the highest elevation. It's obviously the coldest continent on Earth. It has an incredibly rapid rise from sea level, ocean level, where there are glaciers and there are penguins, and then all the way up to the plateau, the South Pole. And there's even active volcanoes on the coastline of Antarctica.
- 8.35** And the only way to get there, it's the only continent that you can only go to, well, I suppose if you're super wealthy, you could go there, but very few people do that. I did meet two people on my first trip to the South Pole, which is a completely boring, desolate, and isolated place, one of the most remote places you could ever hope to visit on Earth. There's only 200 people there, and there's no one in the surrounding 700 nautical miles. The whole continent during the summer, which starts in the middle of December, and so the summer in the southern hemisphere is winter in the northern hemisphere and vice versa.
- 8.36** So when I was there, I only go during the summertime, December through January, February. If you miss the last flight in February, you're stuck there until November when the first planes can land. Because when they land, if the temperature is as low as it is during, say, August or July, it could be a negative 100 degrees Fahrenheit, even colder than that. At those temperatures, planes have hydraulic fluid and their gasoline, type of jet fuel, it freezes, and the whole plane, the interior of the plane, can basically be completely destroyed. So the plane's gas turns into hydraulic fluid, and that ruins these multi-hundred, tens-of-millions-of-dollar engines. So they won't land there.
- 8.37** If you have a problem, you are basically on your own. So you have all these medical exams you have to go through, psychological exams you have to go through, if you're staying long enough. You have to do dental exams to have any teeth pulled that could even be a problem. There's not a dentist for 3,000 miles in any direction. But another type of research, I went there to build a cosmology telescope called BICEP. You can read about that in my first book, and I talk about that a little bit in my cosmology course.
- 8.38** But the other thing that they do there is study the planetary properties. They study ecological environments down there. They even do psychological studies on people that spend the whole year there in isolation when no planes can get in or out. 40 people spend their lives there. And they found what is the optimal ratio of women to men and what happens if there's a problem and there's actually a sheriff down there. It's incredible. There's no doctors, but there's a guy or a girl, and they have a gun. It could be a pretty spectacular place if you don't mind not seeing anybody else for about six to eight months.

- 8.39** And it's completely dark, basically from the beginning of June till the middle of September when the Sun starts to come up again. Completely dark. So these collectors here, these are scientists that are collecting meteorites on the ice shelf of Antarctica. Because if you see a rock just sitting in the middle of nowhere, the only other structures in the back there, there are some mountains there, but it's very hard to get a rock just from nowhere. So what they'll find is they'll find chunks of rock, and they'll trace that to their origins in meteors.
- 8.40** ALH stands for Allan Land Hills. Now, that particular meteorite played a big role in science and even in the movies. If you ever saw the movie Contact with Jodie Foster, which is about first contact with an alien civilization, there's a scene with former President Bill Clinton, and he's on the White House lawn, and he's talking about this discovery. And he says, and I won't do his Arkansas accent, although I love to do it. He talks about this rock, or he says, "This discovery speaks to us across the ages." And he's talking in the movie about the fictitious discovery of an alien civilization.
- 8.41** And when they cut into this meteorite, they found these objects, these structures, which were for a very long time thought to be microbes. So microbes from space. And actually, this particular rock came from Mars. So it was thought that this asteroid, fragment, meteoroid, originated from the surface of Mars, and it contained deep inside of it microbes. That'd be pretty spectacular. And for about 10 years, people debated, studied, tested this object. And to this very day, there are some that still claim it does contain microbes, that these are actual microbe fossils, or byproducts of the respiratory process of fossil's life, fossilized life.
- 8.42** I would say the jury is pretty much out on that. Pretty much, people have decided it's not evidence of life on Mars, that it's some other explanation that's not consistent with being the byproduct or actual evidence of life on Mars. It doesn't mean Mars life didn't exist in the past, but it's something to take into account. As I showed you before in this particular sample that I own, you have this beautiful pattern, these Widmanstätten patterns. Very hard to say, German word. And it implies that the original parent body, the asteroid of it, was very large and it had fragments in it, and it also had different chemical constituents inside of that asteroid.
- 8.43** So there could be complex chemistry, molecular chemistry, possibly even trapping in it. If you combine movies together, you could combine the movie Contact with the movie Jurassic Park, and this could be like amber, and there could be a little fly in there. We could take its DNA from the dinosaur it bit on the surface of—no, none of that is actually the case. But it is interesting to note that from looking at specific scientific predictions, we can learn about the composition, the history, and even the potential future of life in our solar system.
- 8.44** We know about life and so-called OOL stands for origin of life on Earth. Do they play a role? There is a concept that life could be delivered from other bodies in the universe, in our galaxy, particularly, maybe our solar system. There's a Moon of Saturn that's

particularly promising and will be explored called Enceladus. Obviously, Mars—there are many rovers that are there looking for evidence of life on Mars at this very moment. We never saw any evidence for life on the Moon. That doesn't mean, though, just because there's no evidence yet, that there's no existence yet. So evidence of absence doesn't mean absence of evidence or vice versa.

- 8.45** So these meteors are forming in collisions. The question is, could these collisions be like a seeding process, a process by which materials from other parts of the galaxy or solar system end up on Earth to explain the origin of life on Earth? Not the origin of life in the universe. It could be that life originated somewhere else a long time ago, far, far away, and was brought here. But that would still be an extremely important mystery to solve. How did life begin on Earth?
- 8.46** Maybe life only began on Earth, though. And then, perhaps the converse process is taking place. Because just as this rock came from the Moon via an impact on the Moon's surface to the Earth, so too does material from Earth make its way to Mars. There are meteorites from Earth on Mars. It's inevitable. And there might even be components that have some life forms on it because Mars and Earth are 4 billion years old and have been sharing the same solar system for literally billions of rotations of orbits around the Sun.
- 8.47** That's plenty of time to spend, to spread material, and possibly, if that material is hardy enough, that it could make its way to survive and be inculcating another life or another potential ecosystem on other planets. Maybe not on Mars, but maybe on other planets in our solar system. So it works both ways. Impacts cause material to come to Earth, but also spread material from the Earth, and not only from Earth and Mars, but from every other body in our galaxy, potentially.
- 8.48** And we'll see about the time scales involved, and they're very short on the time scale that Earth has been around for, for information and even life forms to make their way to and from Earth. It's very fascinating to think about. What I want to point out is the role of cosmic collision. The Moon was formed from a bigger body that preceded the Earth, called Theia. Theia was hit and impacted the early Earth, and that broke off a fragment of the outer crust of the Earth. And that fragment, over millions or perhaps a half a billion years, cooled down, formed into a spherical shape and became the Moon, the original source of this guy.
- 8.49** Now, what's different about the rocks from the Moon is that they don't have much iron at all, if any at all. And that's because the Earth doesn't have as much iron in the crust. It has some. Otherwise, we couldn't be able to have iron in our blood because we're eating stuff that grows organically in the Earth's surface. But it's nowhere near as much as in the core. The core of the Earth has a liquid and a solid iron core, two different types of core surrounding each other.

- 8.50** So there are simulations of this process, of this collision, which seem to match the formation and the prediction that there shouldn't be much iron in these Moon rocks, either the meteorite version that I have or the versions brought back by the astronauts. Simulations agree with this model, and the findings as well. So that was one collision. That's probably the oldest collision that's relevant to our existence. Many scientists think that, without the Moon, life couldn't exist. The Moon produces tides, which exchange material, organic material, from the subcrust zone to the crustal zone.
- 8.51** And in the Earth's ocean, they provide nutrients and the chemical processes that provide mixing, which then can be responsible in part for the evolution and different processes that take early primitive single-celled organisms up through the macroscopic and much, much more important organisms like us. The next collision that took place had to do with comets. Now, what happened to the Earth? Well, the Earth has also been bombarded by comets for perhaps 100 or more million years. It took place after the formation of the Moon, but before the next collision that we'll talk about, which was the collision in the Yucatan Peninsula that caused the extinction of the dinosaurs.
- 8.52** So that cometary impact, during what's called the heavy bombardment phase of the Earth's early existence, was unique in the solar system, where the only solar system that has liquid water in it, that liquid water was brought to the Earth by the pummeling of the Earth by comets. But if you think about it, even though the Earth's surface is 70% covered by water, the amount of water is pretty minuscule. The total water relative to the Earth's radius is much, much less than the radius of this plastic Moon rock container, compared to this iron sphere.
- 8.53** In other words, if we took all the water in the Earth, the oceans, every lake, every river, all the fresh water, all the saltwater, and made it into a sphere, that sphere would have a smaller diameter than this plastic container compared to this if this is the Earth's rock. So, yes, all this water is distributed, 70% covers the surface, but it's very minuscule. But if it was a little bit more, 30% more, the whole planet would be covered with water, and we wouldn't be able to create any technology that uses things like molted metal, metallurgy, semiconductor transistors.
- 8.54** In other words, if the Earth had a little bit more water, less than 50% more than it does now, it would be covered in water. If it had a little less water, we might not be able to live because the amount of fresh water is almost microscopic. Even though it looks like this is huge, it's less than half the diameter of a pea compared to the radius of something like that. It seems like we're in this Goldilocks zone. In fact, scientists do call it the Goldilocks zone. Just enough water that we can have the properties that we'll see in just a moment that are deemed necessary for life. To have liquid water is believed to be one of the prerequisites for life. It makes me thirsty just thinking about it.
- 8.55** But this water originated from a comet, many comets bombarding the Earth. Luckily, they bombarded the Earth after Theia hit the Earth to form the Moon. Because if it came before that period, that water would have boiled off and been completely

obliterated by the impact that created the Moon from this pre-Earth body and its impact with this object called Theia. So, it's almost miraculous, we have to say, that there's something incredible about the fact that not only do we have these two collisions, one to create the Moon, to create the conditions of recycling, of convection, of transportation of nutrients in the tidal zones of the Earth, the Moon also has craters on it.

- 8.56** Guess what? Every single crater is a bullseye that we missed or a bullet that we missed getting hit by, thanks to the Moon's gravitational force. Luckily, the Moon is just the right distance away from us, that it's not too close, and that we would get hit by almost the same number of meteors and impacts, some of them quite large, much larger than New York City, and so we wouldn't be here either. And then following it, how lucky are we that we had enough comets to fill up our oceans, but not too much, to have fresh water and the water cycle? That is also a really incredible gift to life on Earth and seemingly a necessity for life on Earth.
- 8.57** And then the last major collision. All these had to occur in just the right order, in order for us to be here. It's not proof of some divine intervention necessarily, but it's amazing to think about the improbability of us being here, able to study how improbable it is that we are here studying how improbable it is that we're here. The very famous impact, where a much, much larger impactor than this hit in the Yucatan Peninsula of Mexico and blasted an enormous crater that you can't see because it's under the ocean right now, pulverized the crust of the Earth, heated up the temperature of the Earth, but not too much. Not too much.
- 8.58** It ended up darkening the skies for years and years, killing off a lot of vegetation, which was the staple, the base of the food pyramid for these guys. So eventually, they would die off, and that would then leave room for us, the rise of the mammals, so that we could then dominate the planet as we have, in just 66 million years. Dinosaurs were around for 150 million years before that or more. And even predinosaur organisms, things like crocodiles and things, they still exist. Sharks in some ways still exist. But dinosaurs themselves were the dominant apex predators on land, and then when they died, we took over— we meaning mammals, not we meaning rodents or something like that.
- 8.59** How do we know this took place? Well, there's actually corroborating evidence in many different ways. But one of them is that there's a layer of iridium, which is a metal that would be found in these asteroids and meteors, and they have been found in even these little ones that you have in small amounts. And at a period of time that geologists traced to 66 million years ago, there's a layer around the entire Earth. It's also when most of the fossils can be dated using radioactive dating. This is carbon—this is corroborating evidence that seems to indicate a mass extinction event took place.
- 8.60** Again, not too much that it destroyed all the life on Earth and destroyed the Earth itself. It came after both the late heavy bombardment, the great bombardment that

delivered the water that the dinosaurs and the mammals need. And after the Moon came to exist, which provided the tides and the mixing and the solubility of water and saltwater in the intertidal zones to provide the first nutrients that may have provided the first incubators for life in the universe. Sorry, the first life on Earth.

- 8.61** So the effects on Earth are dramatic, can lead to mass extinction. Question arises, can we do anything about it? Can we prevent such an event from occurring? The dinosaurs didn't have a very advanced space program. Too bad for them, but good for us. Had they, maybe they would have deflected it, they would still be here. Now, can we do anything about it? There's basically several different components that go into a strategy to preserve life on Earth from such an event.
- 8.62** One would be first to spend way more money than we do. We spend almost no money tracking such objects that could cause devastation. On Earth, we only get notice maybe a few years or a decade away. A lot of people think, well, we could just find it, send Bruce Willis up there, blow it up, and then we'd be in good shape. You have this giant asteroid coming to the Earth, just blast it apart with some nuke, and everything's fine. But the problem is, once you blast it apart, you don't get just nothing left. You get billions of fragments. Now you don't have one city-sized meteorite, but now you have many, many house-sized meteors, maybe billions of them, and that's almost just as bad.
- 8.63** It's like a shotgun versus a howitzer. And so you'll see that that might be hard to do. My colleague and friend Phil Lubin, at the University of California, Santa Barbara, has developed an innovative strategy using lasers to gently nudge away objects when they're very, very far away, coupling fairly high-powered lasers in space, powered by solar panels that receive this 1,000 watts per square meter, day and night, providing imaging. So he would have a telescope that would detect it and a spectrometer to determine the composition, what it's made of.
- 8.64** And then you would have a laser tuned to that particular chemical composition that could either blow it up, disperse it when it's maybe light, not light, when it's maybe thousands of astronomical units away from the Earth. And then with a very, very high-powered laser, at the focus of a telescope, using a telescope in reverse, basically—do you see that? You see the laser light is now expanding out. The opposite of a telescope in reverse is a beam that's going to expand to be a very large diameter, but still with very, very highly focused and concentrated light beam.
- 8.65** So that could be used to just gently nudge the asteroid. And all you have to do is change its elliptical orbit by a little bit and effectively pressure it out of an orbit that would impact the Earth, maybe steer it towards another planet, where there's nobody living there, or maybe steer it into the Sun, and then wouldn't have to worry about these impacts at all. OK, so again, meteorites available on my website, briankeating.com/PA, and you guys can pick up yours here, and then you get

information about what they're made of, how to see them, what their composition is, and you can learn more about those as well.

- 8.66** So once we know the existence of us, we know that life is possible. Is life inevitable? Is it almost impossible not to have life in a universe like ours? In other words, is there a probability, not just a possibility—of course, there's always a possibility—is there a high probability of the existence of life in the universe outside of Earth? We know about life. Some say we don't even know if there's intelligent life on Earth. We're not going to debate that. We're just going to say, we assume that life exists in one place. That's enough to prove that life could exist elsewhere.
- 8.67** We've discussed the discoveries by the amazing team on the Kepler space telescope and follow-up observations by the transit-type telescope called TESS, transit exoplanet searching spacecraft. Many of these are looking for particular forms of exoplanets where they're perhaps just like the Earth. Now, admittedly, we're biased, but that's the only planet we know that has life. So at least we should look for as many planets like the Earth that are orbiting stars like the Sun as there are. You could have a star like the Sun that has no planets like the Earth. You could find a planet like the Earth that's not around a star like the Sun.
- 8.68** Those are all possibilities. There are many possibilities. There are exoplanets orbiting around pulsars, which are neutron stars that look nothing like our Sun. And yet you could have an Earth-like planet near such an object. Now, the question is, does it exist in what's called the Goldilocks zone? Not too hot, not too cold. For what? For liquid water. Liquid water is thought to be a prerequisite for most forms of life. Every form of life that we know about utilizes liquid water in one form or another.
- 8.69** If a planet's too far from a Sun-like star, it'll be too cold; you'll have an icy planet. Too close, you'll have a planet with no water in liquid form, maybe none even in steam or vapor form. Why is water so important? I mean, it's one of the most abundant molecules in the universe. It's made of the most abundant atom in the universe, hydrogen, and the third or fourth most abundant element in the universe, oxygen. So its constituents are very common, and many objects in the universe have water in solid or liquid form. Well, actually, we don't know too many that have a liquid form, but we know many that have an ice form. All the comets are basically icebergs floating through space.
- 8.70** So, comets bombarded the Earth during this heavy bombardment after the formation of the Moon, before the extinction of the dinosaurs. Again, that timing is crucial. If you had the water forming in the ocean, and then it was hit by an object the size of the planet Mars, we would not be here, and there would be no water on Earth anywhere. Some comets are actually found to have organic, meaning molecules that have carbon, that were typically called organic because they were produced in organs. The very first organic compound to be synthesized was urea, which is in the primary component of urine in mammals.

- 8.71** So they thought for a long time that only organs, namely living creatures, could produce organic compounds. Now we know we can synthesize them, like they did with urea 150 years ago. But some of these objects have organic compounds in them. Amino acids have been discovered, many other things. Water is what's called a universal solvent. Many different chemicals can dissolve in it, and only when you have this dissolving can you have chemical reactions—chemical reactions that can take place to then sustain and generate life and modify it in forms of evolution.
- 8.72** There are many theories, going back to Charles Darwin and even probably before, of a primordial soup, some early Earth environment with the right chemistry, the right temperatures, the right electrical properties, lightning, and so forth, that could take inorganic, non-living material, convert it somehow into living material. It was first proposed in a short essay by Charles Darwin. He called it his warm little pond theory, that if you had just the right things, like ammonia, salt, phosphoric acid, light, heat, electricity, that, oh, what wonderful things would happen, that living creatures could be produced, again, without any preexisting creatures.
- 8.73** So this would be the formation of life from non-living chemicals. And the chemical reactions I've translated there, just for those aficionados watching at home. And this recipe can produce things like amino acids. And it was actually attempted in the 1950s to actually make Darwin's warm little pond, but in a warm little pond in a laboratory at the University of Chicago. And later, these two experimentalists, Stanley Miller and Harold Urey, came to UC San Diego, where I teach, and they continued this research.
- 8.74** But basically, they tested the hypothesis. They put some chemicals, then they shocked it with electricity, replicating lightning. They kept the right temperature. And eventually, they found a lot of different amino acids, which are precursors of building blocks of the types of proteins that later make longer polymers and perhaps make DNA and things like that. A question that's been asked for thousands of years, but in particular quantitatively in just the past 100 years, is, what is life? To say if life exists elsewhere, we first have to have a good definition of what life is. Unfortunately, there's no one definition that everyone accepts.
- 8.75** Many people believe that an organism has to metabolize energy, produce energy. It has to reproduce. And once it reproduces, it has to transmit information about its genetic makeup to other entities. Some say DNA is responsible for life or it's a prerequisite. Then there's a question of intelligent life. And then there's also another question of technological life. This racoon knows how to get into my garbage can every Sunday, but it's not technologically sophisticated. It doesn't build tools, doesn't have prehensile thumbs. It can't do, construct technology the way that we can. And only we can.
- 8.76** Dolphins are incredibly smart, maybe have some capabilities that are shared in common only with humans. Bonobos and chimpanzees share 97% of their genetic makeup with us as people. And yet they have no iPads or iPhones as far as I know. So, how do you go from a single-celled organism shown on the left to a trillion-celled

organism or more? So for cellular life, like we are made of, not things like viruses necessarily—talk about that separately—but we need to have some sort of chain that has information, that can also transmit that information, that's resilient and sustainable, but is also self-replicating.

- 8.77** There are many candidates for how this could be, but there's always this problem of the chicken or the egg. How do you get DNA or RNA, as we'll talk about, without something that produces it, namely a protein or an organ or a cell that already existed? These are famous questions of the priority of what had to come when. Just like there had to be a specific order for the water on Earth to form and the Moon to form and the dinosaurs to die, perhaps for us to be here, so too you have to ask the question of, what is the first structure, the structure that produces the information or the information needed to produce the structure itself?
- 8.78** My kids joke, you can find out which comes first, the chicken or the egg. Just order a chicken and egg on Amazon, see which comes first. So far, that hasn't worked out. Have not been able to publish that theory yet. So replication and reproduction and information storage is accomplished, now we know, through DNA. That was very difficult to suspect and understand for many decades until Rosalind Franklin and Watson and Crick and Wilkins discovered that it had this double helix structure.
- 8.79** There's another hypothesis that actually DNA is not the essential ingredient, actually RNA. So RNA is a single-strand helix, not a double helix shown here. But we think that perhaps some suggest that RNA came first, and RNA has properties that DNA doesn't have, in that it can catalyze reactions. It also has reproduction. It has the same four-letter base pair; it's different letters, but basically the same digital code that can encode information about life. Some people suggest that the early life on Earth form from RNA still doesn't answer the question of, where did the RNA egg come from first?
- 8.80** So both of them are nucleic acids. They form the nuclei of cells. They transmit information. They have slightly different properties. One is single-stranded, versus DNA is double-stranded. They have four letters. There's a letter U called uracil, which is replaced in DNA by thiamine, but that doesn't seem to be super important. The question is, how could either one of these form without a structure to produce it first?
- 8.81** OK, so once we know that there are worlds, exoplanets, in the habitable zone where life could be supported theoretically because of the possibility of liquid water on the surface of those planets, we can measure their mass, we can measure their volume, and therefore we can know their density. Compare their density to the density of Earth and see, how similar are they? And there are many hundreds of them that have been found through various techniques that are exactly like the Earth, and many of them are around Sun-like stars. You have Earth-like planets around Sun-like stars.
- 8.82** I told you also about the ability to measure the properties of an atmosphere surrounding an exoplanet. As it transits a star, sometimes you get very lucky and you

see the blockage of the light from the Sun, from the star, blocked by the rocky planet's core. But then some of the light goes through the atmosphere, allowing you to do chemical spectroscopy and see what's in that atmosphere. And if you saw things in the atmosphere that were only possible on Earth from either the byproduct of respiration and agriculture, if you saw things even like oxygen, carbon dioxide, you'd start to think that there could be life there.

- 8.83** And then if you saw things that are much more advanced, very advanced chemicals that would be demonstrated by their presence, be demonstrated by the spectroscopy of the atmosphere, that could be the byproduct of technology. You saw vapor from copper sulfate that's only produced in laboratories on Earth. It doesn't occur randomly or naturally. You might be able to say, well, there must be some technology there that's producing it. And that would be incredibly exciting. So there are missions that are planned in the next 10 years or so to go way beyond what Kepler did.
- 8.84** Now that you know there's a habitable planet in an environment around a Sun-like star, now let's zoom in on the atmosphere of that planet, and learn as much as we can about it. And maybe we could detect radio waves or technology signals from those planets as well. So far we haven't. And so for the last 70 years or so, people have been consumed, after learning about the solar system, learning about the planets, that there could be life on those planets. Even as early as 100 years ago, when Mars was first looked at through a telescope, people perceived the patterns on the surface of Mars that seemed to indicate that there were canals there. In other words, that there must be entities there building navigation canals in the oceans of Mars which don't exist.
- 8.85** And that became very, very good fodder for science fiction, to write books like *The War of the Worlds* and others about Martians invading and life forms on Mars that we know don't exist right now. But as technology improved, and as we learn more and develop more technology on Earth, we start to assume that there could be other species of civilizations, of organisms that had similar technology, or worse, more powerful technology, that they could be here right now.
- 8.86** Most famous of which is the flying saucer that crashed, allegedly, in Roswell, New Mexico, in 1947. A lot of this came about right after the atomic age, when there was a lot of existential fear of humanity's technology overwhelming our capability to control that technology. A lot of it is psychological in these perceptions, but they occur even to this day. A framework to estimate how many different civilizations could there be, possibly, in our solar system—sorry, how many different civilizations possibly could there be in the universe or in our Milky Way galaxy? It's parameterized by something called the Drake Equation, which multiplies together various factors.
- 8.87** How many planets there are, how many stars there are, what are the rates of these things, how long do technology civilizations last for? And it gives a framework. The unfortunate thing is, in science, again, your goal is to produce not an estimate, but an estimate of how ignorant or wrong you might be. And this does not do that. So it's

overly optimistic, and you can get almost any number you want out of it. So it's an interesting way to think about it.

- 8.88** But when we look at our galaxy alone, we take into account the speed of light and the size of our galaxy, as big as it is. It's only, quote, unquote, 160,000 years across, traveling at the speed of light. But that's nothing compared to even the age of the Earth. The age of the Earth is 4 billion years old. So it's plenty of time to travel, even at a much slower velocity, about the velocity that the Voyager spacecraft that we made on Earth is traveling right now. It's about 20 kilometers per second, so a little bit faster than that. It only takes 1.6 billion years to cross the whole galaxy. And that's in the worst-case scenario that we are at the edge of the galaxy or near the edge of the galaxy.
- 8.89** And the only other civilizations at the other edge of the galaxy, still they would have had at least three times the age of the Earth and 10 times almost the age of the universe to get to us. But they're not here. So Fermi asked this question, if the universe is teeming with life, where are they? There are many answers to this question. We're going to go through a few of them, but I invite you to construct your own and to think about ways to evade this. In other words, for the paradox to not be a paradox, for us to be able to answer it. And we'll go through some of those today.
- 8.90** So why hasn't—if they exist, where are they? That's the fundamental question. And so to make sure that we don't have a false negative, that we don't rule out the existence of alien technology, we're not looking for dolphins swimming in an ocean of an exoplanet. We can't see that yet. We might be able to see evidence of life or organic molecules or what have you, but we can't see them specifically. What we're looking for is technology, radio waves, laser beams, craft traveling through space and time. Can we detect that?
- 8.91** So there's two projects in particular. There are many projects looking for alien life, but in particular, these two are some of the best-funded examples of this. They're called breakthrough projects. One is called Breakthrough Listen, where these very large radio telescopes are scanning as many stars as possible, looking at millions of stars in our galaxy, and analyzing, do you see something special? Do you see Netflix on some other planet? Do you essentially see something that's not random, that's not accidental, that's not noise? You see a coded message coming to you from another star system.
- 8.92** Then we should be able to detect it out to many, many dozens or thousands of light years even. And again, it's plenty of time for a civilization that's known about our existence. Now, we've only been making radio waves that could be detected for about 100 years, maybe 70 or 80 years. So that means, if another civilization is looking for us, they would just hear about us now if they're 70, 80 light years away. That wouldn't be enough time if they're beyond that to even know that we exist using radio waves alone.

- 8.93** They could detect our greenhouse gas emission and warming of the planet. And some say that's a technology signature because agriculture is a form of technology. And that's been around for 10,000 years. So you could put out a signal even 10,000 years that's passive or the active signals that we've been putting out since the 1936 Olympics, which are broadcast around the world by radio and television that is going in all directions in our cosmos.
- 8.94** So some other civilization would have heard about us if they're within 80 so, 90 or so light years away. That's not enough time for us to hear back from them. That'll take twice that time. But if there's something within 10 light years, that's plenty of time to start sending messages back and forth. The question is, what would we do if we found out? It's like, what do we do if we found out a comet's coming? You didn't think so many existential questions would be asked in an astronomy class, but these do come up, and I find it incredibly fascinating.
- 8.95** The inverse of this is to send technology to another stellar system where there is an exoplanet. So there's an exoplanet near Proxima Centauri or Alpha Centauri. And the same Russian billionaire, Yuri Milner, is funding a \$100-million project to send tiny little chips, little cellphone camera-sized sensors on a light sail, pushed by lasers, the inverse process of this DE-STAR program, my friend Phil Lubin, to push these tiny cloud of swarms of thousands or tens of thousands of cellphone cameras that will have their own power source to fly to this exoplanet.
- 8.96** Right before it crashes into it or gets shot down, who knows, it will send information back to Earth. That will take four years of radio travel time from the star system. It will take approximately 16 to 20 years for the little tiny microsatellites to get there, traveling at 20% of the speed of light. It's a long-term play. But as it's said, if you don't even make an effort, the probability of success is zero. You don't get a chance to understand the universe unless you're willing to wait a very long time because the universe is huge.
- 8.97** The last topic we'll cover is, why hasn't an early-developed alien civilization already colonized the whole galaxy like Star Wars, where they have cantinas and exoplanets, and luckily, everyone speaks English. That's cool, even though it's a galaxy far away from here. So, just one civilization would have plenty of time to take over their whole galactic neighborhood and eventually colonize the whole galaxy. So why haven't they done that? We don't see that. Maybe they are doing it and they just haven't gotten here. That's a possibility. Maybe they have already seen Earth, found us to be too difficult to deal with, and decided they don't want to interfere with us, or they don't want to be associated with us.
- 8.98** So they could be out there, in other words. The paradox could be answered by the fact they're just not interacting with us. When you go to the San Diego Wild Animal Park, they keep you in these buses, and so you can see the giraffe in the distance and the lion. Sometimes they'll come up to you, but they'll arrest you if you jump out and start

screaming and punching the chimpanzee in the face. So other civilizations could be doing the same thing to us, watching us from a distance but not interacting with us. Maybe we're some wildlife refuge of the galaxy.

8.99 Maybe there is no other option. But again, this is also terrifying. Maybe there are no aliens in the universe and we're it. Maybe they're not interested in colonizing. Maybe they don't want to have anything to do with us. So this is a fascinating thing. Again, these are the types of questions that I love about astronomy. You stop asking them because you have to get a job and pay for your kids and take them to school and pack their lunches and pick them up and do all the things that we have to do in our quotidian daily lives. But that's what makes astronomy so fun.

8.100 And I hope you have enjoyed the ride through space and time, through civilizations, from early, primitive, prehistoric times, which were focused, very intently, on studying the pattern of the skies, the heavens, the constellations, the planets, thinking that they were gods that needed to be propitiated, to be pleased, or they would exact their wrath upon us. We built modern telescopes that are modern versions of ancient pyramids, and the soothsayers and the astrologers have given way to precise scientific predictions about the past, the history of the universe, the current state of our universe, other universes and beyond.

8.101 I want to give you just a couple more final perspectives by Carl Sagan, who, with his wife, who I interviewed in my podcast, Ann Druyan, they wrote the book *Contact*, about this encounter, a fictional encounter, of an alien civilization with the Earth. But Carl was an amazing person. He was an excellent scientist, but he also had the mind of a poet and an artist. And he, at one point, commanded the Voyager spacecraft to take a picture of the Earth on Valentine's Day, 1990, when it was past the orbit of Saturn.

8.102 And you can't even see the picture, the so-called pale blue dot, on this image here. But there's a single pixel of this camera that was built in the 1980s to be launched—sorry, it was built in the 1970s, to be launched in the late 1970s to fly by Saturn in 1990—and turned around the camera to face the Earth to take a picture of it. At that distance, the Earth is only one pixel in this, maybe it was only a few megapixel camera, so it wasn't that advanced. But he wrote about that, and the pixel is blue, and there are these streaks like you see sometimes in clouds, and that's because of dust and other optical artifacts in the camera.

8.103 But he says, "Look at that dot," the pale blue dot. "That's here, that's home, that's us. On it, everyone you love, everyone you know, everyone you've ever heard of, every human being whoever was, lived out their lives. The aggregate of our joy and suffering, thousands of confident religions, ideologies, and economic doctrines. Every hunter and forager, every hero and coward, every creator and destroyer of civilizations, every king and peasant. Every young couple in love, every mother and father, hopeful child, inventor and explorer, every leader, teacher of morals, every corrupt politician, every

superstar, every supreme leader, every saint and sinner in our history, lived there on a moat of dust suspended in a Sunbeam."

- 8.104** That we fight over the fraction of a dot he talked about as being so emblematic of the futility, but also the majesty of what it means to be the only conscious technological life form in the universe that we know about. There's simply no evidence for other existence. So this might be us. This might be it. We might be it. We talked about the technological impact. I didn't even mention how developments in astronomy lead to developments in technology that we know and love and use. Cellphone cameras, those all came from astronomical discoveries. The cell phone itself came from radio astronomy in the 1960s, doing radio-astronomical research.
- 8.105** And so we've now understood incredible details about chemistry, origin of life, and the transmission of information in a vast, vast cosmos, and even come down to do things that we take for granted; like the GPS that got me here today would be impossible but for Einstein's theory of general relativity, which was created by one of the most important astronomers of all time. In the future, you'll hear about new telescopes. I want you to be informed about them. Soon we'll have a space telescope called the Nancy Roman space telescope, looking at dark energy, exoplanets, and objects that are transients in the universe.
- 8.106** In Chile, not far—oh, sorry. This is my observatory that I'm pleased to co-lead with my collaborators from around the world, called the Simons Observatory. We're looking for the fluctuations in the cosmic microwave background that might tell us what was the force that gave the universe the bang that set into motion the Big Bang that we observe today. We're searching for gravitational waves, neutrinos, and even the hypothesized Planet Nine.
- 8.107** At the South Pole, near one of my experiments called BICEP, is another one called IceCube, which is a neutrino observatory which doesn't use a mirror. It uses one kilometer of ice underneath the Earth's surface crust at the South Pole as a detector of the byproducts of neutrinos interacting with matter. And that detector has revealed some of the most powerful explosions in the universe since the Big Bang itself.
- 8.108** And I want to conclude with this fundamental tension that I always felt but never really articulated until maybe now. And that's this distinction between what I call Whitman or Feynman. So, Walt Whitman is a phenomenal poet. You may have heard *Leaves of Grass*, and he was one of the foremost poets of the 19th century, and really helped to soothe the nation in the aftermath of the assassination of President Abraham Lincoln. Then Richard Feynman was one of the towering physicists of the 20th century, some say all time. They had different approaches to things that we've encountered in this class.
- 8.109** So Walt Whitman wrote a poem, and it's called *When I Heard the Learned Astronomer*. It's a poem of despair and dejection of the application of formulae, of equations, of

reactions, of physics, to the ethereal, the beautiful, the spiritual. He says, "When I heard the learned astronomer, when the proofs, the figures, were ranged in columns before me, when I was shown the charts and diagrams, to add, divide, and measure them; when I was sitting, heard the astronomer where he lectured to much applause in the lecture room, how soon, unaccountable, I became sick and tired; till rising and gliding out, I wandered off by myself, in the mystical moist night air, and from time to time, looked up in perfect silence at the stars."

- 8.110** He's basically saying, "Don't bother me with your equations and your formulae and your telescopes and your tubes and this thing and that thing. I don't want to know. It's beautiful. Don't distract me. The night sky is poetry." And then Feynman, 50 or 60, 80 years later maybe, he made the retort to good old Walt. He said, "Poets say science takes away from the beauty of the stars, mere globs of gas atoms," as we've discussed.
- 8.111** "I too can see the stars on a desert night, and I feel them. But do I see less or do I see more? The vastness of the heaven stretches my imagination. Stuck on this carousel, my little eye can catch one-million-year-old starlight, a vast pattern of which I am just a part, what is the pattern, or the meaning, or the why? It does not do harm to the mystery to know a little bit about it, for far more marvelous is the truth than any artists of the past imagined it. Why do the poets of the present not speak about it? What men are poets who can speak of Jupiter if he were a man, a god, but if he is an immense spinning sphere of methane and ammonia, must be silent?"
- 8.112** In other words, we know about the composition of these objects. Does that make them more or less beautiful? I think, to me, the answer is obvious. Knowing more about the universe makes me appreciate it more. Appreciating it more makes me have gratitude for the mere fact that we exist, and we are the only conscious technology-producing organisms that we know about. That means we have to guard and take care of the universe. But most of all, we have to continue exploring it in our beautiful cult of astronomers. I hope you've had a good time enjoying the astronomical journey from Earth to the heavens and beyond.
- 8.113** I hope you'll take my other courses on Peterson Academy as they become available. I hope you'll stay in touch with me as well. Join my mailing list. You'll get these meteorites if you're watching at home, and you in the audience will pick them up right now. Thank you very much.
- 8.114** Hi, Professor. Question regarding the Big Bang theory and if it's a singularity, the known universe. The question is, is there an order of the known universe ruled by the force of gravity? That means universe is composed by galaxies, and galaxies are composed by solar systems, what compels meteorites or debris or whatever. Or is the universe a messy place?
- 8.115** Well, it's certainly, the universe is incomprehensibly large, and we've only literally scratched a tiny fraction of the surface of everything that exists in our observable

universe. The question of whether or not there might be another universe where there are other potential forces and fields and energy sources is an active form of scientific research. It's something that goes by the name of the multiverse. See, when we were young, there was only one universe. In fact, the word uni suggests one, and verse means rolled or thing. So it was, the entirety of all there was was the universe.

8.116 But as a consequence of the theory of cosmological inflation that I study with my team and colleagues, that theory comes under pressure, in that it's almost impossible in some scientists' conception for there only to be one universe. For there only to be one universe is like saying there would only be one planet, which, by the way, was all we knew about until 1995. So all of us, pretty much, many of us were born in that age. And so we didn't know of any other planets other than the Earth. And now it's ludicrous to say there's only one planet, I mean, besides our solar system.

8.117 And so for that reason, some scientists suggest we should be more humble and suggest that there could be other universes. Now, what do you call that? Well, the universes are like ours, but they're not in contact, they're not in touch, they're not in motion that we can discern with respect to our universe. But they could be discovered in future experiments, or we could inductively assume that they exist based on discoveries that my colleagues and I might make. So stay tuned for that.

8.118 All right, cool. Thank you so much for this course. It's been awesome. So my question is trying to get a visual of what happened after the Big Bang in the expanding universe. And so it's actually a two-part question. Question one is, could I think of this as a visual, like a firework going off, with everything an exact distance away from the same center point? Could we locate that center point? And the second question is, why is Andromeda the only galaxy that's moving closer to us, aside from everything else?

8.119 Yeah, very good question. So the first question is related to a very, very common conception that the Big Bang was a big bang. It was an explosion, when in fact, it was essentially the expansion of space at this unimaginable rate through a process that we're still learning about that may be this process called inflation. But the key distinction is that every point, even the Andromeda galaxy, even the Crab Nebula, you could go as far as you like away from Earth— every particular location in the observable universe sees the same thing that we do.

8.120 They see all galaxies moving away from them, with the exception of the very closest galaxies, which I'll explain in the answer to your second question. But every galaxy sees every other galaxy moving away from it. Therefore, each galaxy can claim to be the center of the universe. But if each galaxy can claim to be the center of the universe, there is no center of the universe. What we should really be thinking about, and it's almost impossible for a human mind that thinks in two and three dimensions to visualize a four-dimensional expansion, where this thing is expanding along lines and along planes, that each one of them can basically claim to be the center of the universe.

- 8.121** It's more akin to, say, a balloon that has been blown up partially with dots on it. That balloon is expanding, and every dot is moving away from every other dot. But there's no center of the balloon. The center of the balloon, that's not what matters. It's the center of the surface of an infinite balloon or an infinite trampoline, if you like, stretched out. There is no surface to something that's infinite. We don't actually know if our universe is infinite in size or not. It could be infinite in size. But the key fact is that the Big Bang occurred everywhere at the same time.
- 8.122** What we call the observable universe is the region over which light, traveling at the fastest possible speed, could have just reached us to this very moment. That is what we call the fictitious realm that is our universe. There could be another universe right next to ours, but separated by so much distance that we can't actually see it yet, but both universes came into existence at the same time. In each universe, each point, each observer will see every other observer moving away from it.
- 8.123** The answer to your second question is, the reason that Andromeda is moving towards it is that we are attracted to its gravitational force. And by Newton's laws, it's attracted to us. So we're experiencing gravitational pull. It's very massive. It's about two or three times the mass of the Milky Way. So it's actually pulling us towards it. Our galaxy doesn't have enough velocity to, say, orbit around it or stay apart from it. But there's only a few galaxies, the Magellanic Clouds, there's a couple of dwarf galaxies, there are the globular clusters, and everything else is moving away from us.
- 8.124** So out of 100 billion galaxies, there's maybe 10, 10, 20, that are actually bound to us gravitationally in a system that's causing us to eventually collide, merge, and make a giant galaxy that some call Milkdromeda. Although that'll take several billion more years to find out what they'll call it.
- 8.125** Well, first, thank you for doing this. You've done a fantastic job of being clear, and I think very interesting in every thought you discussed. I guess my question concerns what I would call settled science, because science is, in some sense, the process of finding out what's not settled. What are the current controversies in astronomy? Are you getting any surprises out of the Webb telescope, where we go, "Uh-oh, we got that one wrong."
- 8.126** Yeah.
- 8.127** What's going on?
- 8.128** You're right. The reaction of a good scientist to data that startles them should be, "That's great." Actually, it is great because if everything were understood, there are branches of science that are very well understood right now. Nuclear physics, for example, the process by which atomic energy fission is made. It's not an active area of research. Fusion is a different matter, and there are even some forms of fission that are important. But nuclear physics, it was very popular in the '50s and '60s after the Cold War, during the Cold War; it's less popular now.

- 8.129** The reason was, it became a victim of its own success. The theoretical understanding of subatomic processes became extremely well understood theoretically, simulations and more. Then experimental techniques and testing bombs and so on. That was very well classified. And so it became very well-settled science. Yes, there could be some surprises, some technological approaches that may be more important than others. But cosmology, astronomy, and the search for life, these are full of surprises. As I said, only a few decades ago did we learn there are other planets outside of our solar system.
- 8.130** We even learned more about our solar system itself in the last 20 years to say that certain planets are no longer planets. Just 15 years ago, 20 years. So it's an extremely active field that is full of new comprehension, new understandings. And in fact, as you hinted at, it's often said that science is the belief not in the knowledge and wisdom of experts, but the ignorance. In other words, if we had settled on the pre-Aristotelian idea that the Earth is flat— it certainly looks flat. That's what experts said. There was consensus. There was scientific consensus that 97% of scientists believe that. But of course, we know that's totally wrong.
- 8.131** We also know that it's not a perfect sphere, as I talked about. We're learning continuously about its processes, its age. Scientists, again, thought the age of the Earth was a few thousand years old. Then they thought 100 million. Now they know it's 4.3 billion. We've developed more and more understanding. Is water really needed for life? Is DNA needed for life? In my field, did the universe start expanding, and the Big Bang, that not only ruined the previous conception that the universe was static and unchanging, but it also had another series of questions that it then prompted.
- 8.132** What caused the bang to bang? Why did the universe not completely contract upon itself in a very finite short amount of time? Why is there structure in the universe to make galaxies, stars, people, universities? These are all things that were asked only after the previous scientific idea was put to bed. The static universe gave way to the Big Bang, but the Big Bang has its own challenges. I often say that the flaws lead to new laws. You learn about something from the things that don't quite fit and don't quite make sense. There are many of them in science, and that's what makes it so exciting and also ensures full employment for scientists because there'll always be things to uncover where we thought we were correct, and it just wasn't so.
- 8.133** We've talked a lot about what makes up the celestial beings and stuff like that. I guess my question would be, what's in between or what do we know about just the void of space in between?
- 8.134** Yeah, it's a very interesting question. Usually that comes up when people ask, well, what is the universe expanding into? Well, it's not really, as I said, expanding into, but the distance between things is getting bigger. So if you've ever seen astronauts on a space station or in a SpaceX capsule or whatever, you think, inside the capsule, they're there. It's warm. There's oxygen, there's food, they can eat stuff. But imagine you go right outside the spacecraft. What's in there? And let's go even farther. Let's go out to

the intergalactic space, between all the galaxies, not just outside of the space station or beyond the orbit of the Moon.

8.135 Let's go really far out there, and we take ourselves a little tiny box, a box that's about one centimeter on the side. And we ask, what's inside of that box? Really, what's inside of that box is about 400 photons from the Big Bang, from the cosmic microwave background that we discussed. There's maybe 100 neutrinos inside that box. There's an awful lot of, maybe, dark matter particles, perhaps. There's even more energy in the form of dark energy, but that's it. You won't find many chunks of rocks and stuff.

8.136 So now the question is, well, what's between the two photons? Let's say, pick two photons. What's between them? Well, the space. That's what it is. That's what space is. Space is an emptiness. That doesn't mean there's nothing possible because we know about the production of particles that can come out of a vacuum and all the other things. But it's essentially space. It's the same space that's right outside the space station. It's not some mysterious substance that we're expanding into, and there's some boundary between where our universe ends and begins.

8.137 As we said, there is no boundary because there's no center of the universe either. It's a concept that the human mind likes to impose upon things to better understand reality, but that's not the essence of what the true nature of space actually is.